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Scotchman

**CPO-350
COLD SAW**



**Please download our .pdf
"Guide to Extend the Life of Your Saw Blade" too.**

**Please read this Manual & the Guide
before operating this saw!!**



www.scotchman.com

MODEL
CPO-350
COLD SAW

S/N B12632 & AFTER

FEBRUARY 2024

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1.0 INTRODUCTION

The CPO-350 Cold Saw is designed to cut solids, tubes, flats and other profiles in grades of material that range from hot and cold rolled steel, annealed tool steels, stainless, aluminum, brass, copper, synthetics and extrusions.

Cold sawing is a process similar to a milling process. In most cases, this, combined with the self centering vise feature, gives a finished cut that does not require any secondary machining or de-burring.

Since milling spindle speeds are used in cold sawing, there are several things that are required to achieve quality results. The selection of the proper pitch (number of teeth) on the blade and the proper spindle speed for the type of material being cut are critical. Proper material clamping and a good quality coolant are also important.

Cold sawing has several advantages over band saws and abrasive saws. Besides the mill quality cut, cold saws have the ability to generate faster cutoff times than band saws.

There are no sparks and excessive noises that are associated with abrasive cutoff saws.

Cold saws also offer the advantage of blades that can be re-sharpened until the diameter of the blade will no longer cut through the material. The self centering vise allows for easy changeover to special clamping jaws, for profiles and extrusions. Having two spindle speeds enables the user to cut a wide range of materials.

By adding the power vise and power down feed options, the saw can be converted to a semi-automatic machine at a very reasonable price.

2.0 SAFETY PRECAUTIONS

1. Any individual operating this machine must be qualified, responsible and well instructed. This manual is not intended to teach untrained personnel how to operate equipment.
2. NEVER operate this machine with the guard disconnected or removed.
3. Wear eye protection, at all times, when operating or observing this machine in operation.
4. Do not wear loose fitting clothing, gloves or jewelry when operating this machine.
5. All electrical connections shall be made by a qualified electrician. This machine must be grounded in accordance with the National Electric Code.

- 6. Disconnect the machine from the power source before performing maintenance or changing blades.**
- 7. Practice good housekeeping. Keep the area around the machine clean and dry.**
- 8. When sawing, always support long pieces and make sure that the material is properly clamped.**
- 9. Keep the guard, as well as all other parts of the saw, in good working condition. Replace worn parts promptly.**
- 10. Do not alter or modify this machine in any way without written permission from the manufacturer.**
- 11. This machine is top heavy and must be anchored to the floor via the holes in the saw base.**

3.0 WARRANTY

Scotchman Industries, Inc. will, within three years of the date of purchase, replace F.O.B. the factory or refund the purchase price for any goods which are defective in materials or workmanship, provided the buyer, at the seller's option, returns the defective goods freight and delivery prepaid to the seller, which shall be the buyer's sole and exclusive remedy for defective goods.

This warranty does not apply to machines and/or components which have been altered, changed or modified in any way or subjected to abuse and abnormal use, inadequate maintenance and lubrication or subjected to use beyond the seller's recommended capacities and specifications.

In no event shall the seller be liable for labor cost expended on such goods or consequential damages.

The seller shall not be liable to the purchaser or any other person for loss or damage directly or indirectly arising from the use of the goods or from any other cause.

No officer, employee or agent of the seller is authorized to make any oral representations or warranty of fitness or to waive any of the foregoing terms of sale and none shall be binding on the seller.

Any electrical changes made to the standard machine due to local electrical code variation must be paid by purchaser.

As we constantly strive to improve our products, we reserve the right to make changes without notification.

This warranty is effective December 1, 2009.

4.0 INSTALLATION AND SET-UP

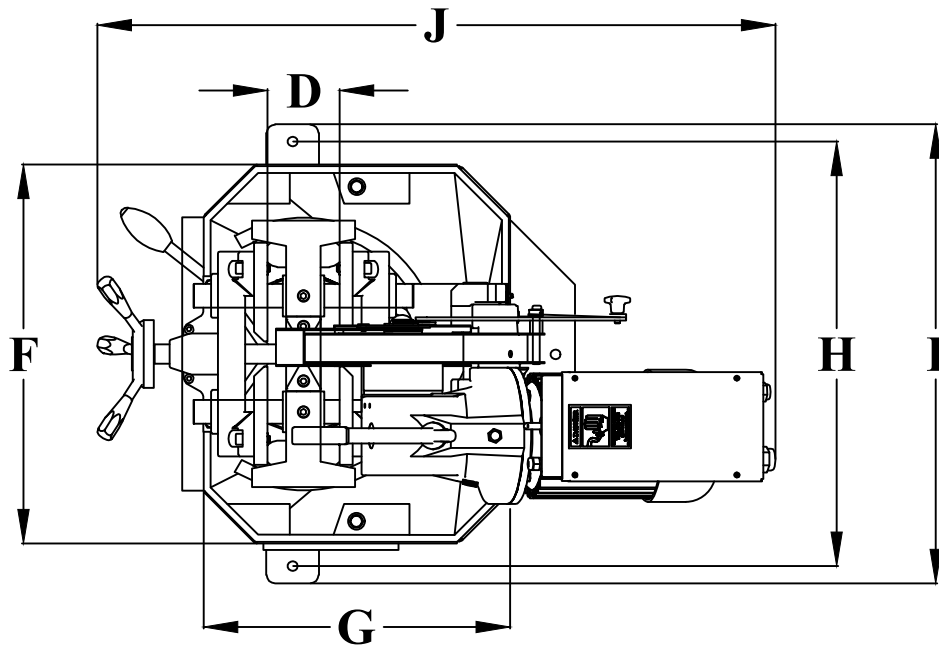
⊠> **CAUTION: THIS SECTION DISCUSSES INSTALLATION, SET-UP AND START-UP PROCEDURES. PLEASE READ IT THOROUGHLY BEFORE OPERATING THIS MACHINE. IF YOUR MACHINE IS EQUIPPED WITH EITHER THE POWER VISE OR THE POWER DOWN FEED OPTION, READ ALL SECTIONS CONCERNING THESE OPTIONS BEFORE OPERATING THE SAW.**

4.1 PHYSICAL DIMENSIONS

SEE FIGURE 1 ON THE FOLLOWING PAGE.

DIMENSIONS	INCHES	CM
A. HEIGHT	67.5	171.4
B. FLOOR TO VISE BED	36.8	93.47
C. FLOOR TO BASE HEIGHT	31	78.74
D. VISE OPENING	5.5	14
E. VISE DEPTH	2.75	7
F. BASE WIDTH	21.65	55
G. BASE LENGTH	17.48	44.45
H. MOUNTING HOLE CENTERS	24.5	62
I. TOTAL WIDTH	26.2	66
J. WEIGHT	660 LB.	299 KG.

TOP VIEW



SIDE VIEW

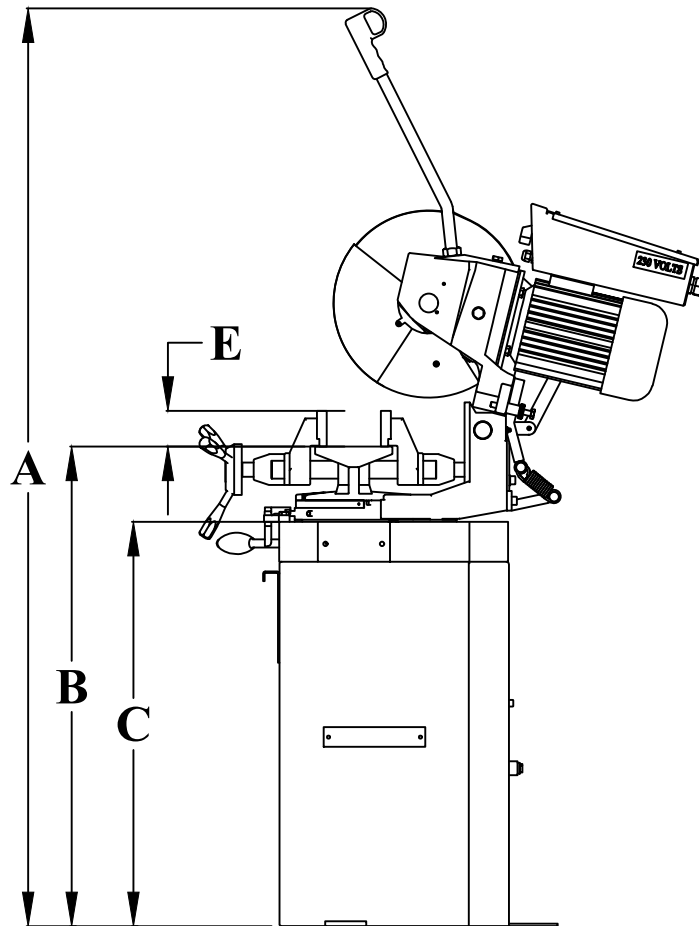


FIGURE 1

4.2 MACHINE MOVING PROCEDURES

SEE FIGURE 2 ON THE FOLLOWING PAGE.

This machine is shipped on a pallet and can be moved to the installation location by means of a fork lift.

**⊠> CAUTION : THIS MACHINE IS TOP HEAVY AND MUST BE MOVED WITH CARE, ON
HARD, FLAT SURFACES ONLY.**

All saws are shipped with the head locked in the DOWN position. Before lifting the machine, release the head on manual and power vise saws by cutting the banding strap and allowing it to move to the UP position. On machines equipped with the power down feed option, release the upper stroke control collar and allow the head to move to the UP position. Lift the machine, using the lifting eyelet provided.

Remove the pallet and place the machine in its final location. This machine is top heavy and must be anchored to the floor via the holes provided in the saw base.

LIFTING EYE

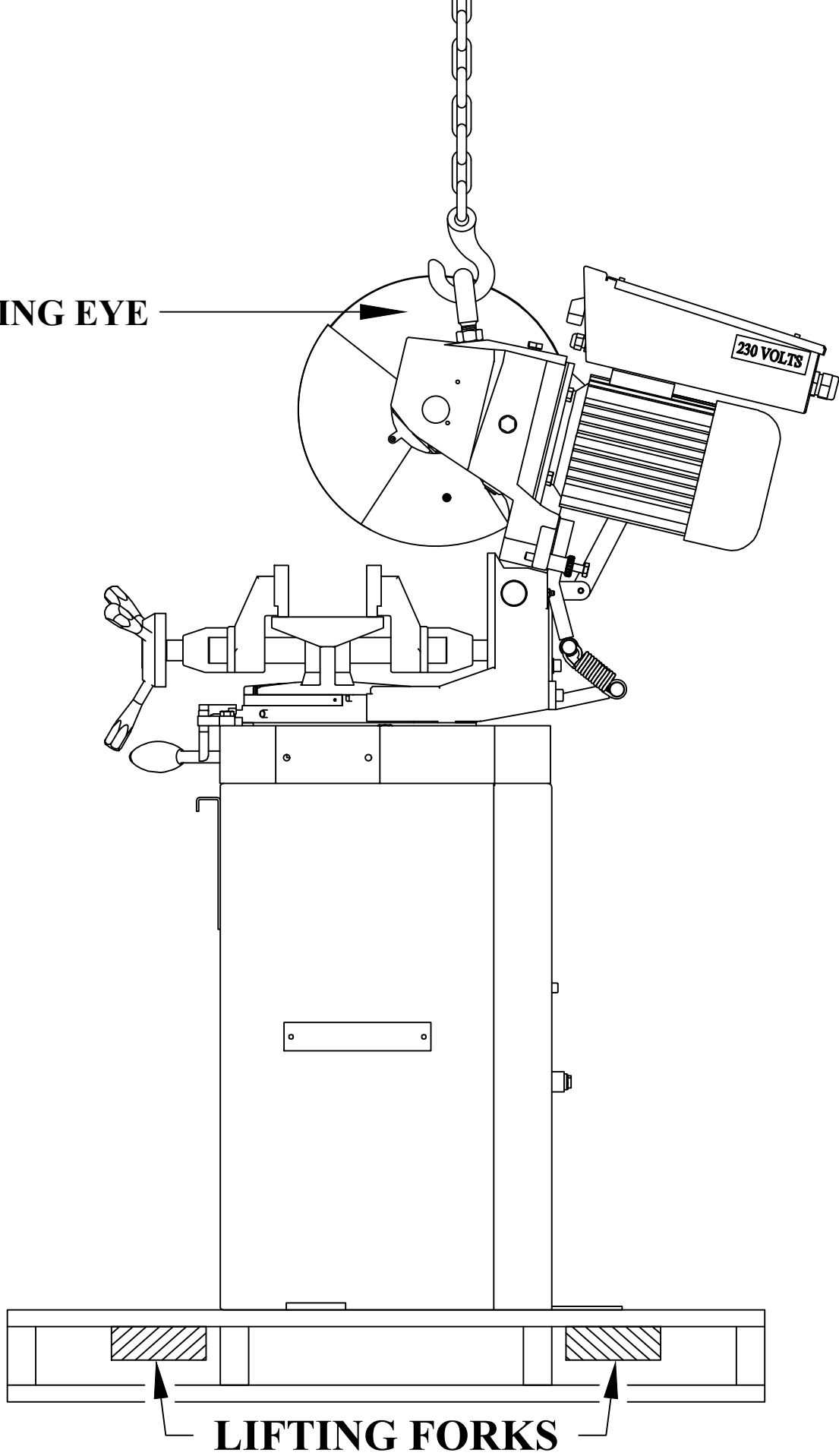


FIGURE 2

4.3 PHYSICAL INSPECTION

Once the machine is located, check it for any possible damage incurred in shipment. Remove the lifting eyelet and install the draw handle.

⊠> **CAUTION: DO NOT USE THE LIFTING EYELET FOR ANY MACHINES OTHER THAN THIS SAW. MAKE SURE THAT THE DRAW HANDLE HAS A JAM NUT ON THE THREADS BEFORE INSTALLING IT ON THE SAW. IF THE HANDLE IS INSTALLED WITHOUT THE JAM NUT, IT MAY CONTACT THE GEARS INSIDE THE HEAD.**

After the draw handle has been installed on manual and power vice machines, remove the cover from the electrical control box and connect the trigger switch wires. REFER TO FIGURES 3-1 thru 3-4.

Remove any other packing material and draw the saw head to its DOWN position to make sure that the guard opens properly. The guard should close completely when the head is up and open freely as the head travels down.

If the guard is not functioning properly, REFER TO SECTION 4.7 FOR THE MANUAL MACHINES OR SECTION 7.8 FOR MACHINES EQUIPPED WITH THE POWER DOWN FEED OPTION.

With the head in the UP position, check the oil level in the gear box through the sight glass in the casting just below the draw handle.

If your saw is equipped with either the power vise or the power down feed option, REFER TO SECTION 7.0 for additional information.

4.4 ELECTRICAL REQUIREMENTS

SEE FIGURE 3-1 THRU 3-4 ON THE FOLLOWING PAGES.

⊠ **CAUTION:** TO PREVENT DAMAGE TO THE MACHINE AND DANGER TO THE OPERATOR, ALL ELECTRICAL CONNECTIONS SHOULD BE MADE BY A QUALIFIED ELECTRICIAN. THIS MACHINE OPERATES WITH LIQUID COOLANT AND MUST BE GROUNDED IN ACCORDANCE WITH NATIONAL ELECTRIC CODES.

All three phase motors are dual speed and will operate on one voltage only. Single phase motors are available in the 44 RPM speed, only. If the machine is not the same voltage as your plant voltage, you will have to replace the motor and rewire the transformer and coolant pump. To insure satisfactory performance, the supply voltage should be (+ or -) 10% of the motor voltage rating. Check the motor data tag for full load current requirements. On machines equipped with the variable speed drive, the drive can be used as a phase converter.

THE ELECTRICAL DIAGRAMS FOR THIS MACHINE ARE ON PAGES 12 THRU 15, FIGURES 3-1 THRU 3-4. For supply lines ten feet (303 cm) or shorter, we recommend 12 gauge wire. For lines longer than ten (303 cm), we recommend 10 gauge wire. We do not recommend supply lines over twenty feet (606cm) in length.

CPO-350 LT (22-44 RPM)

MOTOR VOLTAGE	FULL LOAD CURRENT		HORSEPOWER	
	HI	LOW	HI	LOW
208	7.1	3.6	2	1
230	6.5	3.3	2	1
460	3.3	1.6	2	1
230 1/Phase	14			2

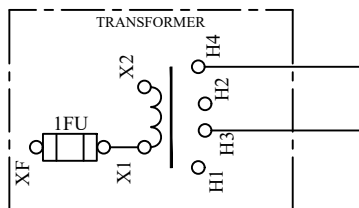
CPO-350 HT (44-88 RPM)

MOTOR VOLTAGE	FULL LOAD CURRENT		HORSEPOWER	
	HI	LOW	HI	LOW
208	9.7	7.1	2.7	2
230	8.8	6.5	2.7	2
460	4.4	3.2	2.7	2
230 1/phase		14		2

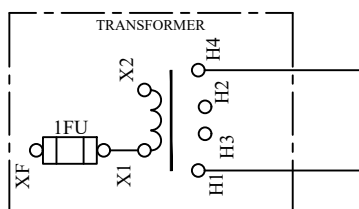
► **NOTE:** SEE SECTION 7.10 FOR THE WIRING SCHEMATIC FOR THE CURRENT ELECTRIC POWER DOWN-FEED SYSTEM.

MANUAL OR PK W/TRIGGER SWITCH

208V



460V



230V

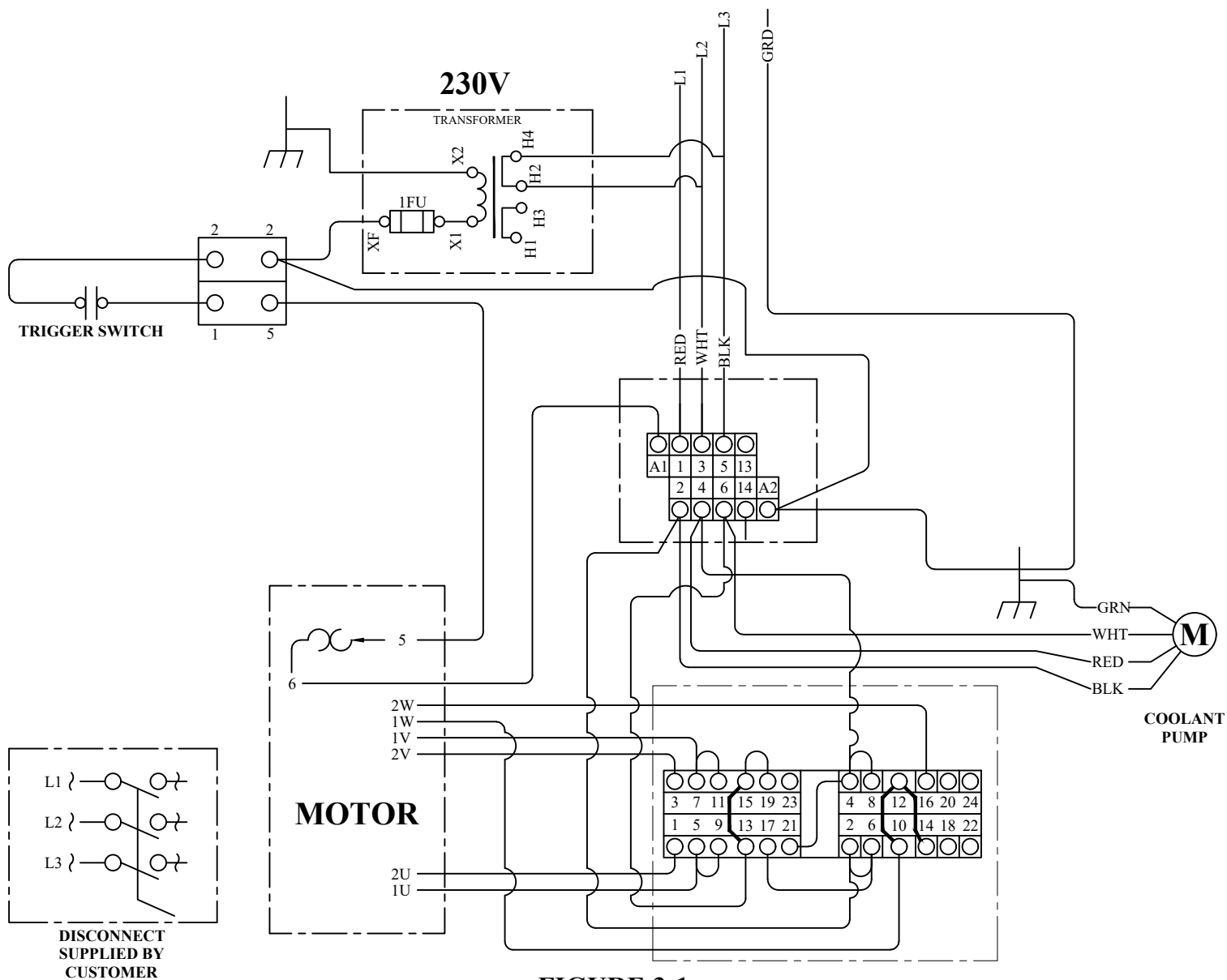


FIGURE 3-1

**PD W/EMERGENCY STOP
(S/N 8001 AND UP)
(SEE SECTION 7.10 FOR POWER DOWN FEED WIRING DIAGRAM)**

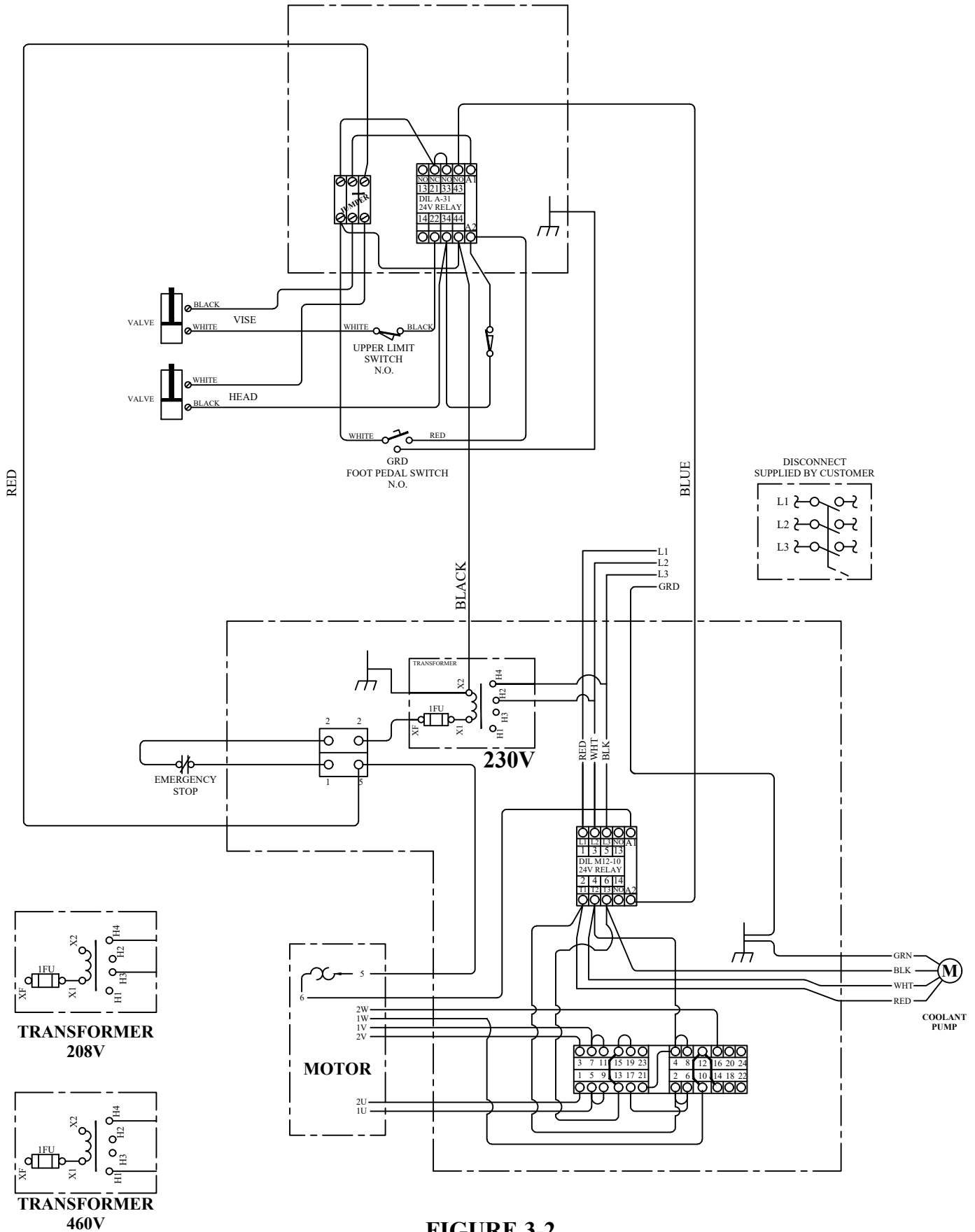
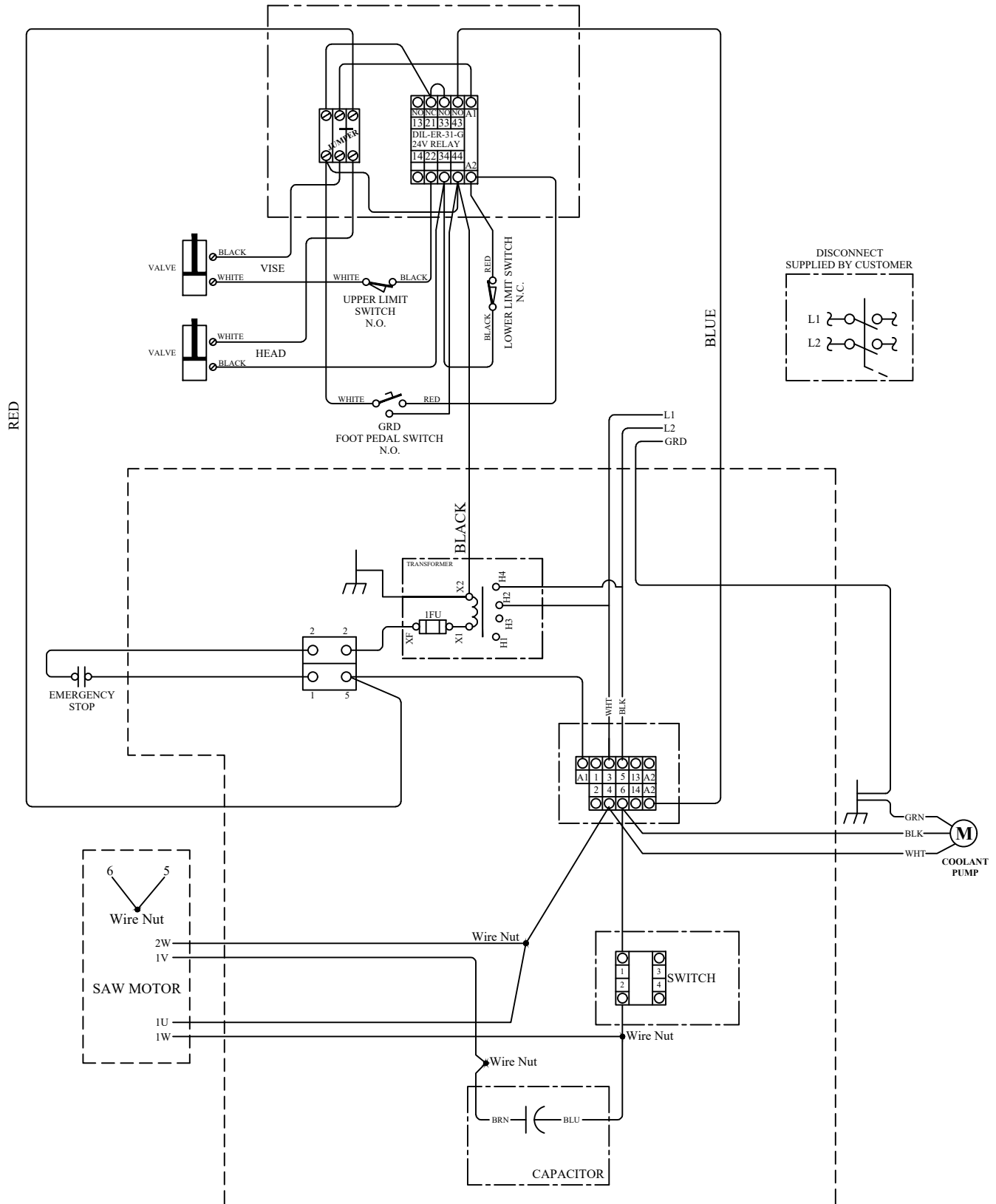


FIGURE 3-2

**CPO-350 1 PHASE PK/PD
(S/N 8001 AND UP)**

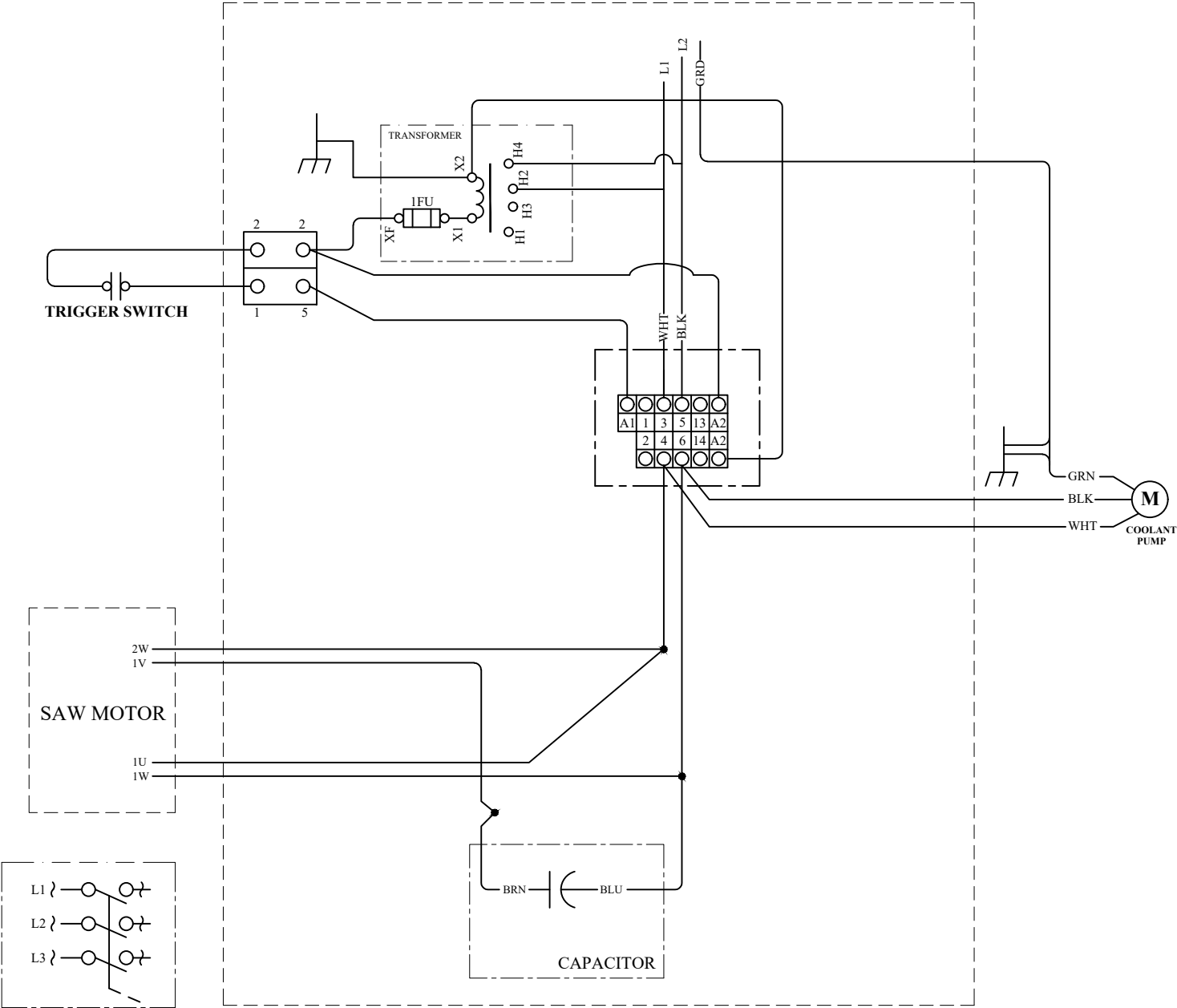
(SEE SECTION 7.10 FOR POWER DOWN FEED WIRING DIAGRAM)



IF RUNNING BACKWARDS SWITCH 2W & 1V

FIGURE 3-3

CPO-350 1 PHASE MANUAL & PK SAWS



IF RUNNING BACKWARDS SWITCH 2W & 1V

FIGURE 3-4

4.5 MACHINE START-UP

Before starting this machine, take the time to review the operator's manual thoroughly, to familiarize yourself with all of the functions of the machine.

We strongly urge you to follow OSHA directive CFR-1910.147 (effective 09-09-90) regarding lock-out, tag-out procedures. Keep in mind that the directive refers to all hazardous energy sources, not just electrical.

On machines equipped with either a power vise or a power down feed, the air supply must also be disconnected and locked or tagged.

Scotchman offers a lock-out switch for this machine as an option, if your plant is not equipped with lock-out capabilities. If you are interested in this option, REFER TO SECTION 7.14 or contact your local dealer or the factory.

Do not install a blade on the saw until after it has been powered and cycled several times.

To power manual and power vise machines, turn the HIGH/LOW switch to either the HIGH or the LOW POSITION and use the trigger switch mounted in the draw handle to start the motor. Always turn the HIGH/LOW switch to the OFF position when the saw is not in use.

To power machines equipped with the power down feed option, turn the HIGH/LOW switch to either the HIGH or the LOW position and depress the foot switch to start the motor. The foot switch can be used to cycle the saw head without starting the motor, by leaving the HIGH/LOW switch in the OFF position.

We do not recommend using the emergency stop switch to turn the machine off during normal operation. If the emergency stop switch is used, it must be manually reset by pulling the switch back out.

Always turn the HIGH/LOW switch to the OFF position when the saw is not in use.

Once the machine has been powered, check the rotation of the spindle. There is an arrow on the guard showing the proper rotation. If the rotation is not correct, the electrician will have to switch two of the three line wires.

If the saw is equipped with either the power vise or the power down feed options:

GO TO SECTION 7 - OPTIONAL EQUIPMENT

There you will find additional information and instructions on the options available for this saw.

4.6 FRL USE AND MAINTENANCE

SEE FIGURE 4 ON THE NEXT PAGE

CAUTION: SHUT OFF AIR SUPPLY BEFORE SERVICING THE FRL

The FRL (Filter Regulator Lubricator) is an important and often overlooked item on a cold saw. It helps to keep moisture from contaminating the pneumatic system and adds a little oil to keep the system lubricated.

REGULATOR

The regulator has the gauge and air pressure is adjusted the knob on top. Pull the knob up to adjust the air pressure and push it down to lock it in place. We recommend 90 psi to 105 psi to operate this saw.

WATER TRAP

Another important function with the regulator is that it has a water trap that is attached to the bottom of the regulator. It should be checked everyday. If the trap is a over half full, it should be removed and emptied. The drawing on the next page has detailed instructions on how to do this. Take care to not damage the o-rings.

OILER

The oiler should add a drop of oil to the pneumatic system every 5 -10 cycles. This can be seen at the top of the oiler. You can see a small tube that the oil drips from. If the oiler needs adjustment, there is a small screw located on the top of the oiler that can be adjusted with a small straight screwdriver.

OILER RESERVOIR

The reservoir for the oiler is attached to the bottom of the oiler. The oil level in the oiler should be checked every day. When the level gets low, the reservoir should be removed and filled about 3/4 full with a quality (ISO 22) air line lubricant designed for automatic oilers (such as our P/N 075759) and installed back on the lubricator. The drawing on the next page has detailed instructions on how to do this. Take care to not damage the o-ring.

CAMOZZI FRL (Filter/Regulator/Lubricator)

DISCONNECT AIR SUPPLY!!

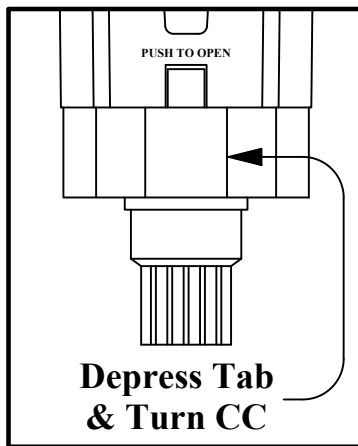
**AIR PRESSURE
ADJUSTMENT HERE**
Pull up knob to adjust pressure
Push down to lock in place

TO CHECK REGULATOR WATER-TRAP FILTER

Remove the water trap by pressing in the tab on the clear plastic part at the bottom and turning it counter-clockwise until it stops. Then pull water trap free. Filter can be removed by squeezing the tabs on top together, to allow for cleaning. When done, snap filter back into place and reattach in reverse order.

- See "DETAIL" below -

DETAIL



OILER ADJUSTMENT HERE
Use a small straight screwdriver to Increase or Decrease the amount of oil used.

Scotchman Air Line
Lubricant is recommended
Qt. 075753 - Gal. 075759

TO ADD OIL TO OILER
Remove oil reservoir by pressing in the tab on the clear plastic part at the bottom and turning it counter-clockwise till it stops. Then pull reservoir free from the oiler. Fill reservoir with oil a 1/2"-3/4" from the top and reattach in reverse order.
- See "DETAIL" below -

DETAIL

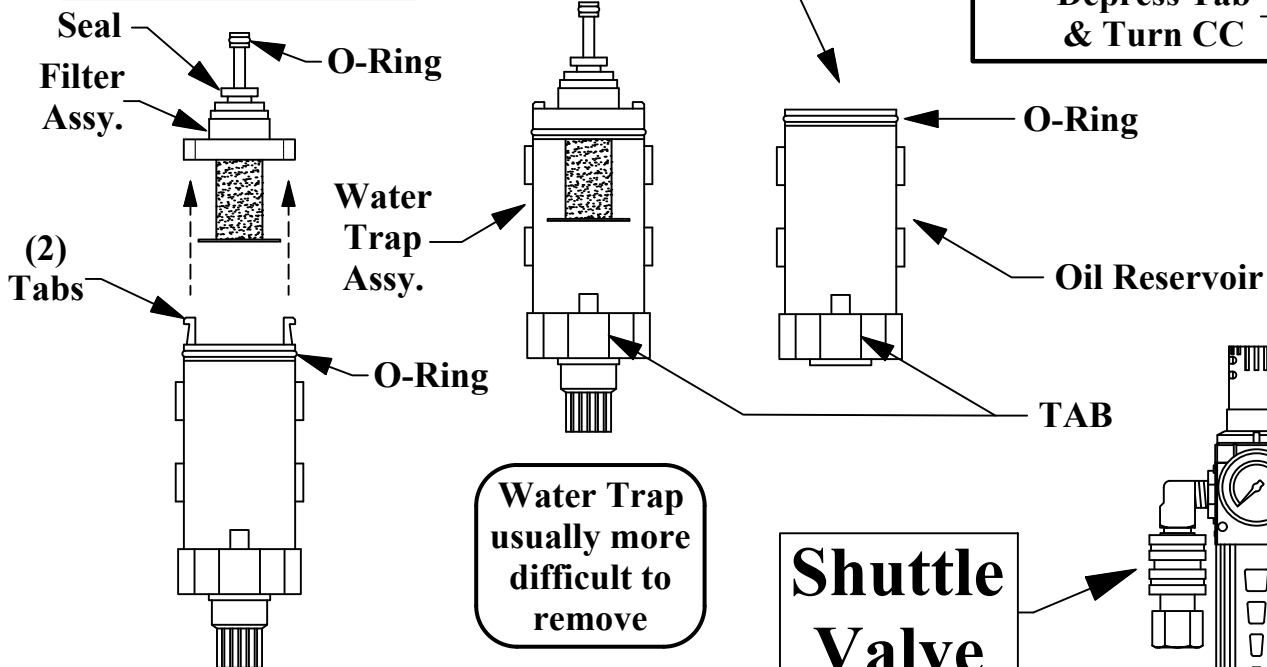
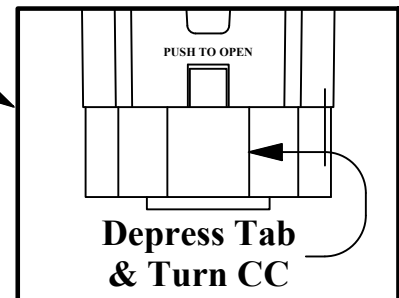


FIGURE 4

4.7 GUARD ADJUSTMENT

SEE FIGURE 5 BELOW.

FOR GUARD ADJUSTMENT PROCEDURES ON SAWS EQUIPPED WITH THE POWER DOWN FEED OPTION, REFER TO SECTION 7.8.

The proper adjustment of the blade guard on this machine is crucial to the operation of the machine and the safety of the operator. If the guard will not maintain proper adjustment, check the guard mounting bolts and rivet joints in the guard and linkage, for wear. Replace worn parts promptly.

USE THE FOLLOWING STEPS TO ADJUST THE BLADE GUARD:

- 1. Turn the power OFF and disconnect from the power source.**
- 2. With the head in the UP position, loosen the mounting bolts (A) on the linkage mount (B).**
- 3. Manually hold the guard open approximately 1/8 of an inch (3mm) at point (C).**
- 4. Adjust the linkage mount (B) down until there is tension on the linkage bar. Re-tighten the linkage mount bolts (A).**
- 5. Manually cycle the head up and down several times, making sure that the guard operates properly.**

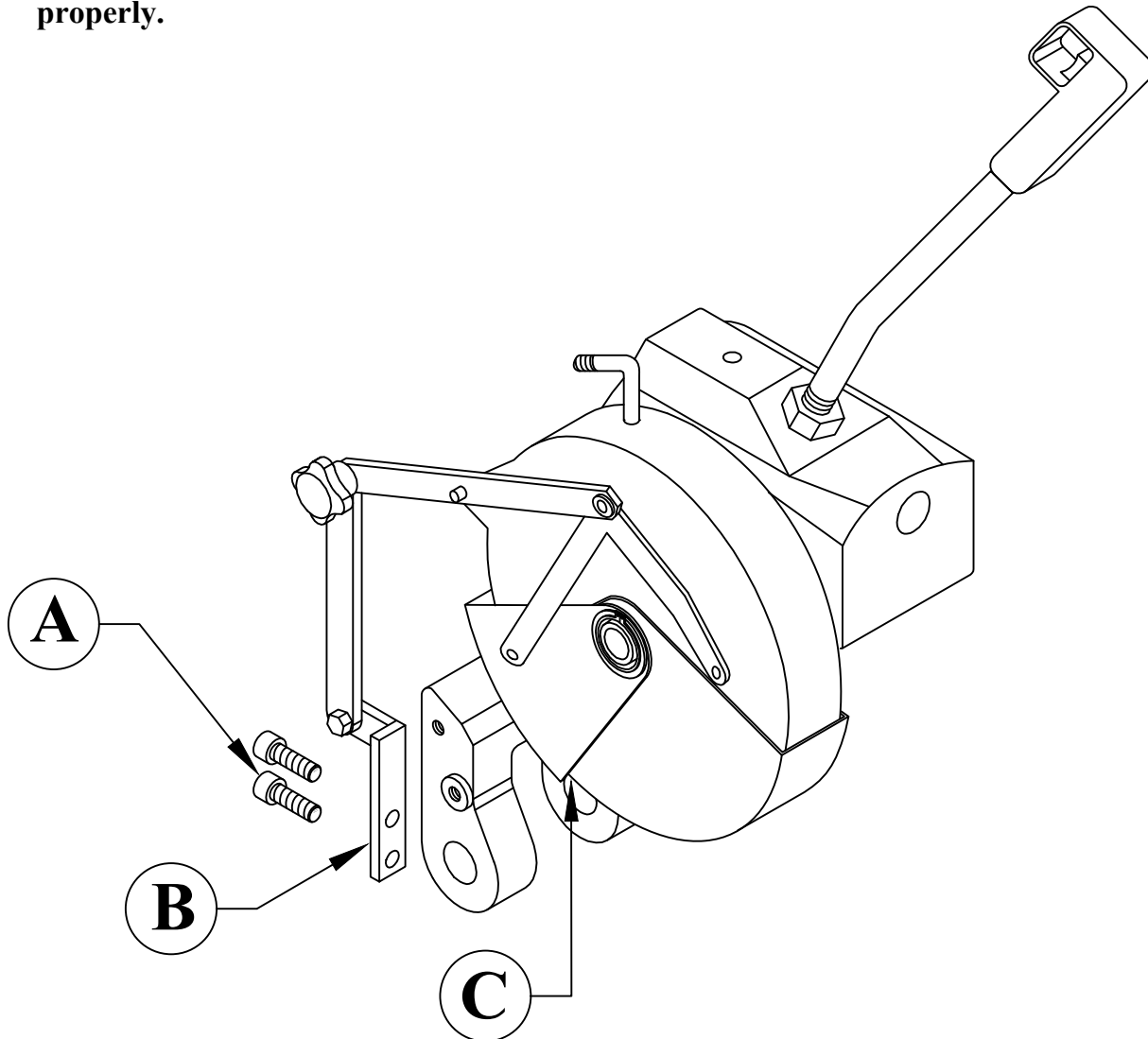


FIGURE 5

4.8 COOLANT SYSTEM

The coolant reservoir has a capacity of eight (8) gallons (30.3 liters). One gallon of coolant is shipped with the saw. For normal cutting, it should be mixed in a ratio of one part coolant to seven parts water. In conditions of heavier cutting, the ratio of water should be reduced to five parts. We recommend using only pure, synthetic, water soluble cutting oils. We recommend pre-mixing the coolant before adding it to the saw. When cutting alloy steels such as stainless steel, we recommend a special mix coolant designed for these applications. For additional information on available coolants, SEE SECTION 10.8.

The drawer located in the upper front of the saw base has a screen in the bottom to keep the coolant clean. Metal chips made from the cutting process will accumulate in the drawer and it can be easily removed for cleaning.

5.0 MAINTENANCE AND LUBRICATION

5.1 LUBRICATION

SEE FIGURE 6 BELOW.

Once a week, grease all of the pivot pins (A) and oil all of the rivet connections on the guard linkage (D). Clean the chips out of the vise at least once a day and apply penetrating oil to the spindle and guide pins (B) and (C). Clear the chips with a brush or similar device or use a shop-vac. DO NOT use compressed air. If your saw is equipped with a power vise or power down feed option, SEE SECTION 7.0 for additional information.

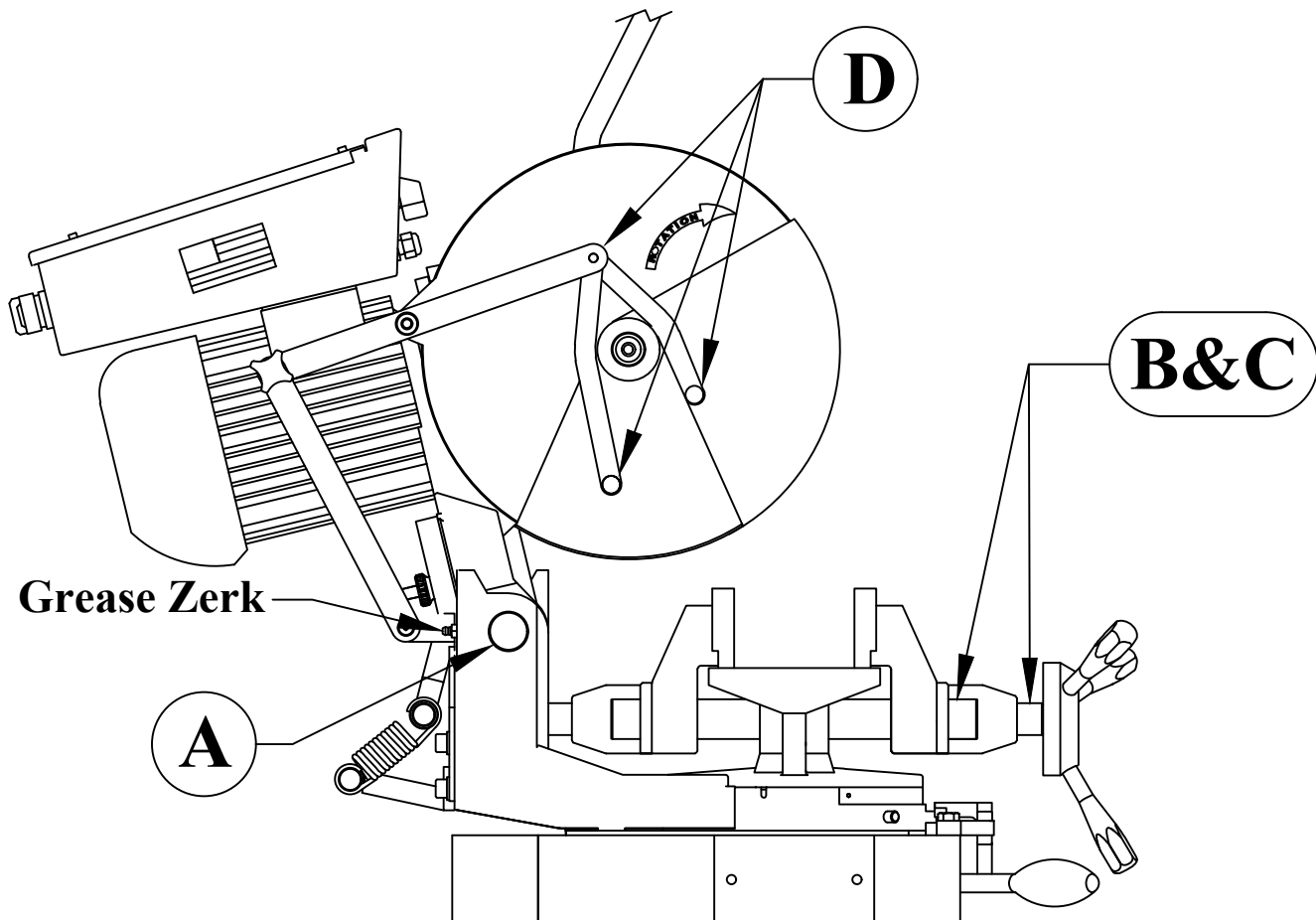


FIGURE 6

5.2 CUTTING OILS AND LUBRICANTS

SECTION 10.8 lists Scotchman's parts numbers for cutting oils and lubricants. Using high quality lubricants and oils will greatly increase the life of this equipment. We recommend using only pure, synthetic, water soluble cutting oil for coolant. For the saw head, use a non-EP additive ISO-460 gear oil specified for worm gears. Oils containing EP additives may degrade or damage bronze gears and must be avoided. Mobile 600W Super Cylinder Oil is highly recommended and available from the factory under P/N 075758.

On saws equipped with the power down feed, use SAE 10W (ISO 32) non-foaming hydraulic oil, such as Mobil DTE 10 or equivalent - Our P/N 060520. On saws equipped with air lubricators, use a high quality (ISO 22) air line lubricant designed for automatic oilers such as our P/N 075753.

5.3 SCHEDULED MAINTENANCE

A program of scheduled maintenance should be set up and documented according to your application and the frequency with which you use this machine. The following is a list of some important things that should be included in a scheduled maintenance program.

1. EVERY 250 HOURS OR 3 MONTHS:

Drain the coolant reservoir and flush it out. Refill the coolant reservoir with new coolant. This will extend the life of the coolant pump considerably. SEE SECTION 10.8 for Scotchmans coolant.

2. EVERY 500 HOURS OR 6 MONTHS:

Drain the gear oil from the saw head and flush with a petroleum product. Refill the saw head with Mobil 600W Super Cylinder Oil (P/N 075758). Check the condition of the pivot pins on the head and on the guard.

Check the complete saw for loose connections in the electrical and air systems. If your saw is equipped with the power vise or power down feed options, SEE SECTION 7.0 for additional information. Since every application is different, each user must design and implement a scheduled maintenance program that fits his applications.

5.4 COOLANT PUMP MAINTENANCE

IF YOUR COOLANT PUMP IS LEAKING OR LACKS POWER, USE THE FOLLOWING STEPS.

- 1. We recommend replacing the pump seal kit anytime that the pump is dismantled.**

FOR PART NUMBER IDENTIFICATION, REFER TO SECTION 9.6.

- 2. Make sure that the power to the machine is off.**
- 3. Remove the four bolts (J) and remove the pump from the machine.**
- 4. Remove the coolant line (H) and the fitting (G). Clean any sludge out of the line and fittings.**
- 5. Remove the three bolts (F) and remove the end plate (C).**
- 6. Remove the screw (E), washer (D) and the impeller (B) from the pump.**
- 7. Clean the sludge out of the impeller, end cap and passageway, from the bottom of the pump to the outlet port.**
- 8. Reassemble the pump, reversing the above steps.**
- 9. Clean out the reservoir and install new coolant.**

FOR OLDER MODELS, PLEASE CONTACT YOUR LOCAL DEALER OR THE FACTORY

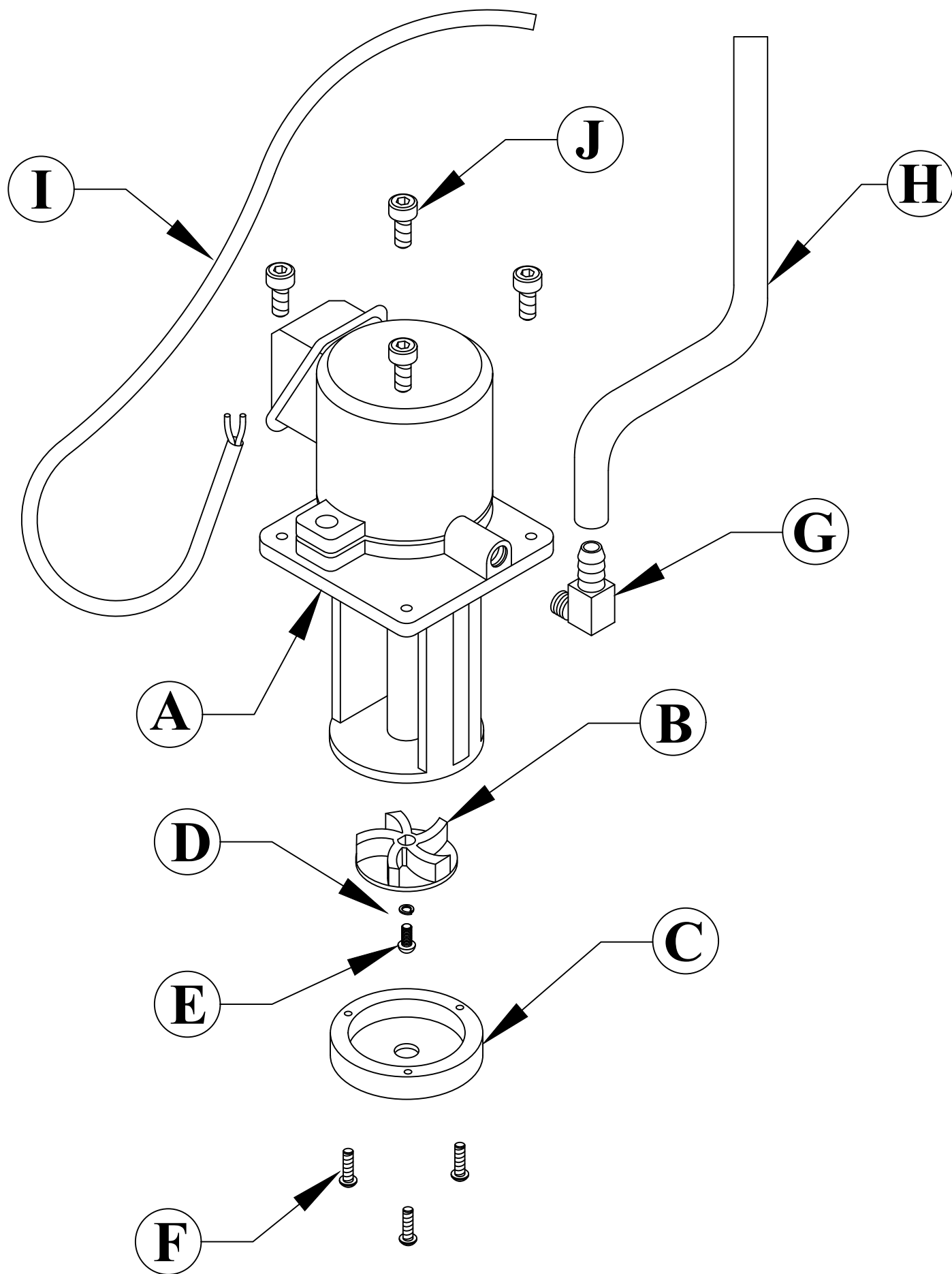


FIGURE 7

6.0 MACHINE OPERATION

6.1 INSTALLING THE BLADE - MANUAL MACHINES

SEE FIGURE 8 BELOW.

FOR INSTALLING BLADES ON MACHINES EQUIPPED WITH THE POWER DOWN FEED OPTION, REFER TO SECTION 7.9.

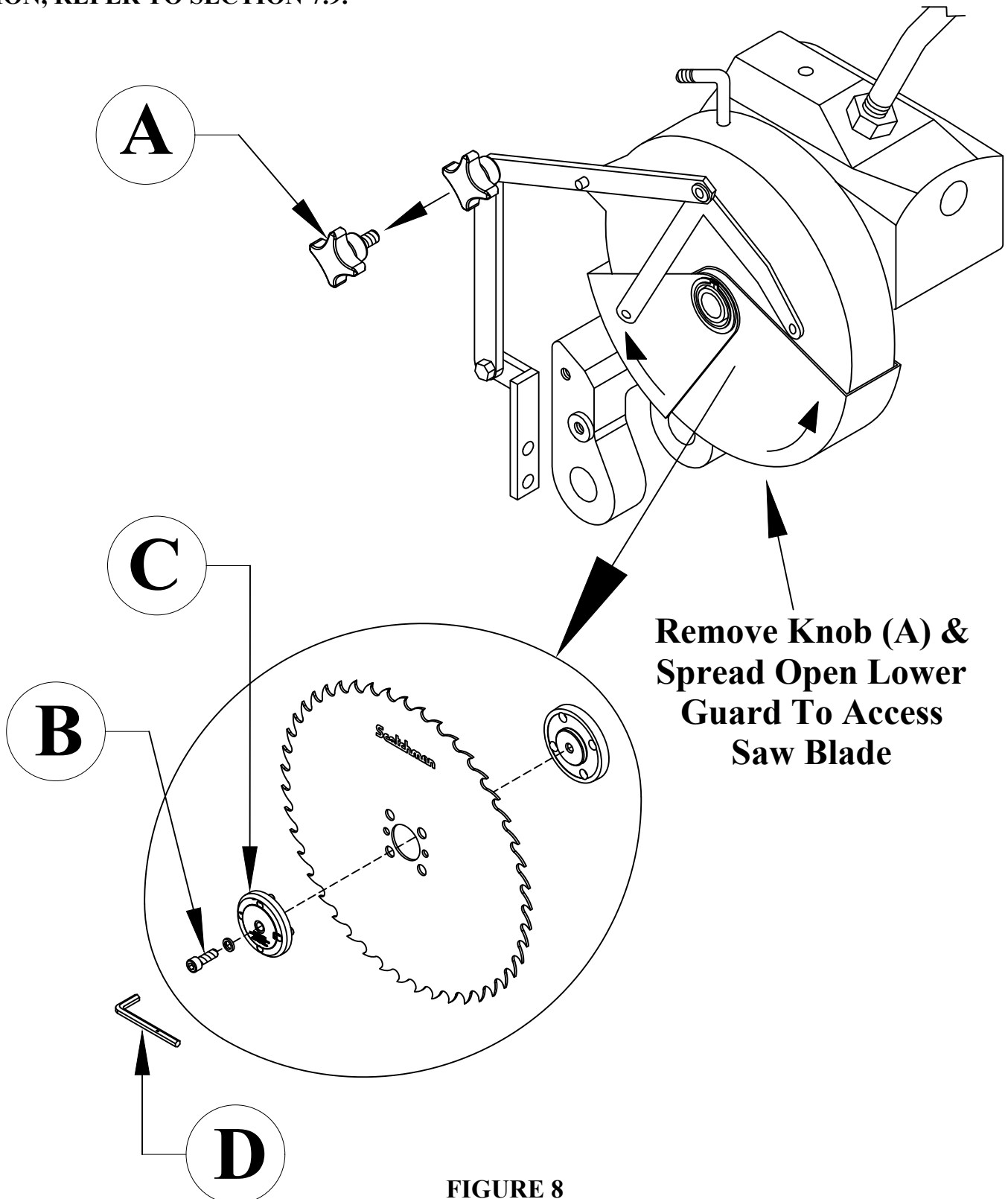


FIGURE 8

⊠ **CAUTION: USE ONLY HIGH SPEED STEEL BLADES DESIGNED FOR THIS MACHINE. DO NOT MODIFY ANY BLADE TO FIT THIS MACHINE. DO NOT USE BLADES DESIGNED FOR THIS MACHINE ON ANY OTHER EQUIPMENT.**

The CPO-350 saw is designed to use a maximum 14 inch (350mm) diameter blade. The arbor size is 40mm with four 12mm pins spaced at 64mm.

BEFORE INSTALLING THE BLADE, make sure that the power to the machine is disconnected and the air supply is turned off (if applicable).

USE THE FOLLOWING STEPS TO INSTALL A BLADE:

(An 8mm hex key wrench (D), shipped with each machine, is required to change blades.)

1. Remove the knob (A) from the guard linkage and manually open the guard.
 2. With the hex key wrench (D) remove the blade bolt (B) through the center hole in the blade guard.
 3. Remove the blade flange (C).
- **NOTE:** It is very important to keep the blade flange, the spindle and the blade clean and free from nicks and chips when installing a blade. Failure to do these things will result in poor performance and possibly broken or damaged blades.
4. Install the blade. Make sure that the pin holes line up to the holes in the spindle.
 5. Replace the blade flange and start the bolt into the spindle.
 6. Before locking the blade in position, the back lash must be taken up. To take up the back lash, rotate the bottom of the blade towards you until it seats against the drive pins.

⊠ **CAUTION: THE BLADES ARE VERY SHARP AND CARE MUST BE TAKEN WHEN REMOVING THE BACK LASH. DO NOT GRIP THE CUTTING EDGE OF THE BLADE BARE HANDED. THE BACK LASH MUST BE TAKEN UP EVERY TIME A BLADE IS CHANGED.**

7. After taking up the back lash, tighten the blade bolt (B).
8. Break in the saw blade. The teeth on new or re-sharpened blades have a sharp edge and should be fed through the first three or four cuts very slowly, before starting normal cutting.

6.2 SAW CAPACITIES

Figure 9 below is a chart showing the maximum capacities of this machine in various materials at the most common angles from 0 degrees to 90 degrees.

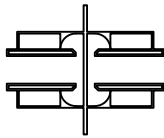
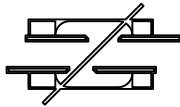
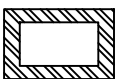
#060331 CPO - 350 03/21				
CAPACITIES WITH MAXIMUM DIAMETER BLADES 350 MM		90° 	45° 	
	INCHES MM	Ø4-1/2 Ø114	Ø4 Ø102	
	INCHES MM	4-1/4 X 4-1/4 108 X 108	3-3/4 X 3-3/4 95 X 95	
	INCHES MM	4-1/4 X 4-1/4 108 X 108	3-3/4 X 3-3/4 95 X 95	
	INCHES MM	4-1/4 X 4-1/4 108 X 108	3-3/4 X 3-3/4 95 X 95	
	INCHES MM	5-1/2 X 4 140 X 102	3-3/4 X 3-3/4 95 X 95	
	INCHES MM	Ø1-3/4 Ø44	Ø1-1/2 Ø38	
	INCHES MM	1-3/4 X 1-3/4 44 X 44	1-1/2 X 1-1/2 38 X 38	

FIGURE 9

6.3 SELECTING THE PROPER BLADE AND CUTTING SPEED

In cold sawing, there is no such thing as a general purpose blade. To achieve the best results from your saw, proper blade selection is important. There are a couple of charts at the top of the next page (one for tube, the other for solids) that will help to determine the proper blade for your application. When sawing flat stock or rectangular solid sections, determine the thickest section that will be cut and use the size shown on the chart to determine the proper blade.

Often it is the case where many sizes and types of material need to be cut and it is impractical to have blade for every size. It is possible to use slightly more or less teeth than what is recommended to cut if the operator is careful and keeps the feed rate slower and consistent. However, there are limits. Using a fine blade (more teeth) to cut a large solid will NOT work. It will instantly load the teeth up and may ruin or break the blade. And using a coarse blade (less teeth) to cut a small solid or tubing can easily break off the tips of the teeth. In the worst case, hooking a tooth can rip a chunk out of the blade or even break the blade.

Selecting the proper blade speed is very important. The CPO-350 LT and HT saws have an RPM range from 22 - 44 (LT) and 44 - 88 (HT). The heavier the material, the lower RPM is best. The lighter the material, the higher RPM is best. The LT saw will cut a wider variety of material. It can cut large solids and still cut thin material, just slower. The HT saw will not cut large solid steel material or thick walled steel tube very well due to the higher RPM.

The HT model (44-88 RPM) is recommended primarily for thin wall tube up to 1/8 of an inch (3mm) wall thickness and nonferrous materials. The low speed (44 RPM) should be used for tubes with a wall thickness of one hundred thousandths (.025mm) and thicker and for solid sections of nonferrous materials. The High speed (88 RPM) is recommended for thin walled tubes and nonferrous tubes and profiles. With either the LT or HT model, a few test cuts will help to determine the best speed. A variable speed motor control option that will greatly increase the RPM range of these machines is available for both models of this saw.

The CPO-350 saw is designed to use a maximum 14 inch (350mm) diameter blade. We recommend using smaller than maximum diameter blades if possible. Using smaller diameter blades reduces the surface feet per minute and smaller blades provide greater rigidity. Smaller diameter blades available from stock for this machine are a 12-1/2 inch (315mm) and a 10-3/4 inch (275mm). For available tooth styles in stock, REFER TO SECTION 11.0.

ROUND TUBING - ANGLE & SQUARE CUT ON THE DIAGONAL

Square Tube Cut Across the Flat - Increase Pitch 1mm - 2mm (less teeth)

MATERIAL	TOOTH #, BLADE SIZE & PITCH			
Wall Thickness	275mm 10-3/4"	315mm 12-1/2"	350mm 14"	Pitch mm
.030 - .060	260	280	320	3.5
.030 - .090	200	220	280	4.0
.090 - .150	160	180	200	5.5
.150 - .250	For Thick-Walled Tube, Please Call Our Factory for a Recommendation			
.250 - .375				
.375 - .500				

ROUND & SQUARE SOLID BAR

Tough Alloys or Stainless - Decrease Pitch 1mm - 2mm (more teeth)

Aluminum & Copper - Increase Pitch 1mm - 2mm (less teeth)

MATERIAL	TOOTH #, BLADE SIZE & PITCH			
Solid Bar	275mm 10-3/4"	315mm 12-1/2"	350mm 14"	Pitch mm
1/2"	260	280	320	3.5
5/8"	180	220	240	4.5
3/4"	180	200	220	5.0
1"	130	150	170	6.5
1-1/4"	110	120	140	8.0
1-1/2"	100	120	130	8.5
1-3/4"	90	110	120	9.5
2"	80	90	100	11.0

A general rule for cutting solid steel is 3 to 5 teeth in the material. 3 teeth is usually better with square or rectangle solids and 5 teeth is better for cutting solid round bar. Consistent rigid down feed pressure and low rpm is best for larger solid steel material as it will reduce vibration and will increase blade life.

6.4 MATERIAL CLAMPING

All work pieces must be clamped securely in the vise. Any slippage of the material can result in broken or damaged blades. The material should be clamped so that the contact surface between the material and the blade is as small as possible. For this reason, when cutting flat stock material, we recommend standing it up and cutting it through the thinnest section, whenever possible.

If the flat stock is too wide to clamp standing up, clamp it in the vise diagonally. We also recommend cutting square tubing through the diagonal section and angle iron with the web up. For examples, SEE FIGURE 10 ON THE FOLLOWING PAGE. This is not always possible when cutting materials at a miter. Some thin walled round sections and profiles will require special jaws to hold them. When trimming or cutting very short pieces that do not extend into both sides of the vise, place a piece of material the same size in the unused side of the vise, to insure uniform clamping. When setting up the saw to miter cut, pull the head down before making the first cut, to make sure that the blade clears the vise jaws.

The steel jaws have slotted mounting holes and can be adjusted for various miters. Always adjust the steel jaws so that they clamp the material as close to the blade as possible, whether miter or straight cutting. All models of the CPO-350, except those fitted with the power down feed option, have this down stroke or cutting depth adjustment. SEE FIGURE 11 ON THE FOLLOWING PAGE. This adjustment is used to keep the saw blade from cutting into the vise spindle and must be adjusted when changing sizes of materials or blades.

If your saw is equipped with either the power vise or the power down feed options:
SEE SECTION 7 - OPTIONAL EQUIPMENT, for additional information.

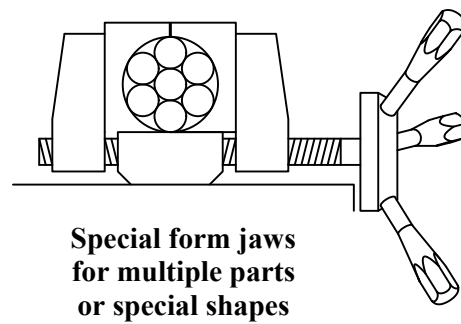
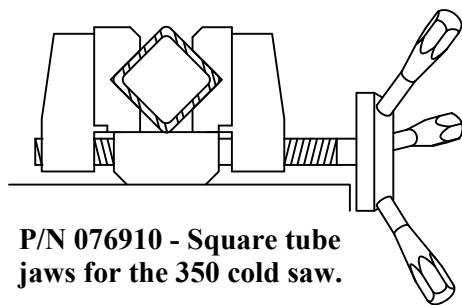
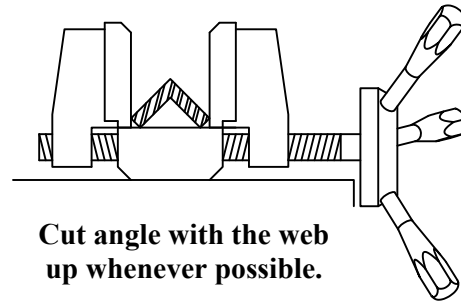
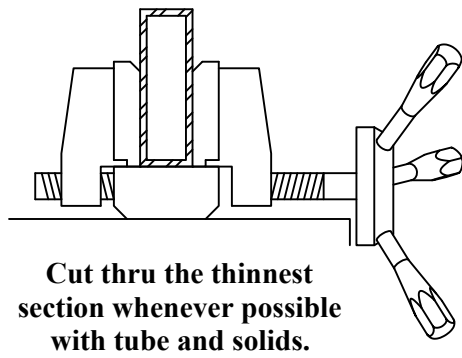


FIGURE 10

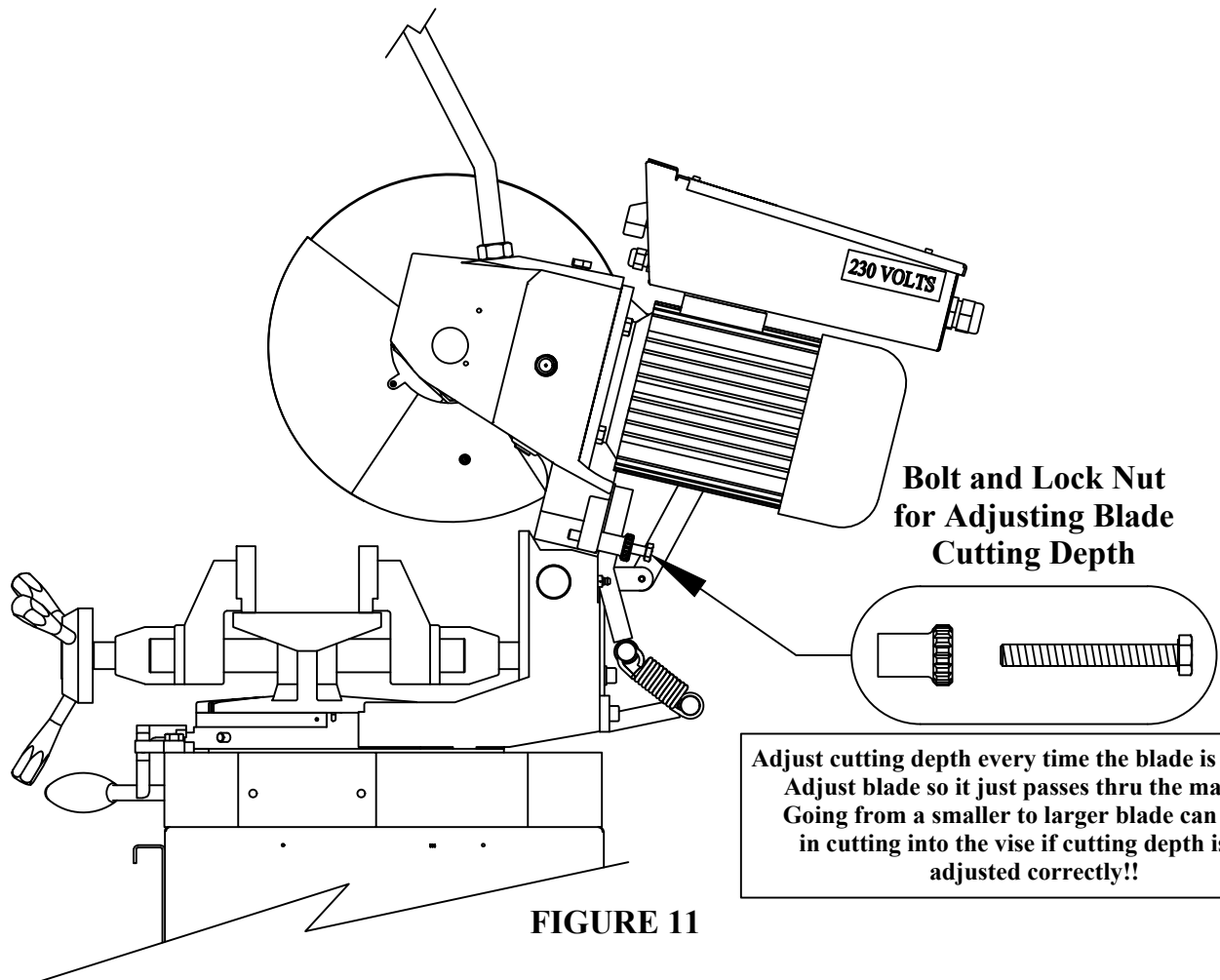


FIGURE 11

6.5 MITER LOCKING DEVICE

SEE FIGURE 12 ON THE FOLLOWING PAGE.

All models manufactured for domestic sales are equipped with a miter locking device which allows quick set-up for mitering at 45 degrees, left and right, and 90 degrees for straight cutting. A miter locking device is available as an option for models manufactured for international sales.

TO USE THE MITER LOCKING DEVICE:

- 1. Unlock the tension handle (A).**
- 2. Push the miter lock release handle (B).**
- 3. Turn the head in the direction that you want to miter.**
- 4. Release the miter lock handle and continue turning the head until one of the pins (G) snaps into the slot. The saw is equipped with three pins. One for straight cuts and one for angled cuts left and right.**
- 5. Then, re-lock the tension handle. When locking the tension handle, do not over-tighten.**
- 6. The miter locking device can be fine adjusted if it does not stop at an exact 45. Loosen the mounting bolts (F) and adjust the complete miter lock, left or right, to the desired position.**

IF YOU WANT TO CUT MITERS OTHER THAN 45 DEGREES:

- 1. Unlock the tension handle (A).**
- 2. Push the miter lock release handle (B) and turn the head to the desired angle by using the scale on the saw.**

NOTE: THE SCALE IS READ OFF TO THE SIDE OF THE VISE AT POINT (C), NOT IN THE CENTER.

- 3. Re-lock the tension handle (A). After a period of time, the tension handle (A) may need to be adjusted if the head will not remain in the position that it was previously set.**

TO RE-SET THE TENSION HANDLE:

- 1. Remove chip drawer to gain access to the tension handle nut in the machine base.**
- 2. Move the tension handle (A) to its unlocked position.**
- 3. Loosen the jam nuts (D) on the adjustment bolts (E) and tighten the bolts finger tight, plus 1/4 of a turn.**
- 4. Work the tension handle several times and re-tighten the adjusting bolts, if necessary.**
- 5. Re-tighten the jam nuts (D).**

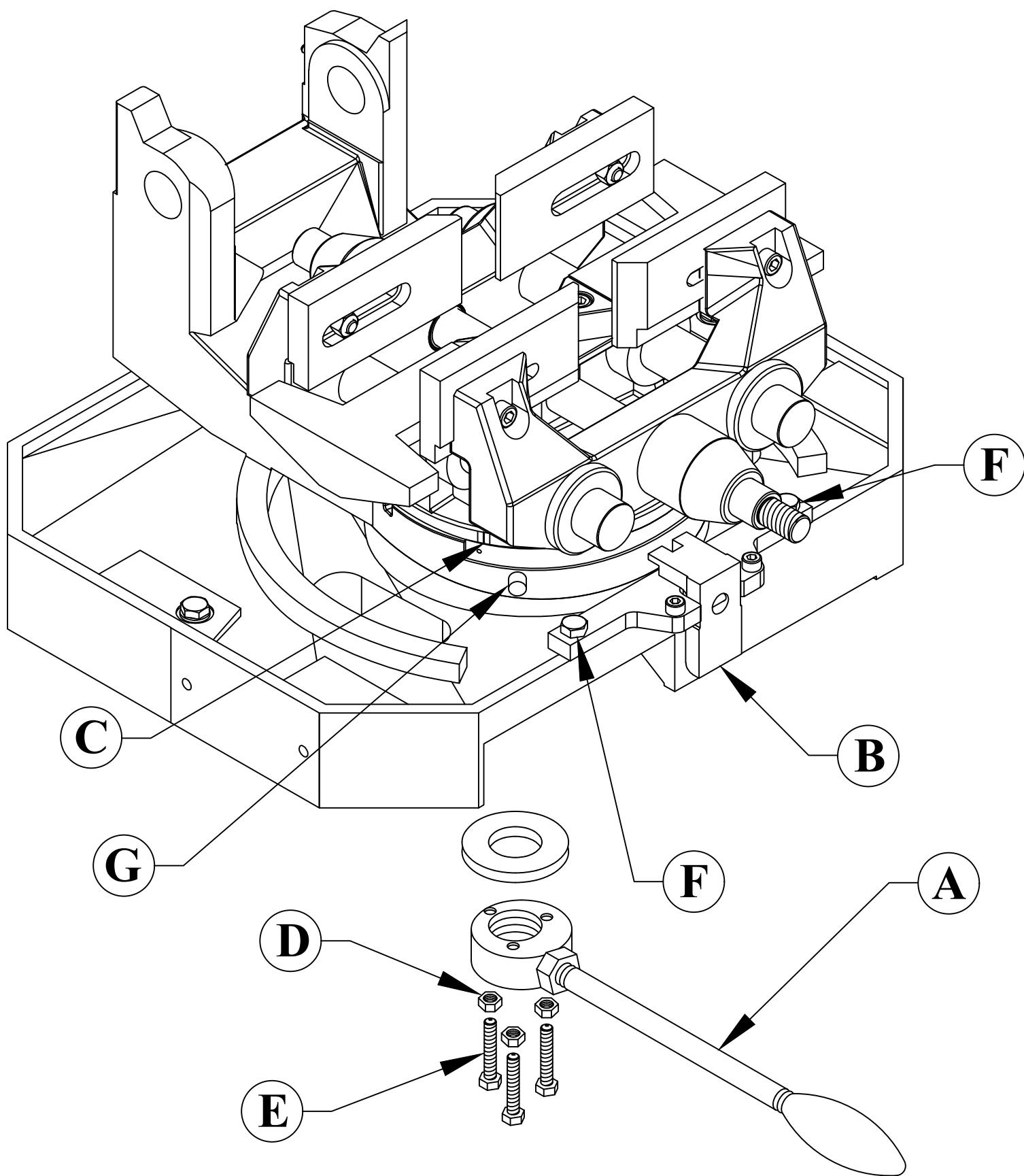


FIGURE 12

7.0 OPTIONAL EQUIPMENT

7.1 POWER VISE

The power vise is an option that is normally ordered with the saw. It is not recommended as a retro-fit in the field. The power vise allows automatic clamping of the material, which improves productivity and reduces operator fatigue. The vise automatically clamps when the saw head is drawn down and releases when the saw head returns.

7.2 POWER VISE SET-UP AND MAINTENANCE

SEE FIGURE 13 ON THE FOLLOWING PAGE.

THE FOLLOWING ARE SET-UP AND MAINTENANCE INSTRUCTIONS FOR THE POWER VISE OPTION (RETROFIT OR FACTORY INSTALLED):

1. Before connecting the air supply to the saw, make sure that the filter/regulator/lubricating (FRL) device (A) is full of oil.
2. Slide the shuttle valve (F) on the filter/lubricator device to the closed position.
3. Connect the air supply to the shuttle valve. Make sure that the vise is clear and that the head is in the UP position.
4. Slide the shuttle valve to open it. Whenever the shuttle valve is closed, it bleeds the air pressure out of the system automatically.
5. Adjust the air pressure regulator (G) to 90 PSI (6.2 BAR) as this is the minimum operating pressure. The maximum operating pressure is 105 PSI (7.2 BAR) .
6. Before powering the saw, pull the head down several times, to make sure that the four way valve (H) and the FRL device (A) are adjusted properly and that the air pressure setting remains constant.
7. The four way valve should activate the vise at the beginning of the down stroke and release it at the top of the return stroke. The four way valve is adjusted with the set screw (B) in the valve arm, that the roller is attached to.
8. The lubricating device (A) should release one drop of oil every 5 to 10 cycles. On top of the lubricating device is a clear plastic dome with a small copper tube inside. The oil should drop out of the copper tube. The lubricating device is adjusted using a small screwdriver (C) on the top of the lubricator.
9. To add oil to the lubricating device, disconnect the air supply and remove the plastic bowl. Push in tab and unscrew bowl from the body. Fill the bowl about 3/4 full with a quality (ISO 22) air line lubricant designed for automatic oilers (our P/N 075759) and screw it back on the lubricator.

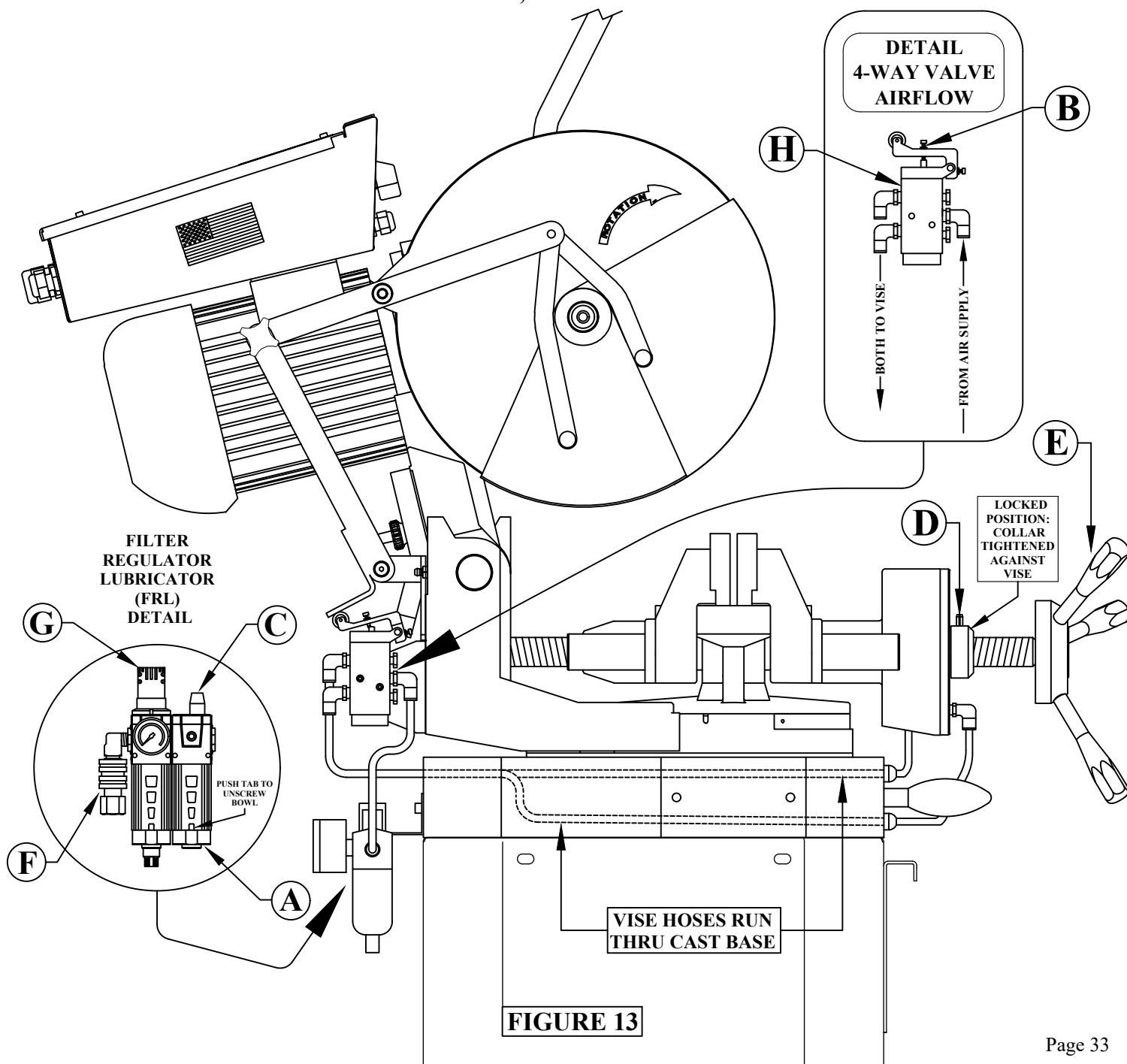
TO ADJUST THE VISE TO THE SIZE OF MATERIAL BEING CUT:

Release the locking collar (D) on the vise spindle (see below). The vise spindle is left hand threaded and the locking collar must be turned clockwise to release it.

Open the vise, using the positioning handles (E), and place the material in the vise.

3. Crank the vise closed to within approximately 1/8 of an inch (3mm) from the material and relock the locking collar (D). Failure to lock the locking collar may allow the vise to vibrate open while cutting, causing damage or breakage of the blade. The power vise has approximately 1/4 of an inch (6mm) of stroke. As with the manual vise, proper clamping is very important and special jaws may be required for some materials.

FOR EXAMPLES OF SPECIAL JAWS, REFER TO FIGURE 10 ON PAGE 29.



7.3 REPLACING THE SPINDLE IN THE POWER VISE

SEE FIGURE 14 BELOW.

⊠ **CAUTION:** DISCONNECT THE MACHINE'S POWER AND AIR SUPPLY

1. **Disconnect the air lines (A).** The air connections on the cylinder are snap connectors. To release the air connections, push the collar around the hose into the fitting and pull the hose out. To reconnect the lines, simply push the air line into the fitting as far as it will easily go.
2. **Remove the support block (B), the bolts (C) and the spacer (I) from the base of the vise.**
3. **Remove the guide pins (D) out through the back of the vise.** The head must be moved to a miter position for the removal of one of the pins.
4. **Carefully lift the complete vise off of the machine and place on a flat work bench.**
5. **Remove plastic cover and remove the lock nut (J) from the spindle boss (K).**
6. **Unscrew the vise boss (K) and the locking collar (M) from the spindle.**
7. **Unscrew the spindle from the front casting (L) and the rear casting (E).**
8. **Install the new spindle and reassemble the vise, reversing the above steps.**

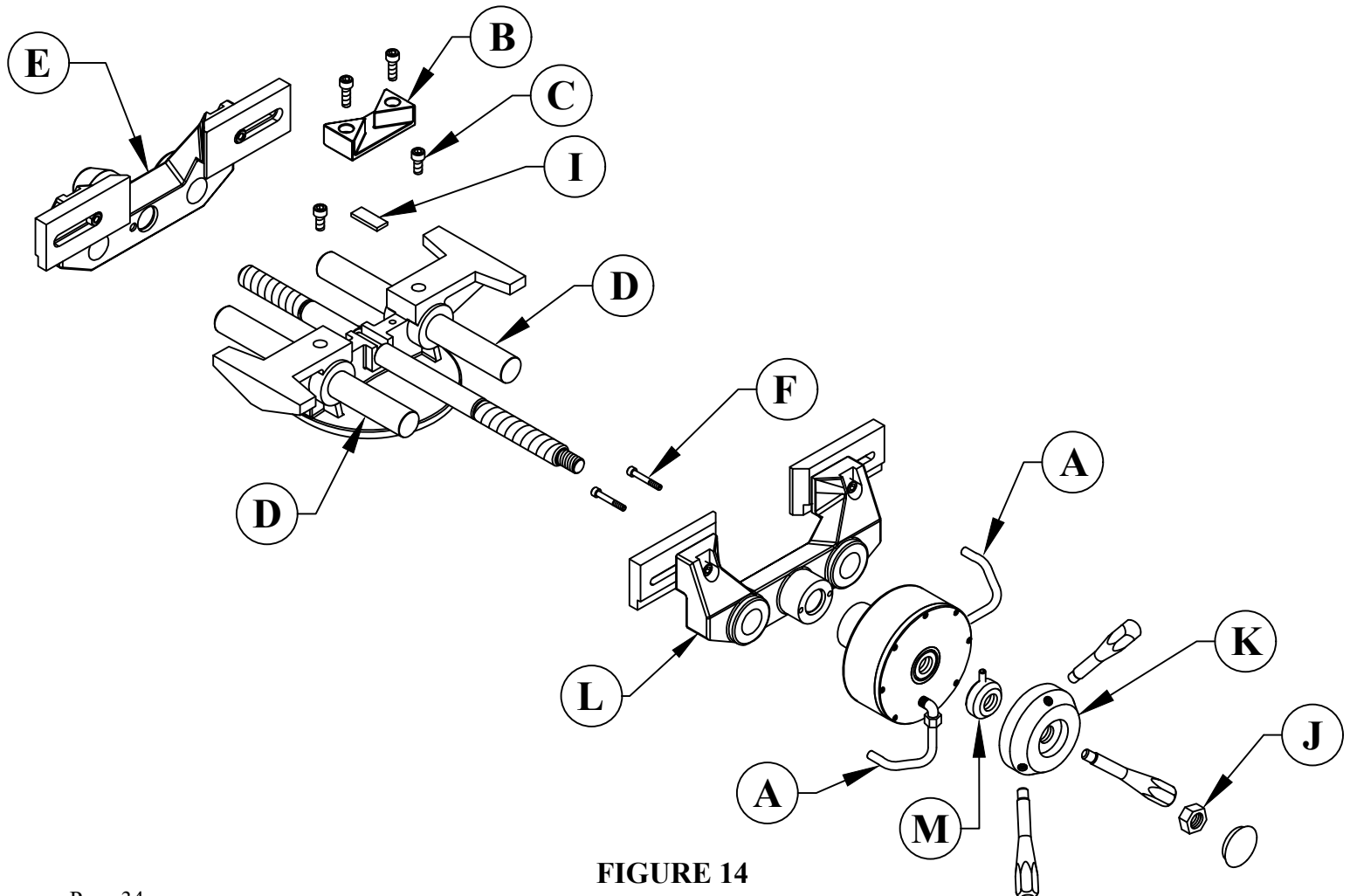


FIGURE 14

7.4 REPLACING THE SEALS IN THE POWER VISE

SEE FIGURE 15 BELOW.

⊠ **CAUTION:** DISCONNECT THE MACHINE'S POWER AND AIR SUPPLY

SEAL KIT IS P/N 076371 - Contains two small and two large o-rings and an oil seal. Locations below.

1. Open the vise to its full open position.
2. Disconnect the air lines from the cylinder (A). (SEE SECTION 7.3 - STEP 1)
3. Remove the jam nut (G), the boss (B) and the locking collar (C) from the front of the spindle.
4. Remove the two bolts (D) from the front vise casting that holds the air cylinder on.
5. Unscrew the air cylinder (A) from the spindle.
6. Remove the eight 4mm screws (E) from the cover (H) and remove the cover and the piston (F).
7. Remove the old seals and clean all of the parts and inspect them for any scratches or nicks.
8. Install the new seals. Lubricate seals, cylinder (A) & piston (F) before reassembly.

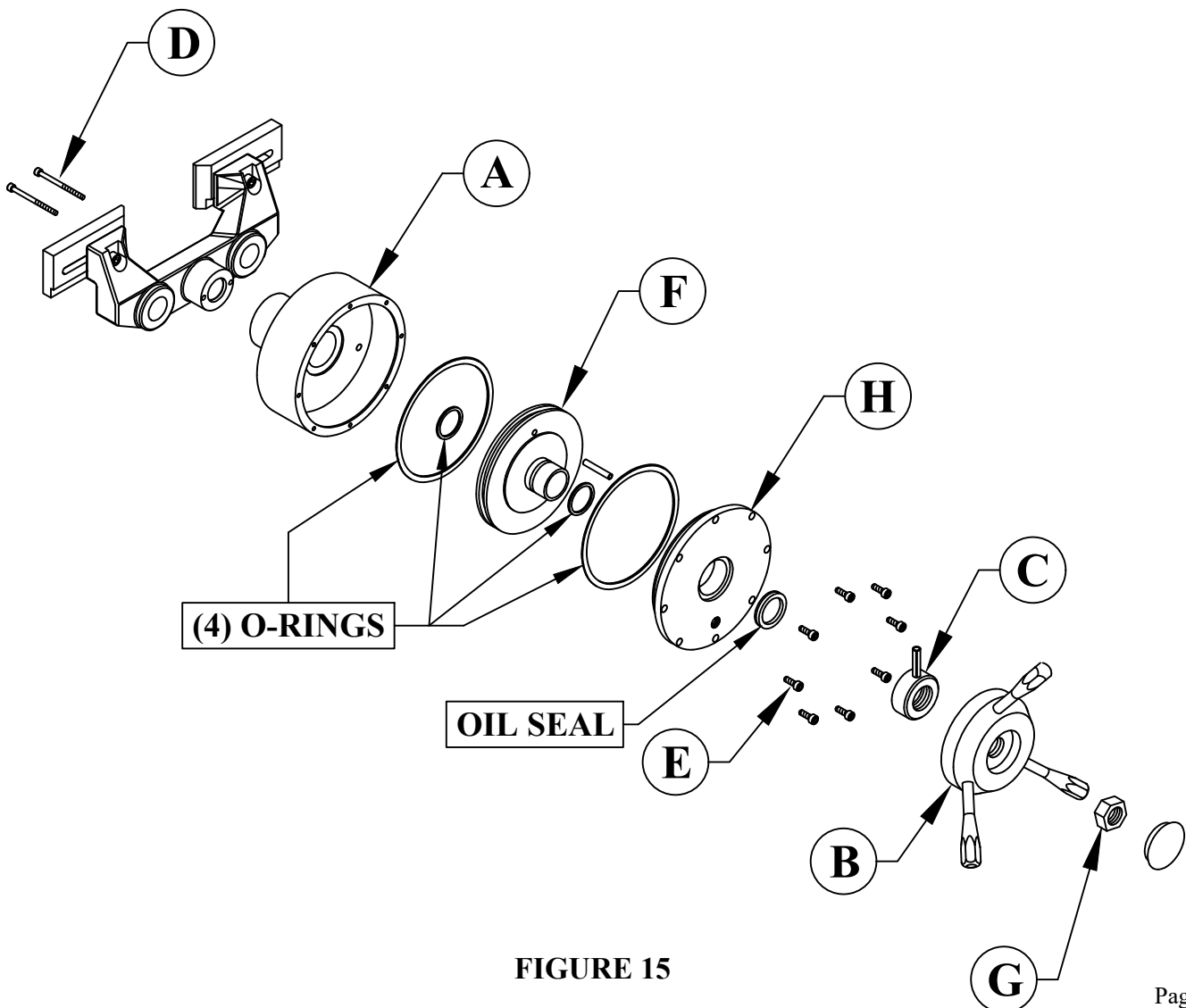


FIGURE 15

7.5 POWER DOWN FEED

The power down feed option, used in conjunction with the power vise option, changes a manual saw into a semi-automatic saw. These options will increase productivity and reduce operator fatigue. The power down feed option will not retrofit to machines with serial number 11940491 and prior in the field. This option can be used on machines with or without the power vise option.

7.6 POWER DOWN FEED SET-UP AND MAINTENANCE

SEE FIGURE 16 ON THE FOLLOWING PAGE.

1. Before powering the saw, check the oil level in the reservoir (A). There is a sight glass in the back of the reservoir. The head must be in the full UP position. The recommended oil to use our P/N 060520 or a SAE 10W (ISO 32) non-foaming hydraulic oil, such as Mobil DTE 10 or equivalent.

⊗ **CAUTION: ALWAYS DISCONNECT THE AIR SUPPLY BEFORE REMOVING THE FILLER PLUG FROM THE RESERVOIR. IF THE FILLER PLUG IS REMOVED WHILE THE MACHINE IS CONNECTED TO THE AIR PRESSURE, THE FLUID IN THE TANK WILL BE PURGED THROUGH THE OPENING UNDER PRESSURE.**

2. Slide the shuttle valve (D) to its CLOSED position and connect the air supply.
3. Slide the shuttle valve to its OPEN position and adjust the air pressure regulator. 90 PSI (6.2 BAR) is the minimum operating pressure. 105 PSI (7.2 BAR) is the maximum.
4. Without powering the saw, manually cycle the head up and down several times, to purge the air out of the lines. The head may cycle irregularly the first few cycles.
5. After cycling the head, shut the flow control valve (B) off. Then, open it one turn.

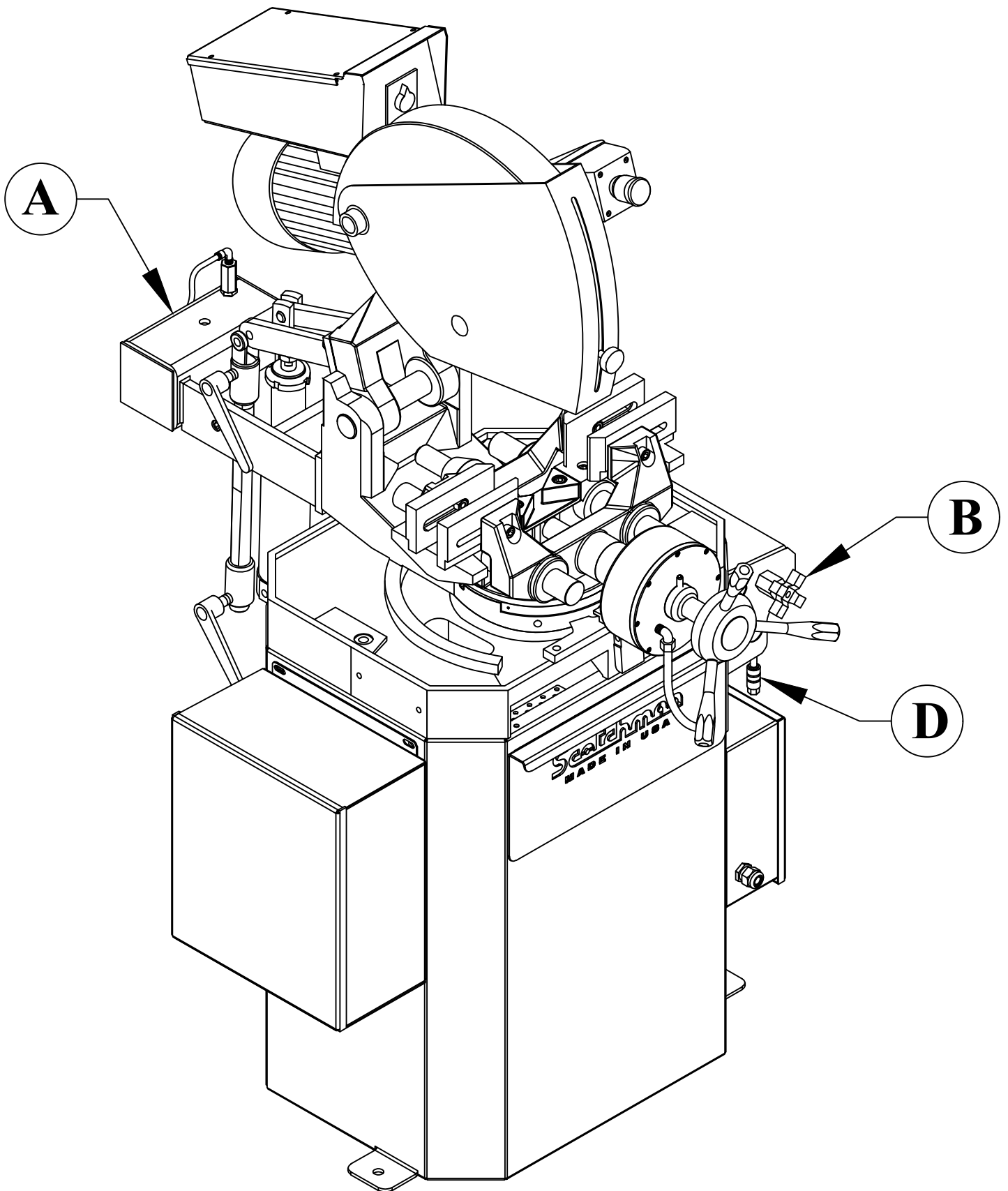


FIGURE 16

7.7 STROKE CONTROL ADJUSTMENT (POWER DOWN FEED)

SEE FIGURE 17 ON THE FOLLOWING PAGE.

Before powering the machine, the up and down strokes of the saw head must be set. The stroke is set by the collars (A & B) on the shaft (C).

TO SET THE STROKE:

1. The saw blade control switch for the saw blade is located in the electrical box on top of the motor.
2. The stroke is set with this switch in the off position - Blade does not turn.
3. The flow control valve (D) should be turned all the way clockwise so the saw head won't move.
4. Take a piece of material that is to be cut and put it in the vise so it is just off to one side of the blade. It's good to hand-tighten the material in the vise to keep it in place.
5. Step on the foot pedal to cycle the saw. The saw head should not move. Open the flow control valve (D) by turning it counterclockwise to allow the head to slowly travel downward. Stop the head with the flow control valve when the blade is just above the material. Loosen the handle (H) on the upper stroke adjustment (A) and move it down to slightly past where it contacts the limit switch (G) and tighten the handle (H).
6. Open the flow control valve (D) to allow the head to slowly travel downward again. Stop it with the flow control valve when the blade passes just below the material. Take the foot pedal around to the back of the saw. Step on the foot pedal and HOLD - Then loosen the handle (E) on the lower stroke adjustment (B) and move it up to slightly past where it contacts the limit switch (F) then tighten the handle (E). Keep clear of moving parts and release the foot pedal. The saw head will then return to the up position.
7. Make sure the blade is high enough for the material to pass below the blade after it has returned to the up position. Step on the pedal and watch as the blade travels downward to ensure that it passes below the material before returning to the up position. Make a slight adjustment if needed.

⊗ **CAUTION : ANY TIME THAT THE BLADE OR THE SIZE OF THE MATERIAL BEING CUT IS CHANGED, THE STROKE OF THE MACHINE HAS TO BE CHECKED. FAILURE TO SET THE STROKE OF THE MACHINE WILL RESULT IN DAMAGE TO THE MACHINE OR TO THE BLADES.**

8. Adjust the flow control valve (D) to the proper cutting feed rate before cutting any material.

The air lubricator should produce a drop of oil every 5 to 10 cycles. Adjustment is done on top of the lubricator with a small straight screwdriver. There is a clear plastic dome on top of the lubricator with a copper tube inside. The oil should drop from the copper tube. The oil in the lubricator should be checked everyday. For information on the FRL device SEE SECTION 4.6.

As with all other functions of the saw, selection of the proper blade, spindle speed and clamping are very important in providing a quality finished product.

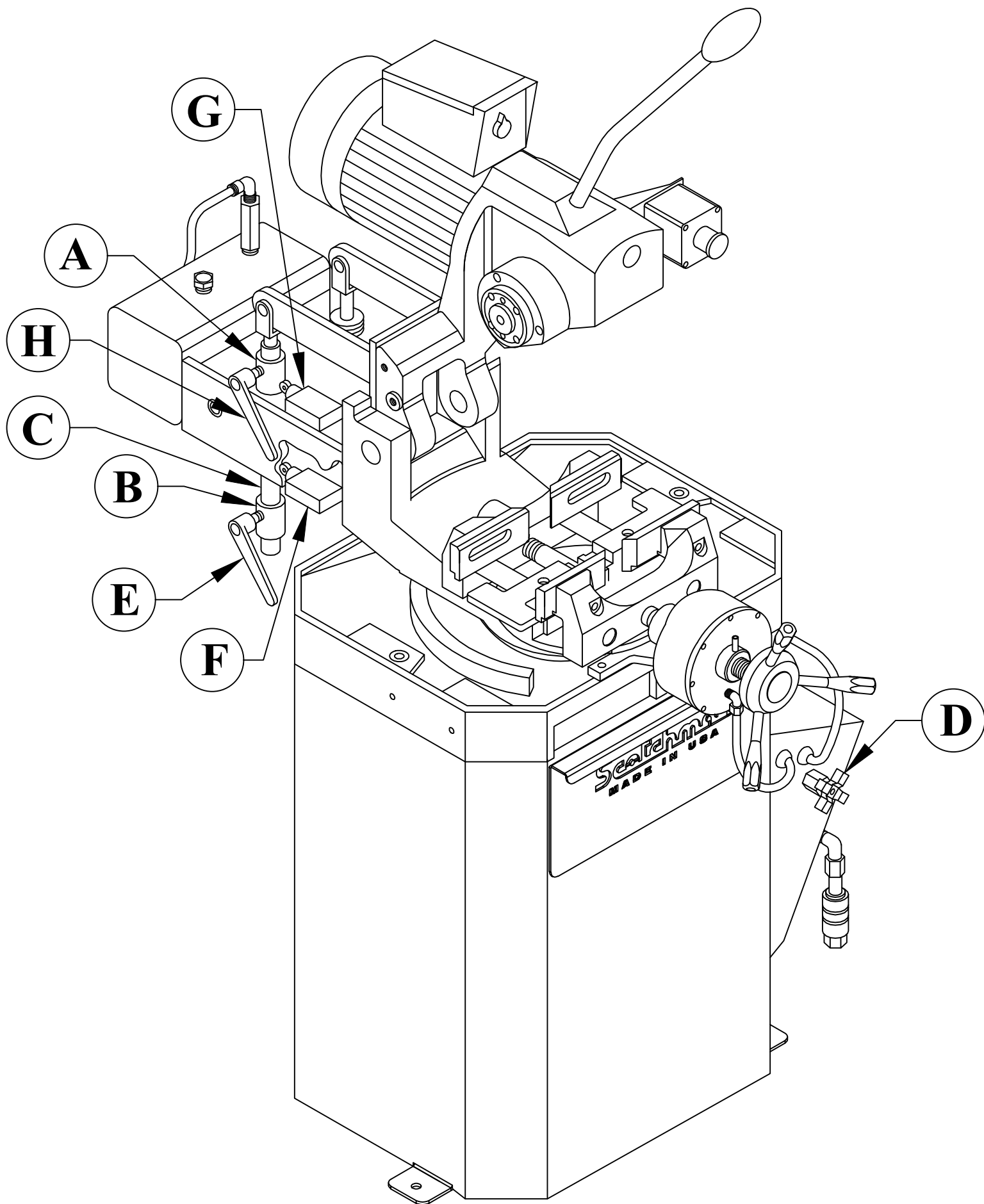


FIGURE 17

7.8 GUARD ADJUSTMENT (POWER DOWN FEED)

SEE FIGURE 18 ON THE FOLLOWING PAGE.

⊠ **CAUTION: THE GUARD MUST BE ADJUSTED EVERY TIME THAT THE STROKE OF THE MACHINE IS ADJUSTED.**

TO ADJUST THE GUARD:

1. Before adjusting the guard, set the up and down stroke of the machine by following the instructions in SECTION 7.7.
2. With the saw head in the UP position, loosen the bolt (A) in the guard stop (B).
3. Raise the movable section of the guard (C) so that it just clears the vise jaws by no more than 1/8 of an inch (3mm).
4. Adjust the guard stop (B) until it contacts the stop on the fixed section of the guard and tighten the bolt (A).
5. Without powering the machine, cycle the head of the saw several times to make sure that the adjustment is correct.

➡ **NEVER PLACE ANY PART OF YOUR BODY NEAR THE BLADE OR THE GUARD WHILE THE MACHINE IS RUNNING!**

DETAIL VIEW

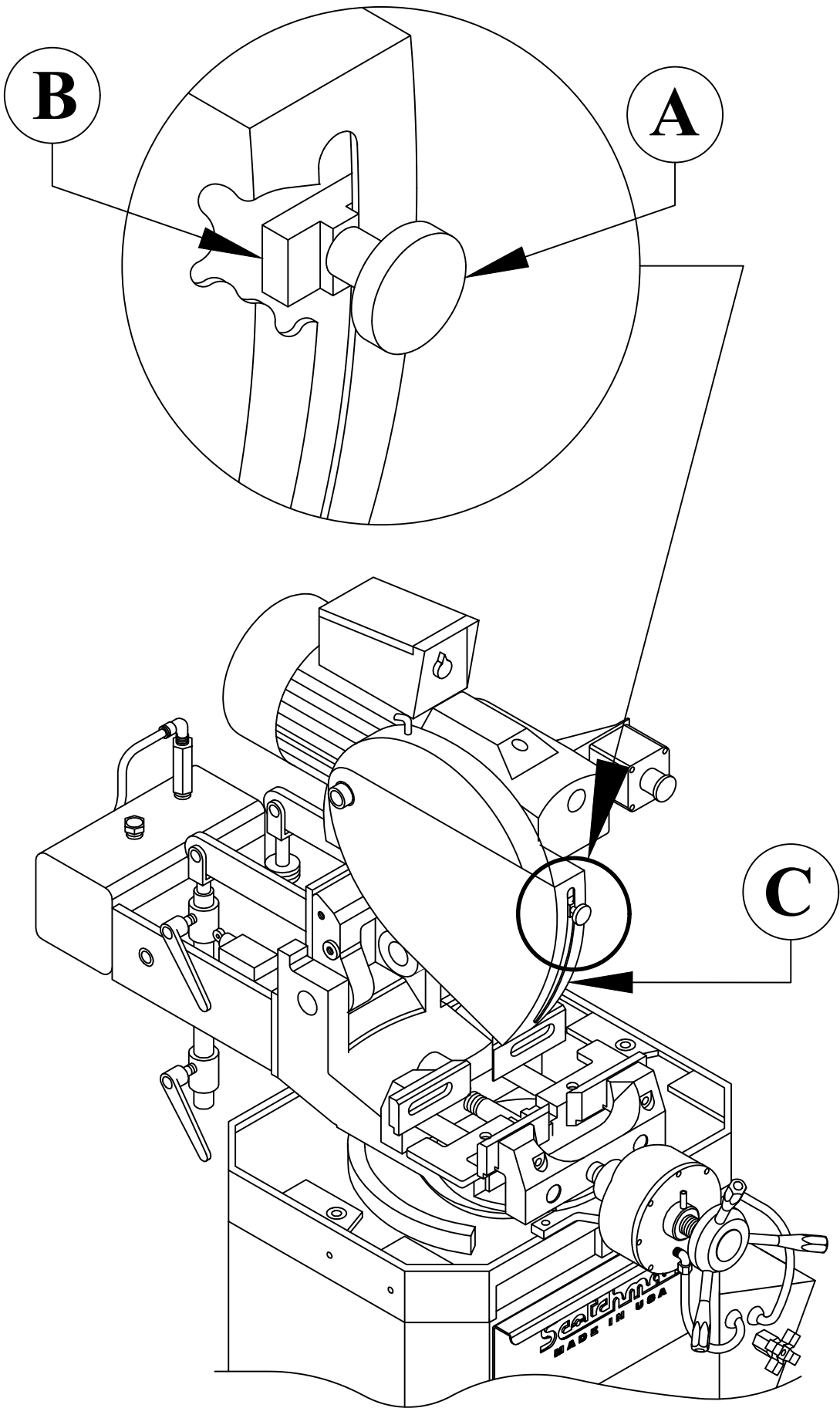


FIGURE 18

7.9 INSTALLING BLADES (POWER DOWN FEED)

SEE FIGURE 19 ON THE FOLLOWING PAGE.

⊗ **CAUTION:** USE ONLY HIGH SPEED STEEL BLADES DESIGNED FOR THIS MACHINE. DO NOT MODIFY ANY BLADE TO FIT THIS MACHINE. DO NOT USE BLADES DESIGNED FOR THIS MACHINE ON ANY OTHER EQUIPMENT.

The CPO-350 saw is designed to use a maximum 14 inch (350mm) diameter blade. The arbor size is 40mm with four 12mm pins spaced at 64mm, also known as 4/12/64 pin spacing.

BEFORE INSTALLING THE BLADE, make sure that the power to the machine is disconnected and the air supply is turned off.

USE THE FOLLOWING STEPS TO INSTALL A BLADE:

(An 8mm hex key wrench (A), shipped with each machine, is required to change blades.)

1. Release the upper stroke control (E) and allow the head to travel to its full UP position.
2. Raise the movable section of the guard (B) up to the OPEN position.
3. Remove the blade bolt (C) through the center hole in the blade guard.
4. Remove the blade flange (D).

► **NOTE:** It is very important to keep the blade flange, the spindle and the blade clean and free from nicks and chips when installing a blade. Failure to do these things will result in poor performance and possibly broken or damaged blades.

5. Install the blade. Make sure that the pin holes line up to the holes in the spindle.
6. Replace the blade flange and start the bolt into the spindle.
7. Before tightening the bolt, you must remove the back lash. To take up the back lash, rotate the bottom of the blade toward you until it seats against the drive pins.

⊗ **CAUTION:** THE BLADES ARE VERY SHARP AND CARE MUST BE TAKEN WHEN REMOVING THE BACK LASH. DO NOT GRIP THE CUTTING EDGE OF THE BLADE BARE HANDED. THE BACK LASH MUST BE TAKEN UP EVERY TIME A BLADE IS CHANGED.

8. After taking up the back lash, finish tightening the blade bolt (C).
9. Return the movable guard to the DOWN position and re-set the upper stroke control.
10. **Break in the saw blade.** The teeth on new or re-sharpened blades have a sharp edge and should be fed through the first three or four cuts, very slowly, before starting normal cutting.

FAILURE TO DO THESE THINGS WILL RESULT IN BROKEN OR DAMAGED BLADES.

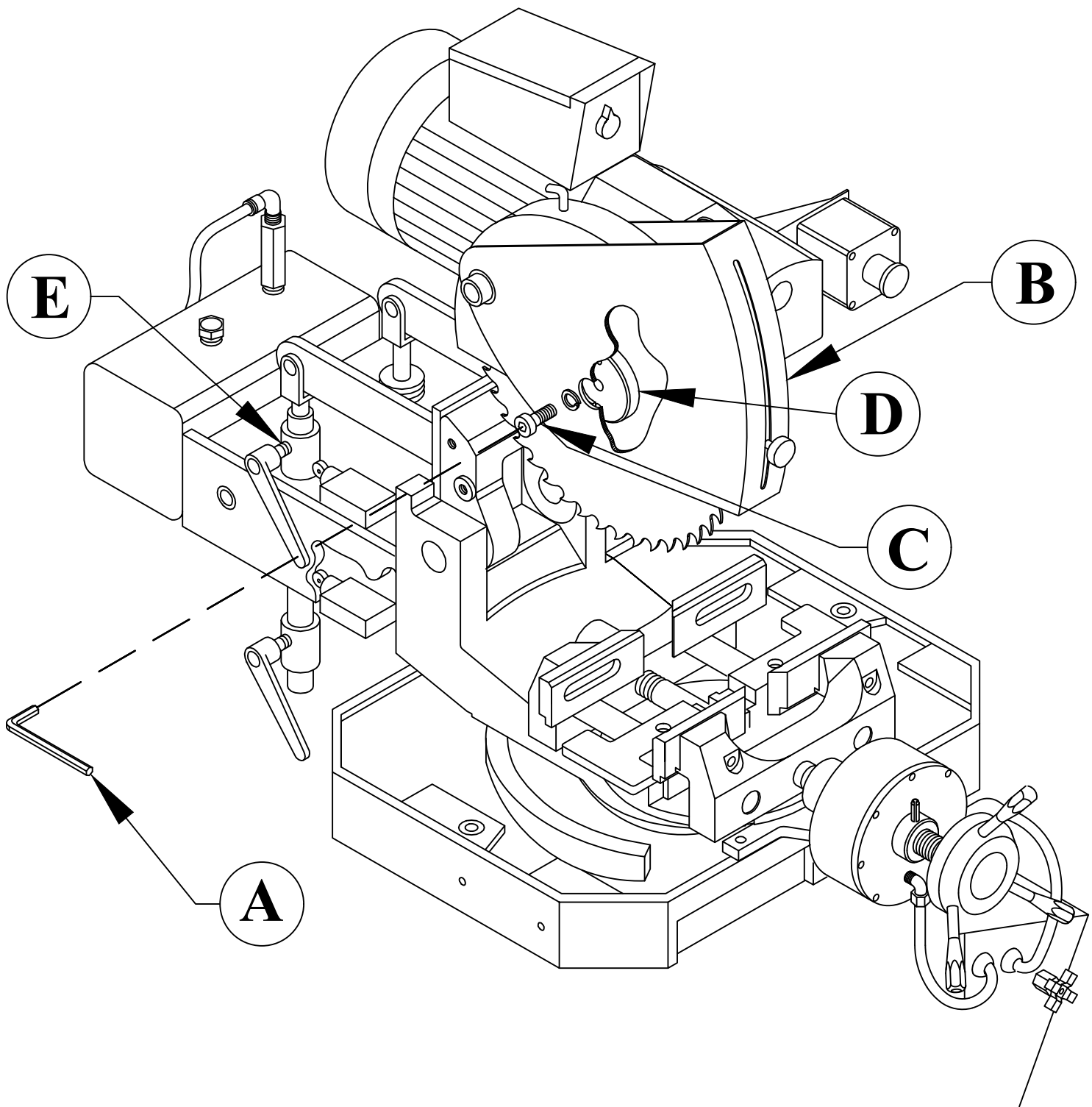


FIGURE 19

7.10 POWER DOWN & POWER VISE TROUBLE SHOOTING

⊗ **CAUTION: SHUT OFF AIR SUPPLY BEFORE PERFORMING ANY MAINTENANCE**

SEE FIGURE 20 ON THE FOLLOWING PAGE.

- 1. THE HEAD FEEDS DOWN FULL SPEED WITH THE FLOW CONTROL TURNED OFF.**
Bad check valve: Clean or replace it.
- 2. THE HEAD FEEDS FAST WITH NO CONTROL, HEAD BANGING UP.**
Low oil level - Add hydraulic oil to reservoir.
SEE SECTION 7.6
- 3. THE HEAD STOPS AND DOES NOT FEED THROUGH THE MATERIAL.**
Insufficient air pressure to machine. Adjust the Pressure Regulator to increase the pressure. SEE SECTION 4.6
- 4. THE HEAD FEEDS DOWN BUT DOESN'T RETURN.**
Check the lower limit switch, then PD solenoid.
SEE FIGURE 20-1
- 5. EXCESSIVE AMOUNT OF OIL IS EXHAUSTING THRU THE PD SOLENOID.**
Hydraulic Reservoir was overfilled or seals are leaking in the cylinder.

6. POWER VISE WILL NOT CLAMP OR UNCLAMP OR IS STUCK.

A. Check air supply to vise. Make sure air gets to the front when clamping and to the side for unclamping. If not, check the limit switches, then the PK solenoid.

SEE FIGURE 20-1

B. Make sure when air is on one side of cylinder, air is not leaking past the seals to the other side of the cylinder. If it is, seals need to be replaced.

C. If cylinder is stuck, the most likely cause is rust from water contaminating the air supply and/or oiler is not putting enough oil in the air supply - See SECTION 4.6 for more FRL info. If this happens, the power vise cylinder must be disassembled, cleaned and the seals replaced.

See SECTION 7.3 & 7.4 for power vise disassembly.

► **NOTE: FIGURE 20-1 ON THE RIGHT SIDE OF THE NEXT PAGE IS THE WIRING SCHEMATIC FOR THE CURRENT ELECTRIC POWER DOWN-FEED SYSTEM.**

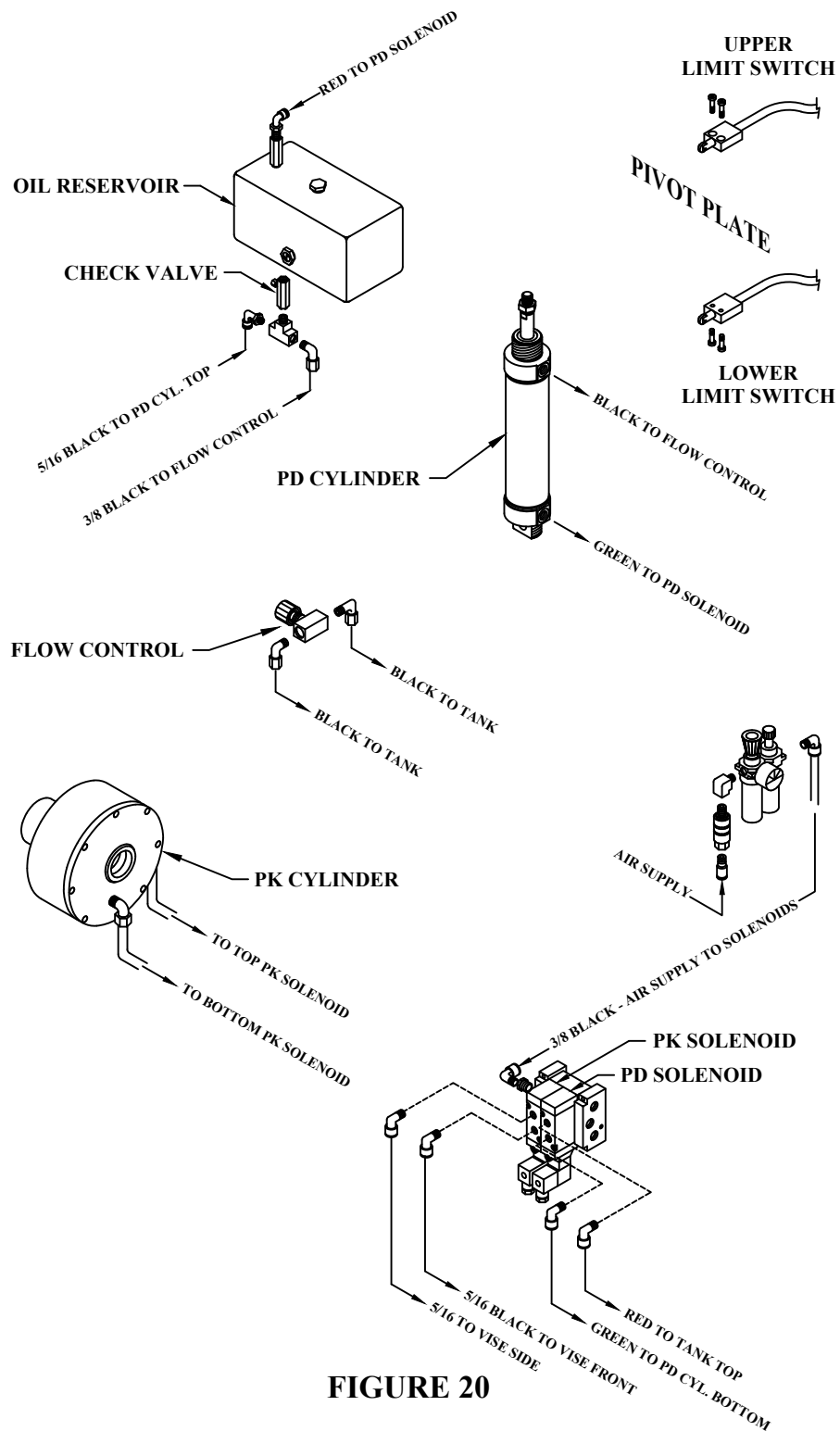


FIGURE 20

POWER DOWN FEED WIRING DIAGRAM

(Ser. #'s 6425 & Up)

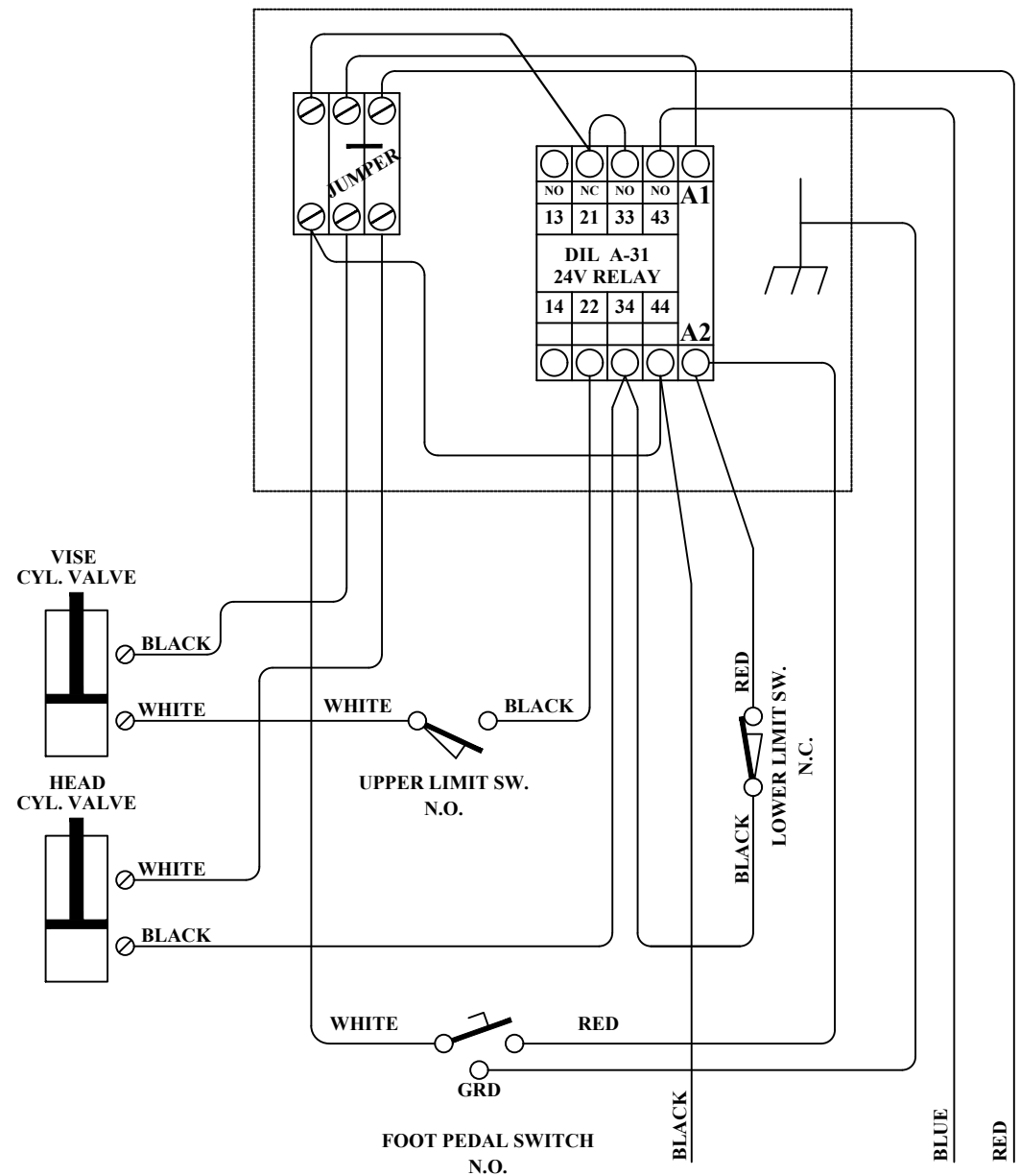


FIGURE 20-1

7.11 MATERIAL SUPPLY TRACKS

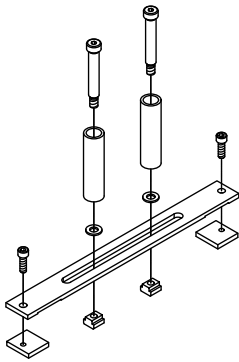
A five foot or ten foot roller supply track is an available option for this saw. It can be installed on the input side or output side of the saw to support longer pieces of material. We have measuring systems that can be bolted to the roller supply tracks that will help increase production. If needed, the supply tracks can also be bolted end to end to make them as long as needed.

SEE FIGURE 21 ON THE FOLLOWING PAGE.

- 1. What is shown is our P/N 029242 - 10' ROLLER CONVEYER W/LEGS that is fully assembled with our optional P/N 076938 - 13" GUIDE ASSEMBLY mounted to it.**
- 2. The conveyors can be bolted together to make longer lengths if needed.**
- 3. Adjust the conveyor so that the rollers are at the same height as the bed of the vise and make sure the conveyor is level. The conveyor height is adjusted via the slots in the legs.**
- 4. For additional stability, we strongly recommend anchoring the conveyor to the floor.**
- 5. The vertical rollers on the Guide Assembly are adjustable for width and can be adjusted to keep the material toward the front or back of the conveyor.**
- 6. Several Guide Assemblies can be mounted to the conveyor if needed.**

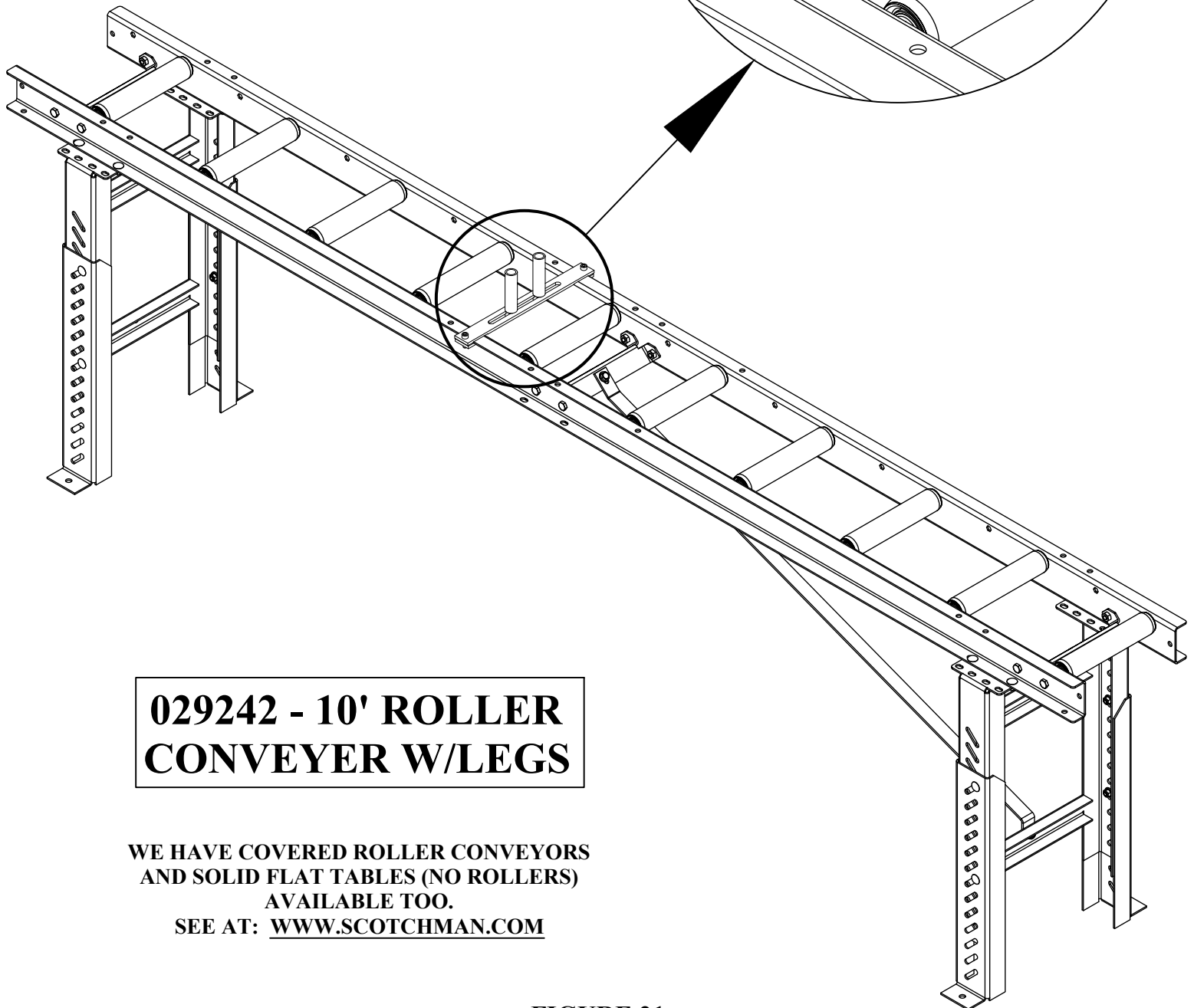
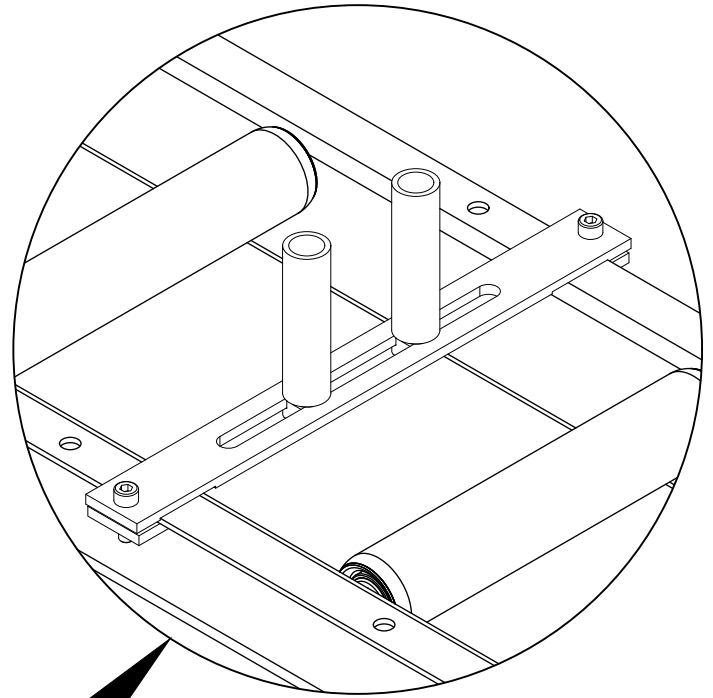
► **NOTE: WE ALSO HAVE COVERED ROLLER CONVEYORS AND SOLID FLAT CONVEYORS AVAILABLE. CONTACT YOUR DEALER OR THE FACTORY FOR MORE INFORMATION OR VISIT WWW.SCOTCHMAN.COM**

EXPLODED VIEW



076938
13" GUIDE
ASSEMBLY

DETAIL VIEW



029242 - 10' ROLLER
CONVEYER W/LEGS

WE HAVE COVERED ROLLER CONVEYORS
AND SOLID FLAT TABLES (NO ROLLERS)
AVAILABLE TOO.

SEE AT: WWW.SCOTCHMAN.COM

FIGURE 21

7.12 SCOTCHMAN MEASURING SYSTEMS

SEE FIGURE 22 ON THE NEXT PAGE FOR THE BELOW.

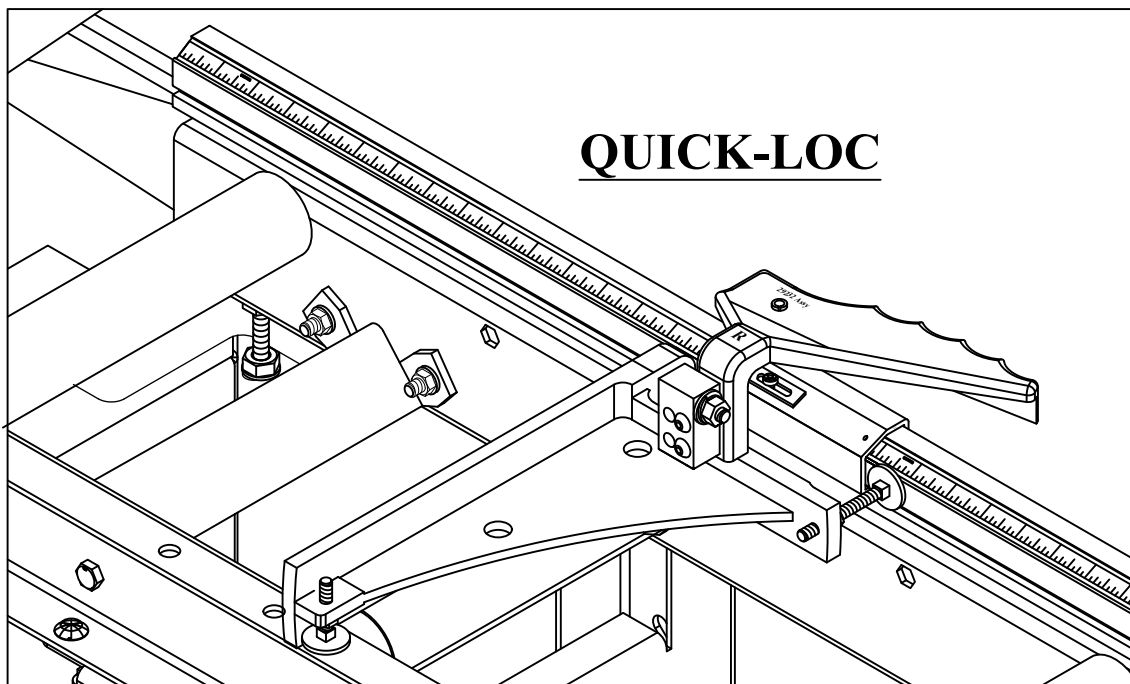
We have manual and digital measuring systems available that can be used with our supply tracks. They are adaptable to almost any type of machinery and are made in the USA.

QUICK-LOC - The Quick-Loc bolts to the conveyor and comes with a measuring tape. It has an aluminum rail with teeth that are in 1/16" increments. The teeth on the stainless-steel stop align themselves to the teeth on the rail, to give you the exact measurement you set and has a squeeze handle that allows the stop to be easily adjusted to any length. Its also guaranteed not to slip.

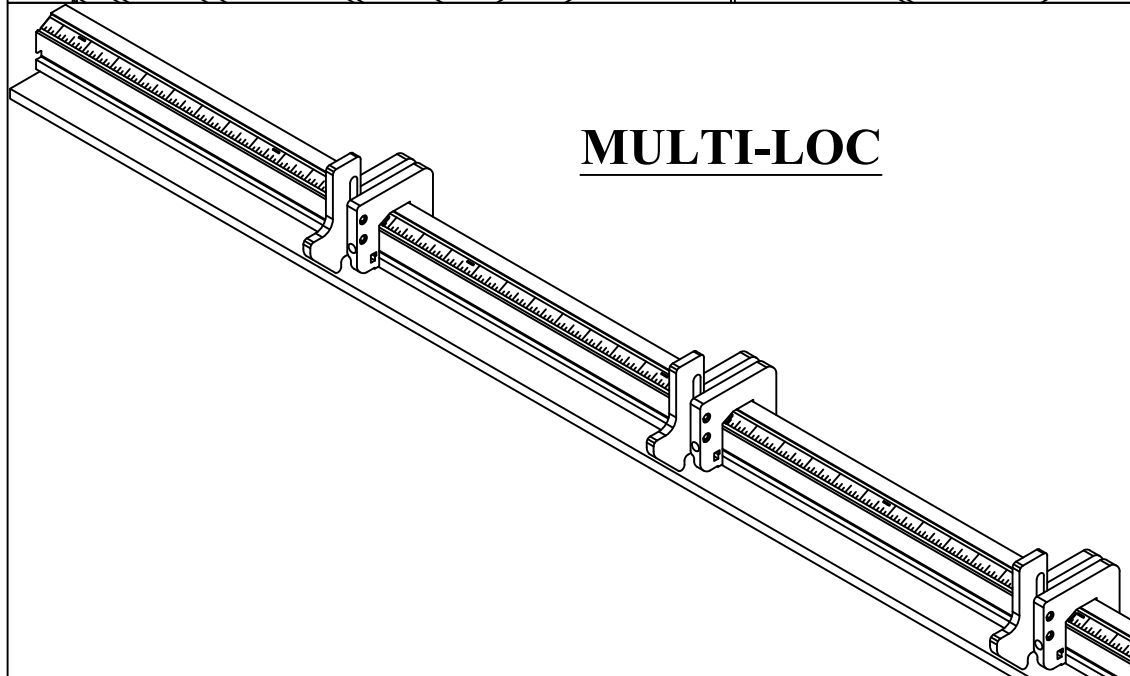
MULTI-LOC - Like the above, the Multi-Loc bolts to the conveyor and comes with a measuring tape and has an aluminum rail with teeth that are in 1/16" increments. This system comes with three stops (Standard or Heavy Duty) that can be positioned along the rail. If more than three stops are needed, you can order more. The standard stops allow parts to be cut as close as 1" apart and the heavy-duty stop 1-1/2" apart. The heavy-duty stops are usually used when a very heavy material is being processed. You can quickly cut a bar in to several different lengths without having to adjust the stop for each different length. For example, if you have a 4 ft. bar and need a 6, 8, 16, & 18 inch long piece, the three stops for the Multi-Loc are placed on the rail at those lengths. The Multi-Loc is best for high production applications.

DIGITAL QUICK STOP - The RG Digital Quick Stop is an entry level programmable stop. It is priced right, yet is very durable. Setup is as easy as entering the desired cut length and pressing go. It can be mounted to move left to right OR right to left. It is available in 8' and 12' lengths.

The above measuring systems can be mounted on either side of the saw and there are several options available for them as well. For more information, please call your dealer or the factory. Or visit our website at: www.scotchman.com



QUICK-LOC



MULTI-LOC



DIGITAL QUICK STOP

FIGURE 22

7.13 SPECIAL VISE JAWS

Special vise jaws for holding square tubing, rectangular tubing and angle iron are stock items. Jaws for holding thin wall round tubes, profiles and bundles are available on a made-to-order basis.

See Section 6.4 for examples.

For prices and delivery on special jaws, contact your local dealer or the factory.

7.14 LOCK-OUT DISCONNECT SWITCH

A lock-out disconnect switch is available for this machine if your plant is not equipped with lock-out capabilities. The switch mounts on the base of the saw and is shipped complete with all of the necessary parts and installation instructions.

It is available as our P/N 078200 - SAW DISCONNECT ASSEMBLY

7.15 PNEUMATIC DIAGRAM FOR PK PD MACHINES

SEE FIGURE 23 ON THE FOLLOWING PAGE.

In the drawing on the next page is a basic illustration of the pneumatic system of a PK PD (power vise and power down feed) saw. If you have a PD only saw disregard the PK cylinder, PK solenoid and hoses. The diagram is intended to show the components and to show hose routings and locations for these saws.

PK (POWER VISE) ONLY SAWS

If you have a PK only saw, this does not represent the pneumatic system for it. The PK only saws have a simple pneumatic roller valve on the saw blade guard that actuates the power vise.

Please see Section 7.2 for a diagram for the PK only saws.

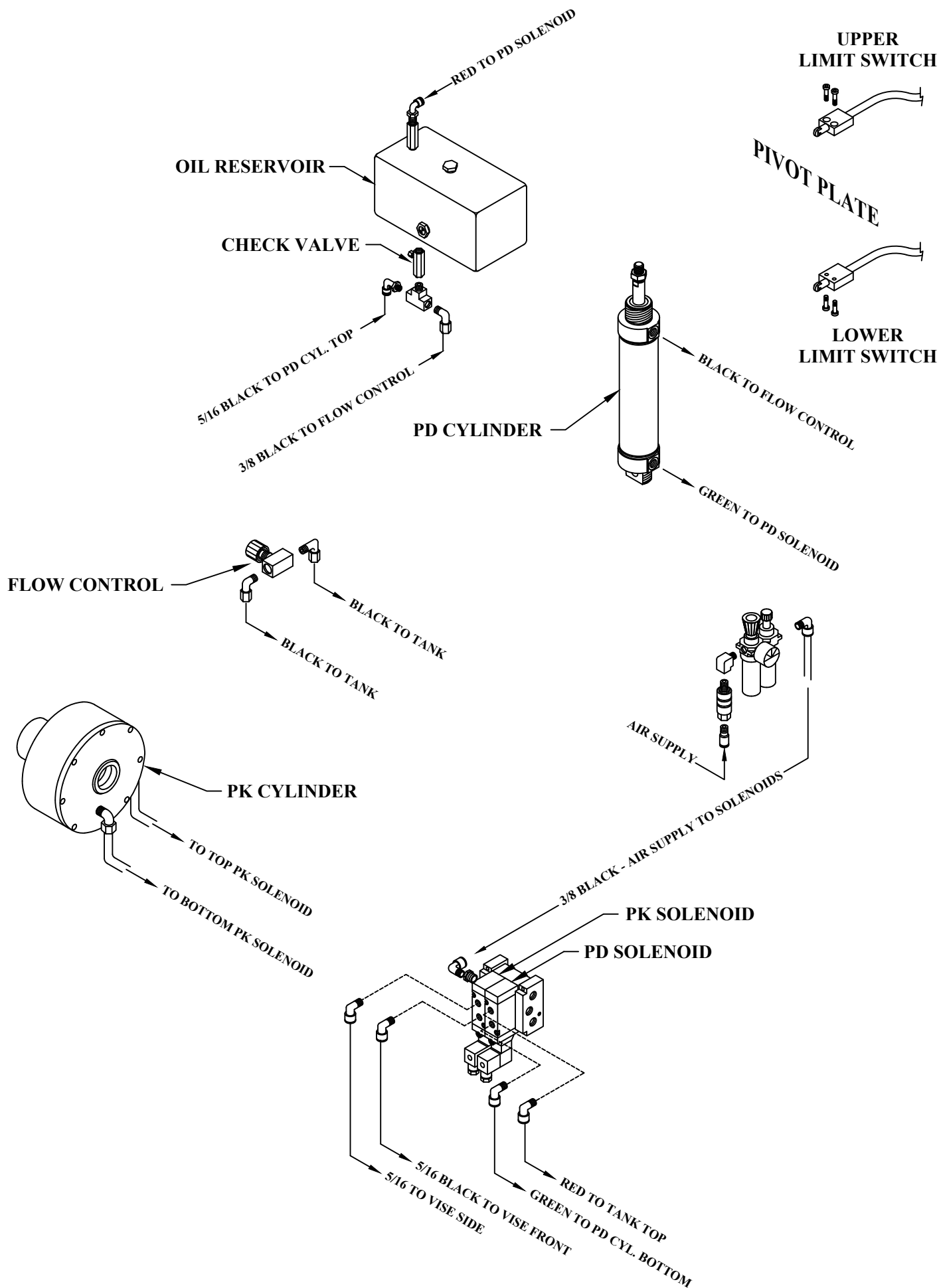


FIGURE 23

8.0 TROUBLE SHOOTING GUIDE

8.1 ELECTRICAL TROUBLE SHOOTING

1. THE MOTOR WILL NOT RUN:

- A. Some models have a lock-out switch in the base of the saw. If your saw has this option, make sure that it is in the ON position.**
- B. On manual and power vise machines, the Hi-Low switch must be in either the Hi or the Low position for the trigger switch to work.**
- C. On machines equipped with the power down feed option, the HI-LOW switch must be turned to either the HI or LOW position and the foot switch depressed to start the motor.**
- D. Also, check the supply voltage to the saw to make sure that it is the same as the motor voltage. If the supply voltage is correct, the switch energizes and the motor still will not run, contact your local dealer or the factory.**

2. THE SAW MOTOR RUNS BUT DOES NOT HAVE ADEQUATE POWER:

- A. Make sure that the supply voltage and phase correspond to the saw motor's voltage and phase.**
- B. Disconnect the machine from the power source and check for any loose or disconnected wires.**
- C. The supply lines to the machine must be of adequate size to handle the load. For recommended sizes and lengths, SEE SECTION 4.4.**
- D. The worm gears in the head may be damaged. With the power to the machine disconnected, check the blade spindle for any free travel. If free play is present, drain the oil from the head and remove the motor. Check both worm gears for wear and replace, if necessary. We recommend replacing the worm gears as a set, if either shows wear.**

For instructions on gear replacement, REFER TO SECTION 8.4.

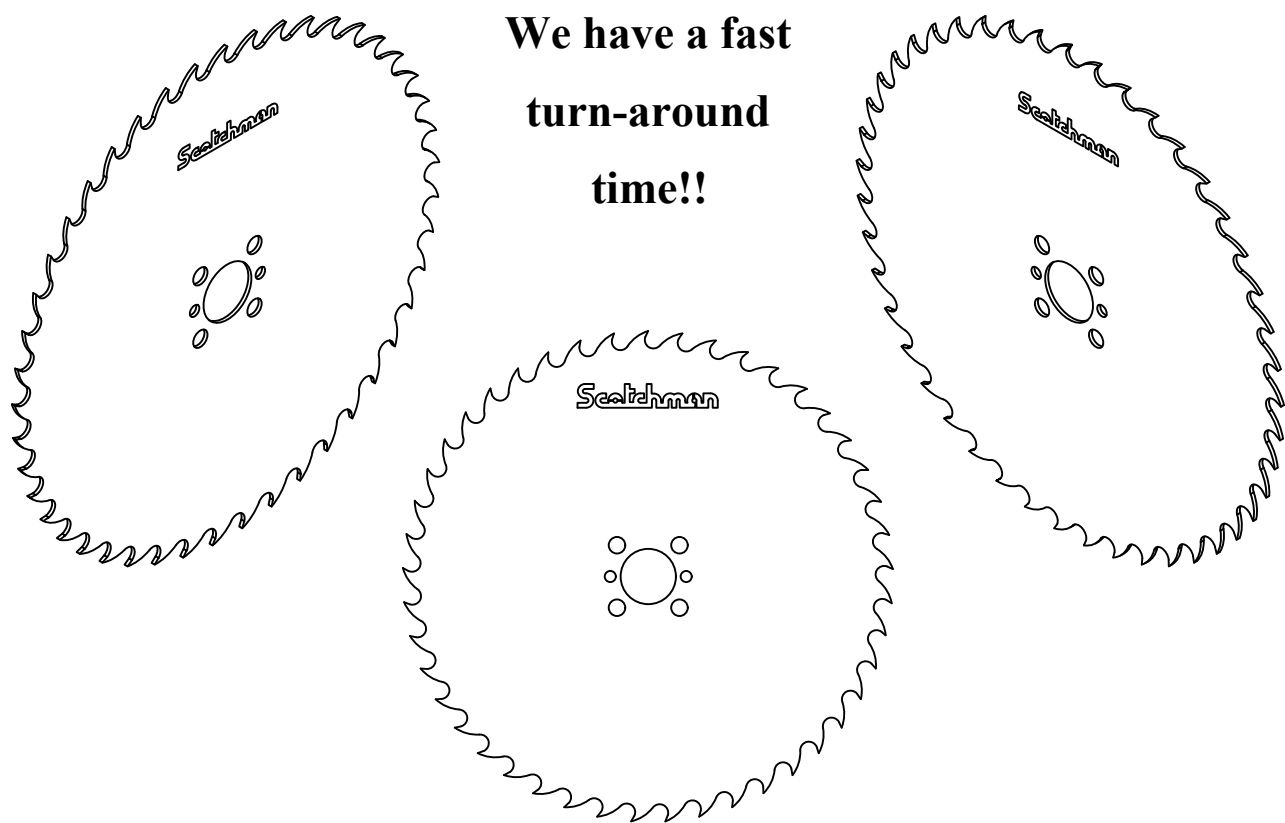
8.2 BREAKAGE OR EXCESSIVE DULLING OF BLADES

- 1. Select the proper blade and spindle speed for the material being cut. For recommendations, REFER TO SECTION 6.3.**
- 2. Always break in the blade before you start normal cutting.**
- 3. Do not apply excessive down pressure on the workpiece. Excessive down pressure will cause the teeth to remove too large of a chip, resulting in premature dulling or breakage.**
- 4. Use a good quality, synthetic coolant and maintain the proper ratio of coolant to water as shown in SECTION 4.8. We recommend our P/N 75752 coolant, or for stainless steel our P/N 075757.**
- 5. Have your blades re-sharpened by someone who has the right equipment for circular cold saw blades. Improper re-sharpening is one of the most common problems encountered in cold sawing. Here at Scotchman Ind. We use only modern CNC equipment operated by experienced people.**
- 6. Keep the blade flange, the face of the blade spindle and the blade clean and free from nicks. Any contamination or nicks on the flange, spindle or the blade will cause the blade to run out of alignment. A dirty flange can also cause many other problems. A clean flange is very important!!**
- 7. Always remove the back lash when installing a blade. For instructions, REFER TO SECTION 6.1. Also, check the condition of the drive pins when replacing the blade. If the drive pins are broken or worn, replace them.**
- 8. Any of the above problems may cause a condition known as pick-up. Pick-up is caused when the material being cut adheres itself to the teeth on a blade. When pick-up is present, you will feel a jerking or jumping motion in the saw head while cutting. This is caused by the blade being pinched as it goes through the material where the pick-up is present.**

Pick-up can be removed by using a fine honing stone or a very fine file. When removing pick-up, care must be taken not to remove any part of the blade - especially the coating. Often it is best just to have the blade resharpened. After the pick-up has been removed, review the above items to determine what caused the problem.

FOR BEST RESULTS, USE ONLY SCOTCHMAN COLD SAW BLADES AND COOLANT

**Send your blades back to our factory to be professionally
resharpend on modern CNC equipment operated by our
experienced employees.**



Ship them to:
SCOTCHMAN IND.
ATTN: BLADE DEPT.
180 E US HWY 14
PHILIP, SD 57567

Make sure to include
your contact information,
company name, and
return address inside the
box with the blades.

8.3 COOLANT SYSTEM

1. IF COOLANT WILL NOT FLOW:

- A. Check the wiring connections to the pump and make sure that the pump is running in the right direction. The pump has an arrow in the casting to indicate the direction.**
- B. Check the level of the coolant in the reservoir. If the level is too low, the fluid will not flow.**
- C. Check the reservoir for contamination or sludge build-up that may be blocking the pump inlet.**
- D. Remove the coolant line from the guard and make sure that it's clear. Also, make sure that the valve on the guard is open.**

2. IF THE COOLANT PUMP IS LEAKING:

- A. Check the connections on the coolant lines.**
- B. If the pump itself is leaking, there is a seal kit available. For instructions, SEE SECTION 5.4.**
For parts numbers, SEE SECTION 9.6.

8.4 GEAR REPLACEMENT

SEE FIGURE 24 ON THE FOLLOWING PAGE.

NOTE: SEE SECTION 9.1 FOR PART NUMBERS

1. Remove the drain plug (A) from the head casting and allow the fluid to drain.
2. Remove the motor from the head.
3. Remove the four bolts (B) from the bearing retainer (C).
4. Remove the worm shaft assembly (D). The worm shaft has a 10mm threaded hole in the end of it for a slide hammer. Inspect the worm shaft, drive gear and bearings for wear.
6. After the worm shaft has been removed, remove the three bolts (E) from the bearing housing (F).
7. Remove the spindle shaft (G) with a slide hammer.
8. Remove the snap ring (H).
9. The brass worm gear (I) can now be pressed off of the shaft.
10. Check the condition of the bearings and the seals before re-assembling the head.
11. If bearing needs to be replaced, press shaft (G) out of bearing housing (F).
12. Remove internal snap ring (K) and seal (M) from bearing housing (F).
13. Press bearing (L) out of bearing housing (F).
14. Check the condition of the key (J) and the key-way in the spindle shaft (G). Replace all worn parts before assembling the worm shaft assembly.
15. Begin assembly by pressing bearing (L) into bearing housing (F). Press only on the outer race of the bearing and make sure it seats against the housing. Then install internal snap ring (K).
16. Install new seal (M) and smear a little oil around the inside lip of the seal.
17. Then press the spindle shaft (G) thru the bearing in the bearing housing. Make sure to support only the inner race of the bearing and make sure the shaft seats against the bearing.
18. Install the spacer (N) on the shaft (G), then the key (J). Align key way and press on new gear (I) and install snap ring (H).
19. Replace the o-ring (O) on the bearing housing (F) and apply a little oil on it before installing.
20. After the gear case is assembled, use an ISO-460 gear oil specified for worm gears. Oils containing EP additives may degrade or damage bronze gears and must be avoided. Mobile 600WSuper Cylinder Oil is highly recommended and available from the factory under P/N 075758.

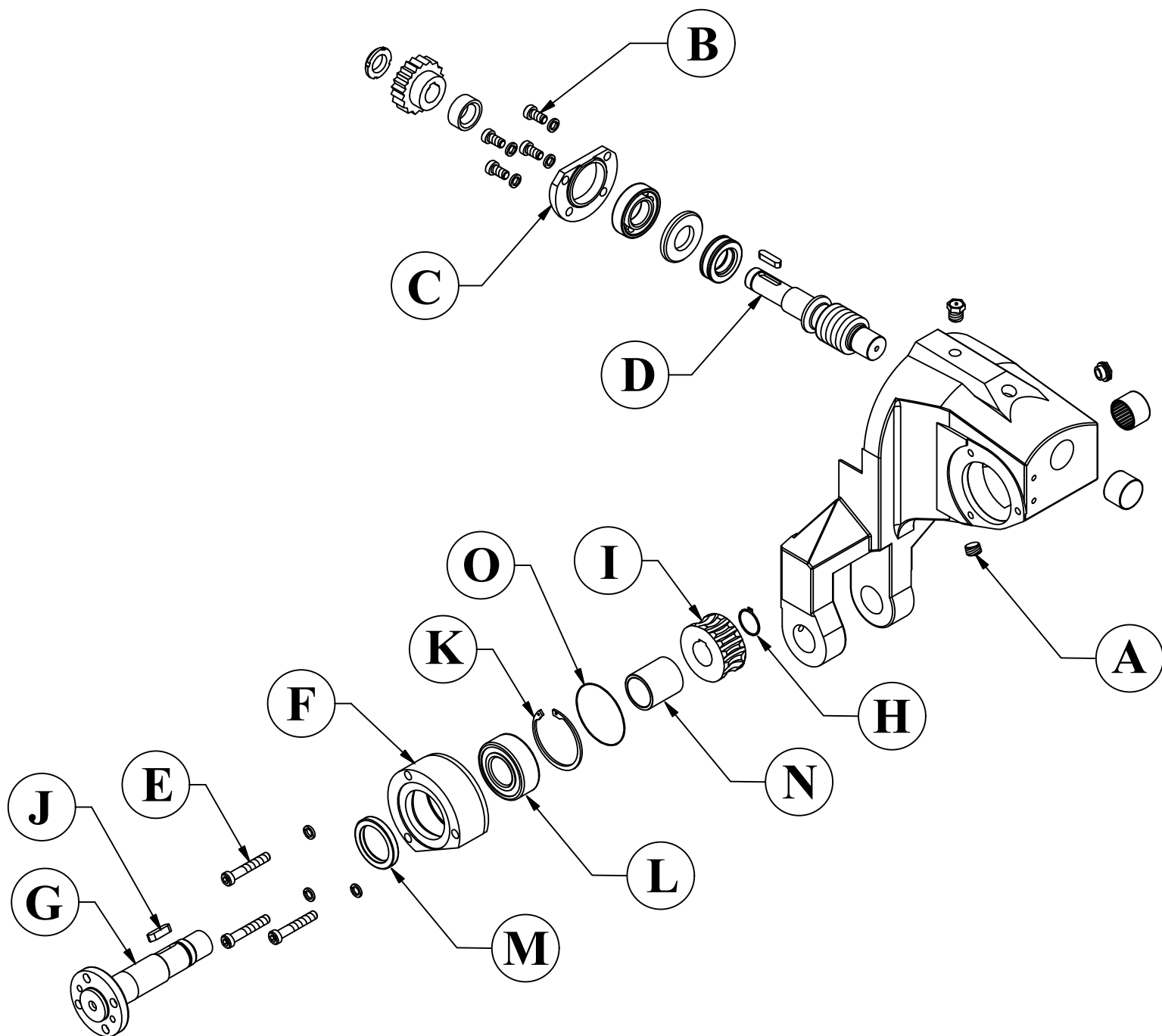


FIGURE 24

9.0 PARTS LISTS

9.1 SAW HEAD

ITEM	PART #	DESCRIPTION	X	075075	Seal
A	077150	Needle Bearing	Y	077628	Bearing Housing
B	077322	Worm Shaft	Z	077624	Saw Spindle
C	077323	Pivot Bearing	AA	077626	Saw Flange (Includes DD & UU)
D	077324	Spacer Ring	BB	221212	M-10 x 30 SHCS
E	077325	Bearing	CC	073110	M-10 Lock Washer
F	077330	350 Lock Ring	DD	073920	Dowel Pins
G	077326	Spacer Ring	EE	073641	M-10 x 65 SHCS
H	077321	Lock Nut	FF	077335	Snap Ring
I	077328	Gear Wheel	GG	077337	O-Ring
J	060250	M-10 SHCS	HH	077340	Key 10 x 8 x 32
K	075080	Key 8 x 7 x 32	II	075081	Snap Ring
L	208020	M-20 Jam Nut	JJ	077864	M-5 x 12 SHCS
M	077000	Draw Handle Complete (Inc. L, M, N, NA)	KK	201110	M-6 x 12 HHCS
			LL	060102	Emergency Stop Mount
N	077002	Handle	MM	060110	Emergency Stop Box
NA	077001	Switch	NN	077619	Complete Head Assembly
O	077152	Sight Gauge	OO	077860	Head Gasket
P	077153	Drain Plug	PP	077630	3/8 BSPT Breather
Q	077622	Head Casting	QQ	073692	Eye Bolt (Not Pictured)
R	077110	Stroke Adjustment Nut	RR	073106	M-6 Lock Washer
S	201235	M-10 x 80 HHCS	SS	077618	350 Worm Shaft Ass'y (Inc. B, C, D, E, F, G, H, I, K)
T	073326	M-8 x 30 HHCS			
U	077333	Worm Wheel	TT	077617	Spindle Ass'y (Inc. U, V, W, X, Y, Z, FF GG, HH, II)
V	077334	Bushing			
W	075076	Bearing	UU	660365	Flange ATTN: Decal

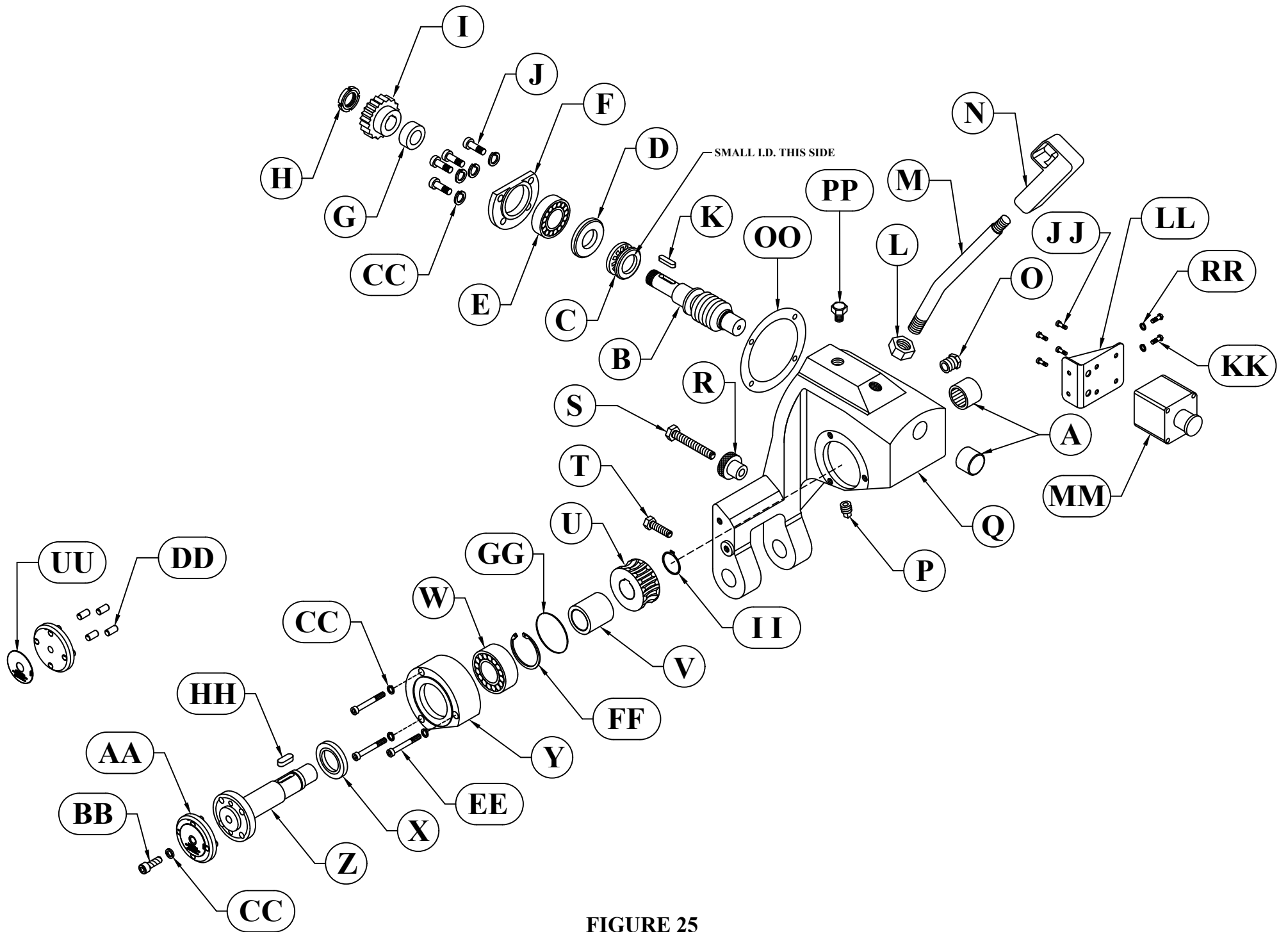


FIGURE 25

9.2 VISE ASSEMBLY

ITEM	PART #	DESCRIPTION
A	221220	M-10 x 40 SHCS
B	077319	Cast Grip Cheek (Rear - Includes F)
C	077610	Vise Jaw (Right)
CA	077612	Vise Jaw Set (Inc. 2 of C & 2 of W)
D	208012	M-10 Hex Nut
E	077309	Guide Shaft
F	077310	Seal
G	077307	Vise Base
H	077100	Dowel Pin
I	077133	Screw End
J	077136	Pressure Plate
K	677879	Tension Nut Assembly (Inc. L, M, K)
L	208010	M-8 Hex Nut
M	073329	M-8 x 45 SHCS
N	210016	M-16 Jam Nut
O	060240	Tension Handle (Includes N& P)
P	077138	Knob
R	077400	Handles
S	077121	M-20 x 1.5 Jam Nut
T	060270	Covering Cap
U	060267	Boss (Inc. T)
V	077318	Cast Grip Cheek (Front)
W	077611	Vise Jaw (Left)
WA	077612	Vise Jaw Set (Inc. 2 of C & 2 of W)
X	077305	Vise Spindle
Y	073460	M-10 x 16 SHCS
AA	077314	Filling Block
BB	077311	Support Block
CC	221212	M-10 x 30 SHCS
EE	060220	Complete Vise Assembly
FF	076910	Diagonal Tube Jaws

9.3 GUARD ASSEMBLY

MANUAL & POWER VISE MACHINES

ITEM	PART #	DESCRIPTION
A	N/A	Guard Shell
B	077165	Spacer Ring (thin)
C	N/A	Hinge Cap (front)
D	077146	Spacer Ring (thick)
E	N/A	Hinge Cap (rear)
F	077166	Nylon Spacer
G	077167	M30 Ext. Ret. Ring
H	077160	M8 Nylon Washer
I	N/A	Coupling Arm
J	077162	Pin
K	077161	8MM Dome Cap
L	077361	350 Pivot Bracket
LA	060490	350PK Pivot Bracket Assy.
M	221210	M10 X 25MM SHCS
N	077366	M8 X 10 X 10 SB
O	077362	Bushing/Bearing Bronze
P	077359	Star Knob
Q	077360	Box/Spacer Threaded
R	077356	350 Guard Lever
RA	060480	350PK Lever Assy.
S	077358	M10 Lock Ring
T	218010	M5 X 10 Set Screw
U	077609	350 Guard Lever Assy.
V	N/A	Coupling Arm
W	073641	M10 X 65 SHCS
X	212012	M10 Lock Washer
Y	060140	9/16 X 85" Coolant Line
Z	077154	Hose Barb
AA	077155	Shut Off Valve
BB	077770	1/8 NPT Hex Close Nipple (for old style valve - not used with 077155)
CC	069998	Coolant Slitter Assy.
DD	077864	M5 X 12 SHCS
EE	N/A	Right Tube (see 069998)
FF	N/A	Left Tube (see 069998)
GG	214012	M10 Regular Washer
HH	214005	M5 Small Washer
II	660350	Blade Rotation Decal
J J	060345	Danger Decal
KK	077869	350 Guard Assy-Sales (Z, AA, BB, CC, DD, HH, II & JJ)
LL	077342	350 Guard Pivot-Up Sales (KK, H, K, N, O, P, Q, R, S & U)

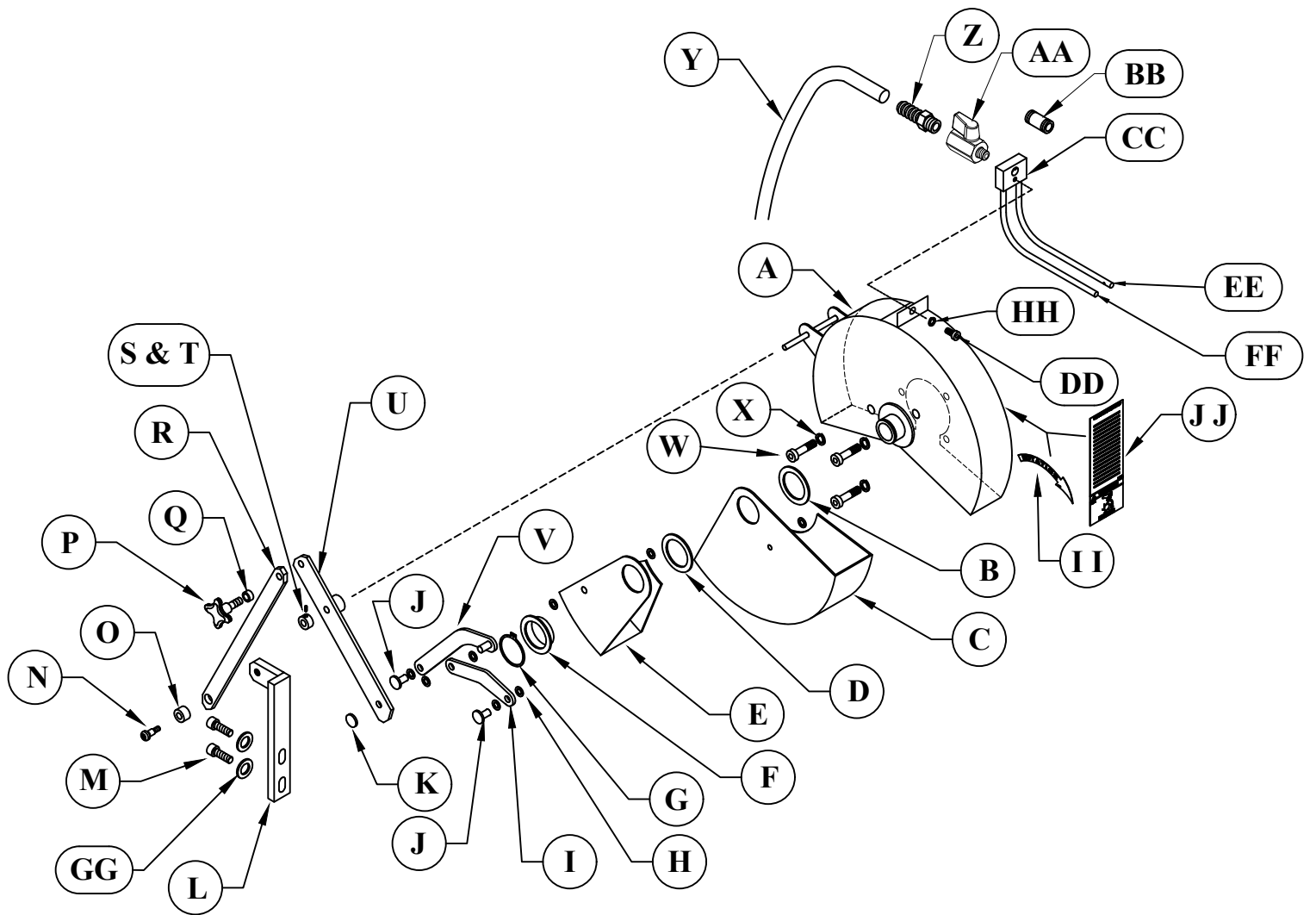


FIGURE 27

9.4 MOTOR ASSEMBLY

ITEM	PART #	DESCRIPTION
A	076883	Fan Cover EMOD
A1	076886	Fan Cover 1 PH
B	073407	M-5 x 6 Slot Head
C	076887	Fan (25mm Bore) FIMEC
C1	076884	Fan (30mm Bore) EMOD
C2	076893	Fan 1 PH Motor
D	077381	End Casting (30mm Bore)
E	075049	Motor Bearing (6205Z) (25mm)
EA	077325	Motor Bearing (6206) (30mm)
F	077191	Retaining Ring
G	077345	Key 8 x 7 x 50
H	203210	M10 X 25 HHCS (4 Req'd.)
I	077370	Key 6 x 4 x 30mm
J	077376	Seal 30 X 47 X 7
K	075050	Retaining Ring
L	076556	Snap Ring (30mm Only)
M	077378	Casting (Front) EMOD
N	077375	350 Motor Pinion Gear
O	077189	M20 X 1.0 Shaft Nut
P	077860	350 Head Gasket
Q	077855	Switch Box Gasket

Complete Motor (Without Switches)

A	076980	44-88 RPM/230 Volt
B	076982	44-88 RPM/460 Volt
C	076974	22-44 RPM/230 Volt
D	076976	22-44 RPM/460 Volt
E	076972	44 RPM/220 Volt/1 PH

Complete Motor Assemblies (With Switches)

A	078026	44-88 RPM/230V/T-S
	078028	44-88 RPM/230V/E-S
B	078030	44-88 RPM/460V/T-S
	078032	44-88 RPM/460V/E-S
C	078034	22-44 RPM/230V/T-S
	078036	22-44 RPM/230V/E-S

D	078038	22-44 RPM/460V/T-S
	078040	22-44 RPM/460V/E-S
E	077020	44 RPM/230V/1 PH/T-S
	077020	44 RPM/230V/1 PH/E-S

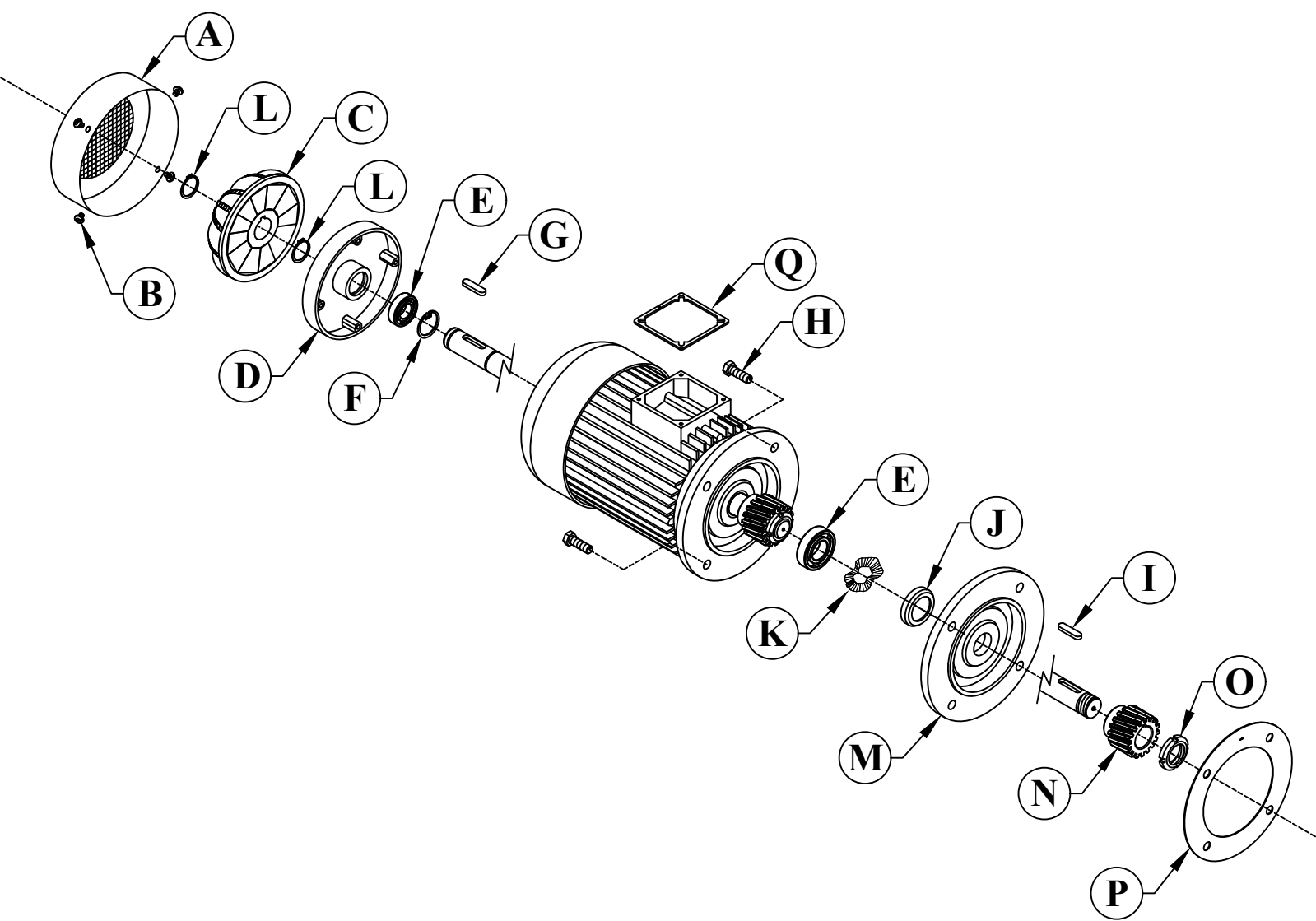


FIGURE 28

9.5 ELECTRICAL UNIT (AFTER JUNE 1996*)

ITEM	PART #	DESCRIPTION
A	060094	Switch Box and Lid Sales
B		Included with Item A
C	221002	M-4 x 12 SHCS
D*	060071	Contactor-24 Volt (S/N <u>34900696</u> & UP)
E	078456	M4/6 Terminal Block
F	078104	End Bracket
G	073440	M-4 x 6 SHCS
H	060222 (New Style - Blue & Oval)	Cam Switch (K&N) USN1427-600E22 (S/N <u>B121460422</u> & UP)
H1	060070 (Old Style - Black & Square)	Cam Switch (K-M) T0-6-4575.64/E
I	060049	Single Wire Ground Lug - (3) Required
J		Included with H, H1, and K
K**	060115 (ON/OFF - Black & Square)	Switch (220V 1PH with PD) T0-1-15511/E
L	011225	Cam Switch Knob - K&N
M	011844	K-M Knob SEL SW KNB-T0
N	060090	Power Cable
O	060095	Pump Cable
P	077183	M13.5 Liquid Cord Conn
Q	060050	Transformer 24V - 208/230/460
Q1	060051	Transformer 24V - 575V
R	075210	Elec. Mount Strip
S	563441	3/4 Liquid Type Connector
T	003122	Danger Voltage Label
U	077564	Fuse
V	060104	Liquid Tight Connector
W	077855	Gasket
X	077864	M5 x 12 SHCS
Y	060069 (has Trigger Switch)	Complete Box & Switch Ass'y T-S 230V
	060079 (has Trigger Switch)	Complete Box & Switch Ass'y T-S 460V
	060078 (has Trigger Switch)	Complete Box & Switch Ass'y T-S 575V
	060010 (has Trigger Switch)	Complete Box & Switch Ass'y T-S 230V/1 PH
Z	060068 (has E-Stop)	Complete Box & Switch Ass'y E-S 230V
	060089 (has E-Stop)	Complete Box & Switch Ass'y E-S 460V
	060088 (has E-Stop)	Complete Box & Switch Ass'y E-S 575V
	060011 (has E-Stop)	Complete Box & Switch Ass'y E-S 230V/1 PH

* Contact Scotchman Industries for machines built prior to June 1996

** Machines built after August 2004

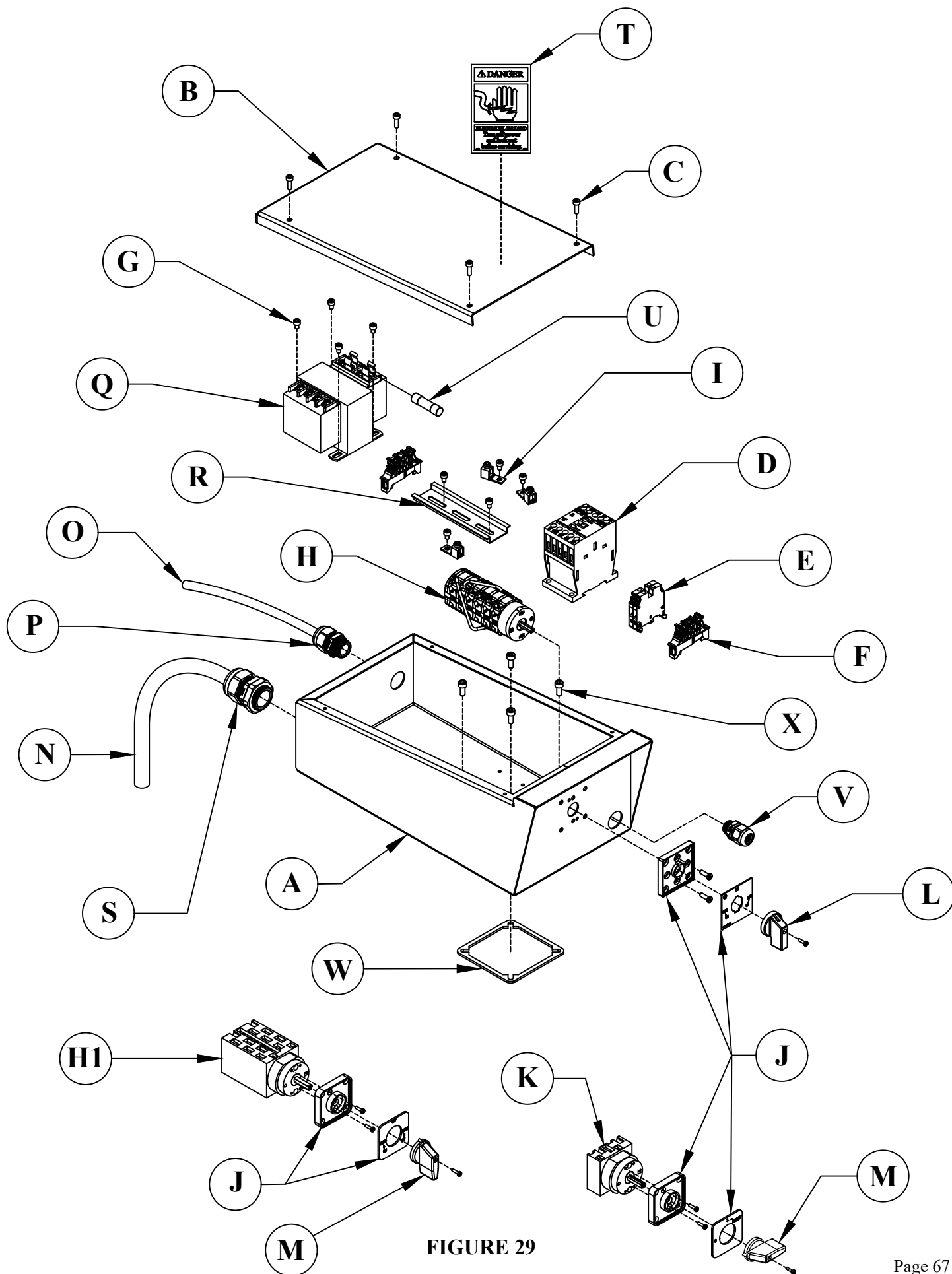


FIGURE 29

9.6 COOLANT PUMP

ITEM	PART #	DESCRIPTION
A	060166	1PH 230V Pump
A1	060205	230V 1PH Pump Assembly
A2	060150	230V 3PH Pump
A3	060200	230V 3PH Pump Assembly
A4	060158	460V 3PH Pump
A5	060201	460V 3PH Pump Assembly
A6	060160	575V Coolant Pump
Above Assemblies include A, G & I.		
B	060152	Impeller (Obsolete)
C	060157	End Cap
D	060151	Pump Oil Seal
E	N/A	Screw
F	N/A	Screw
G	060080	90 Degree Hose Barb
H	060140	Coolant Line
I	060095	Pump Cable
J	077864	M-5 x 12 SHCS
K	077545	Check Valve
L	060149	Pump Screen

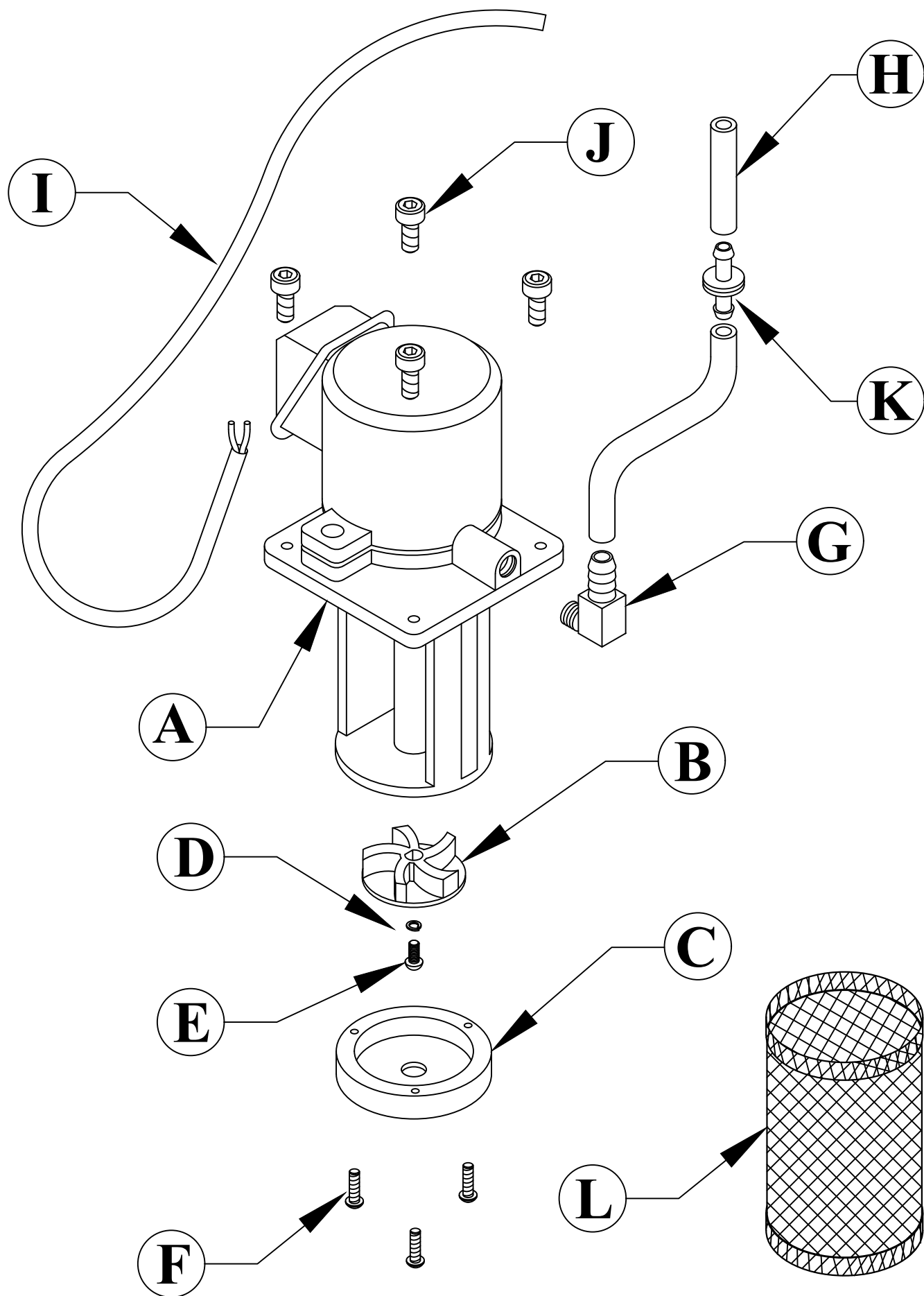


FIGURE 30

9.7 CAST BASE AND PEDESTAL

ITEM	PART #	DESCRIPTION
A	077300	Pedestal (Inc. S, R & V)
B	078600	Pivot Shaft
C	077364	Spring Mount (Lower)
D	077365	Return Spring
E	221210	M10 x 25 SHCS
F	077363	Spring Mount (Upper)
G	077142	Grease Zerk
H	077113	Cast Base (Inc. I & U)
J	073350	M10 x 100 HHCS
K	214012	M10 Washer
L	203210	M10 x 25 HHCS
M	077225	Miter Lock Mount
N	077227	Spring
O	077226	Release Handle
P	077228	Pin
Q	073660	M8 x 12 HHCS
R	077101	Scale
S	077100	Dowel Pin
T	077114	Complete Miter Lock (Inc. M, N, O, P & Q)
V	660255	Drive Pin (Scale-Not Pictured)

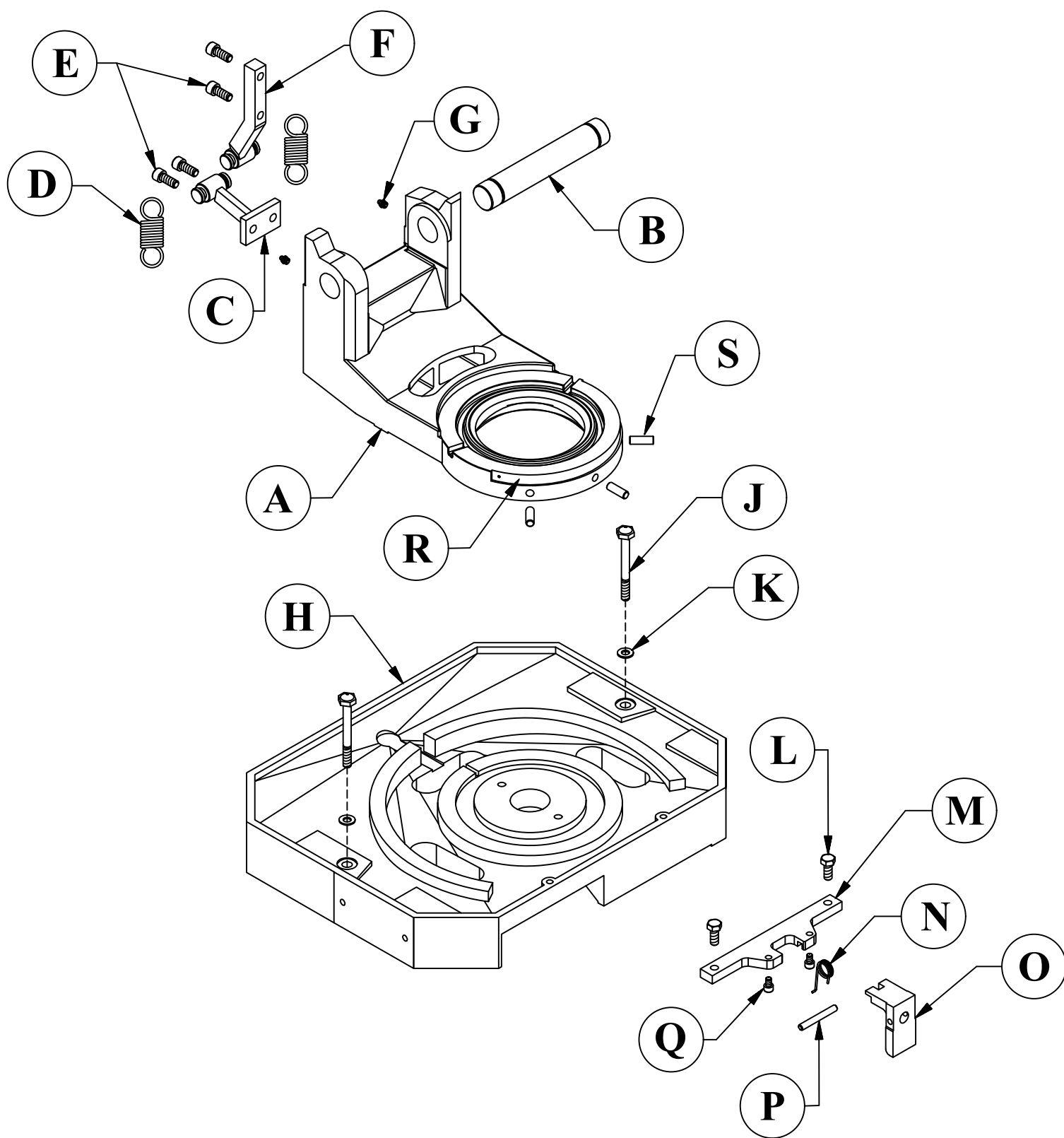


FIGURE 31

9.8 SAW BASE CABINET

ITEM	PART #	DESCRIPTION
1	760115	CPO Base Cabinet
2	760113	Painted Drawer Sales
3	760115.8	Pump Mounting Bracket
4	073420	M-8 x 16 DIN912 SHCS
5	215013	M-8 Greer Nut
6	660110	275/350 Complete Base

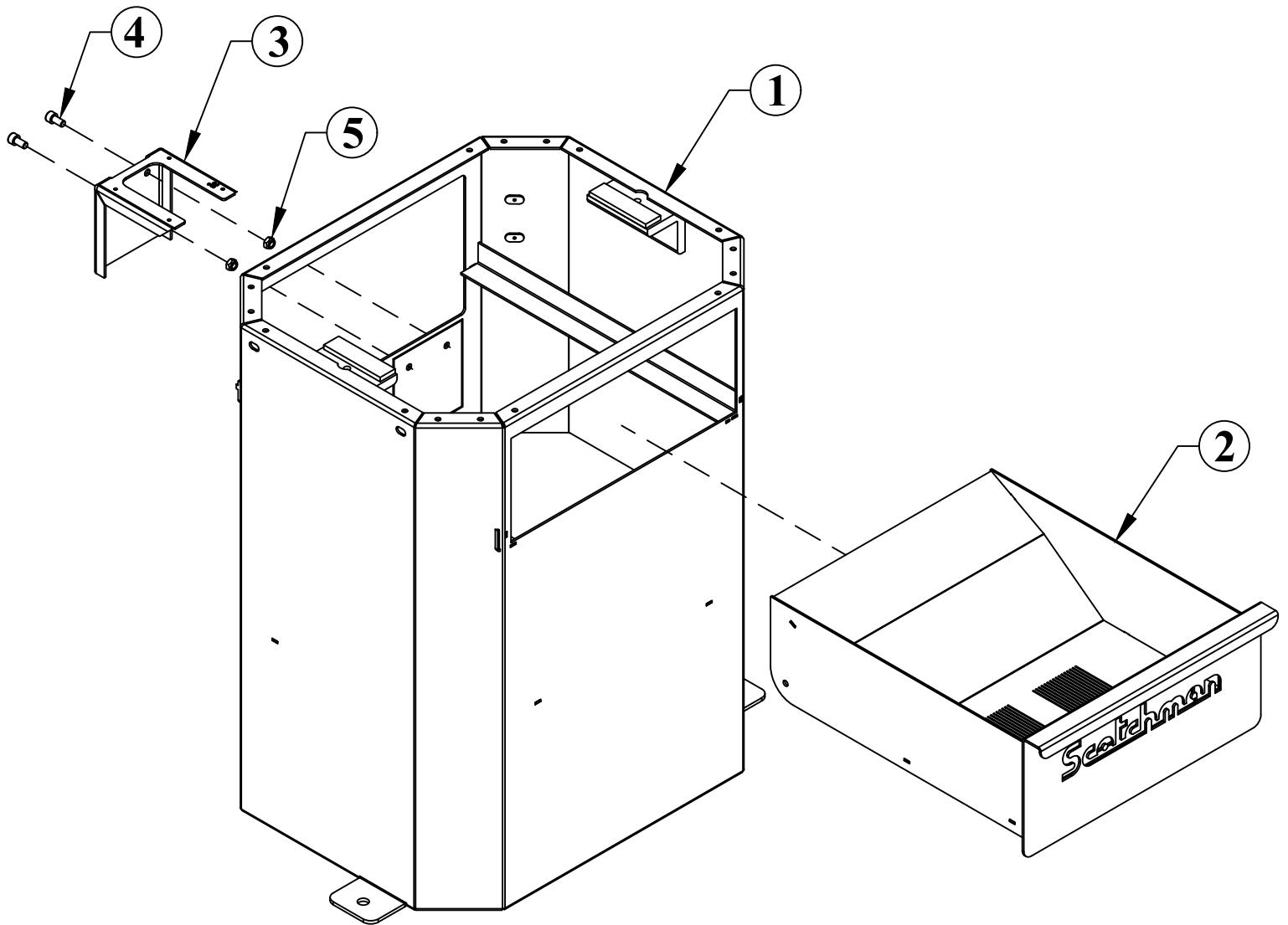


FIGURE 32

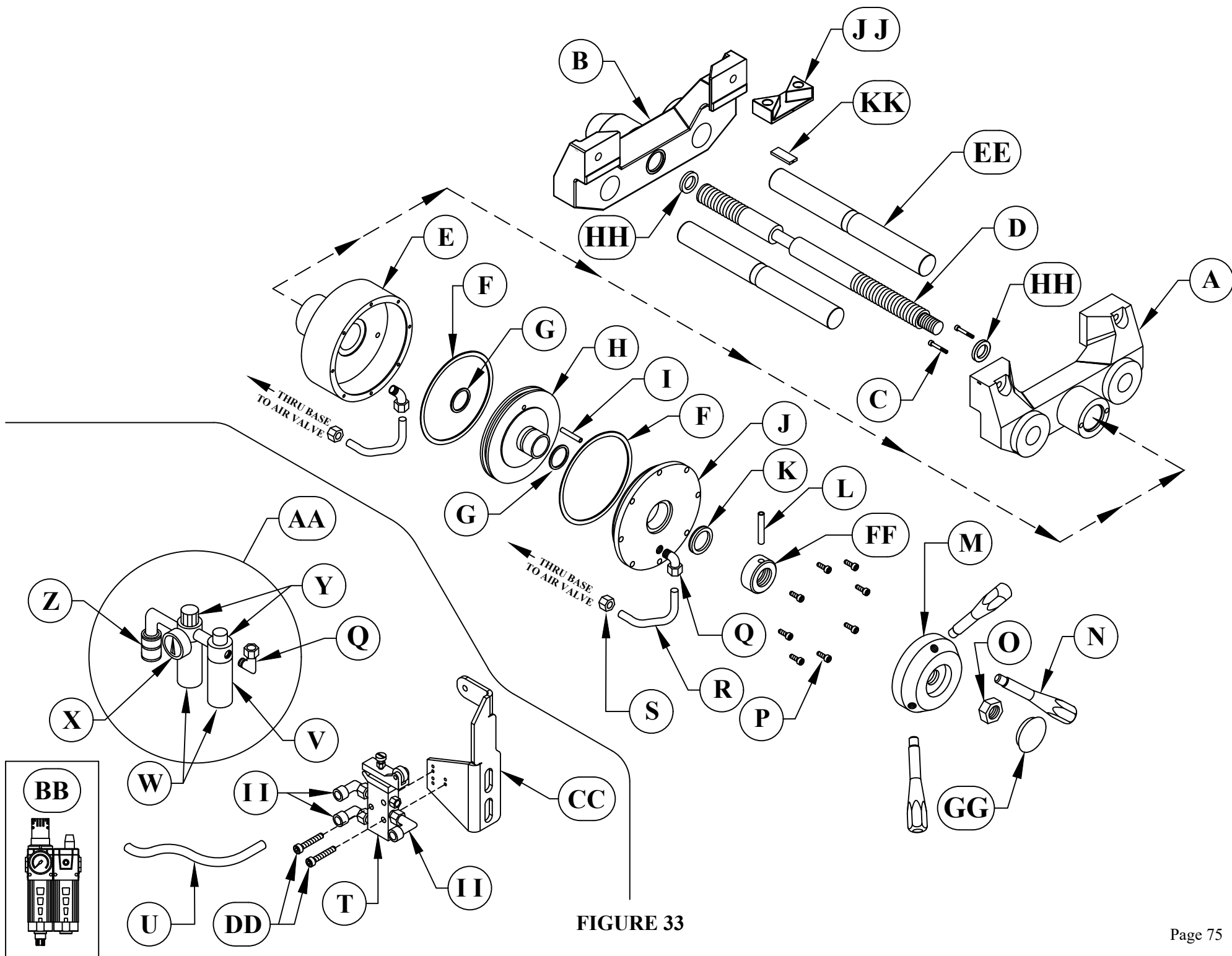
10.0 OPTIONAL EQUIPMENT PARTS LISTS

10.1 POWER VISE ASSEMBLY

ITEM	PART #	DESCRIPTION	V	077542	Lubricator Seal Kit -Parker (Obsolete)
A	077462	Cast Grip Cheek Front (inc. HH)	W	077539	Bowl - Parker (Obsolete when stock is gone)
B	077319	Cast Grip Cheek Rear (inc. HH)	W1	077554	Bowl, Lubicator - Parker (Modular FRL)
C	073457	M-6 x 80 SHCS	W2	077555	Bowl, Filter - Parker (Modular FRL)
D	077460	Screw Spindle	X	077538	Air Pressure Gauge
E	060204	PK Cylinder Body	Y	077540	Filter or Reg. Seal Kit - Parker (Obsolete)
F	077417	O-Ring	Z	077719	Slide Valve
G	077416	O-Ring	AA	077553	Complete Modular FRL Device - Parker (Obsolete) Replaced by BB
H	077411	PK Cylinder Piston			
I	077418	Roll Pin	BB*	077653	Camozzi FRL (Does Not Inc. Q & Z)
J	060450	PK Cylinder Lid	CC	060490	350 PK Pivot Bracket Assy.
K	077419	Oil Seal	DD	073415	M4 x 25 SHCS
L	077408	Roll Pin	EE	077309	350 Guide Shaft
M	060267	275/350 Boss Assy. (Inc. HH)	FF	077409	PK Lock Nut
N	077400	Vise Handle	GG	060270	Hole Plug
O	077121	M-20 Jam Nut	HH	077310	Seal
P	221002	M4 x 12 SHCS	I I	077741	1/8" Male SW X169PL
Q	077742	1/4" Male Swivel	J J	077311	Support Block
R	060501	5/16" Black Tube	KK	077314	Filling Block
S	077183	Cord Connector		077412	Complete PK Cylinder Assy.
T	077430	Four Way Valve		076371	Cylinder Seal Kit (Inc. F, G, & K
U	060501	5/16" Black Tube		060541	350 PK Vise Sales

* THE CAMOZZI FRL REPLACES THE OBSOLETE PARKER FRL's.

We do have some parts for the Parkers. But those parts will become obsolete once stock is gone.



10.2 POWER DOWN FEED ASSEMBLY

ITEM	PART #	DESCRIPTION
A	078510	350 PD Upper Bracket Assy.
B	221210	M10 X 25 SHCS
C	077211	Return Spring
D	078524	PD Cylinder Pivot
E	078525	350 PD Lower Brkt Assy.
F	140415	1/2" x 1-1/2" Clevis Pin
G	078520	PD Stop Rod Assy.
H	123120	1/8" x 1-1/4" Cotter Pin
I	078518	PD Stroke Adjustment Stop
J	078518	PD Stroke Adjustment Stop
K	080063	Stroke Adjustment Handle
L	077715	Pivot Bolt
M*	077670	Cylinder (Includes M, N, P, Q, S, T, U, W, & X)
N	077661	1/4" Elbow 90° (Bottom)
O		Included with Q
P	077662	3/8" Elbow 90° (Top)
Q	077578	M16 x 1.5 Cylinder Clevis
R	678550	Reservoir Decal
S	077505	Bellow Clamp
T	077700	Bellow
U	077512	Nut - Machined
V	078455	Sight Glass
W	210017	M16 x 1.5 Jam Nut
X	660505	Black Zip Tie

*** For older PD saws prior to 5/93 (without ITEM D - PD Cylinder Pivot) use P/N 077517**

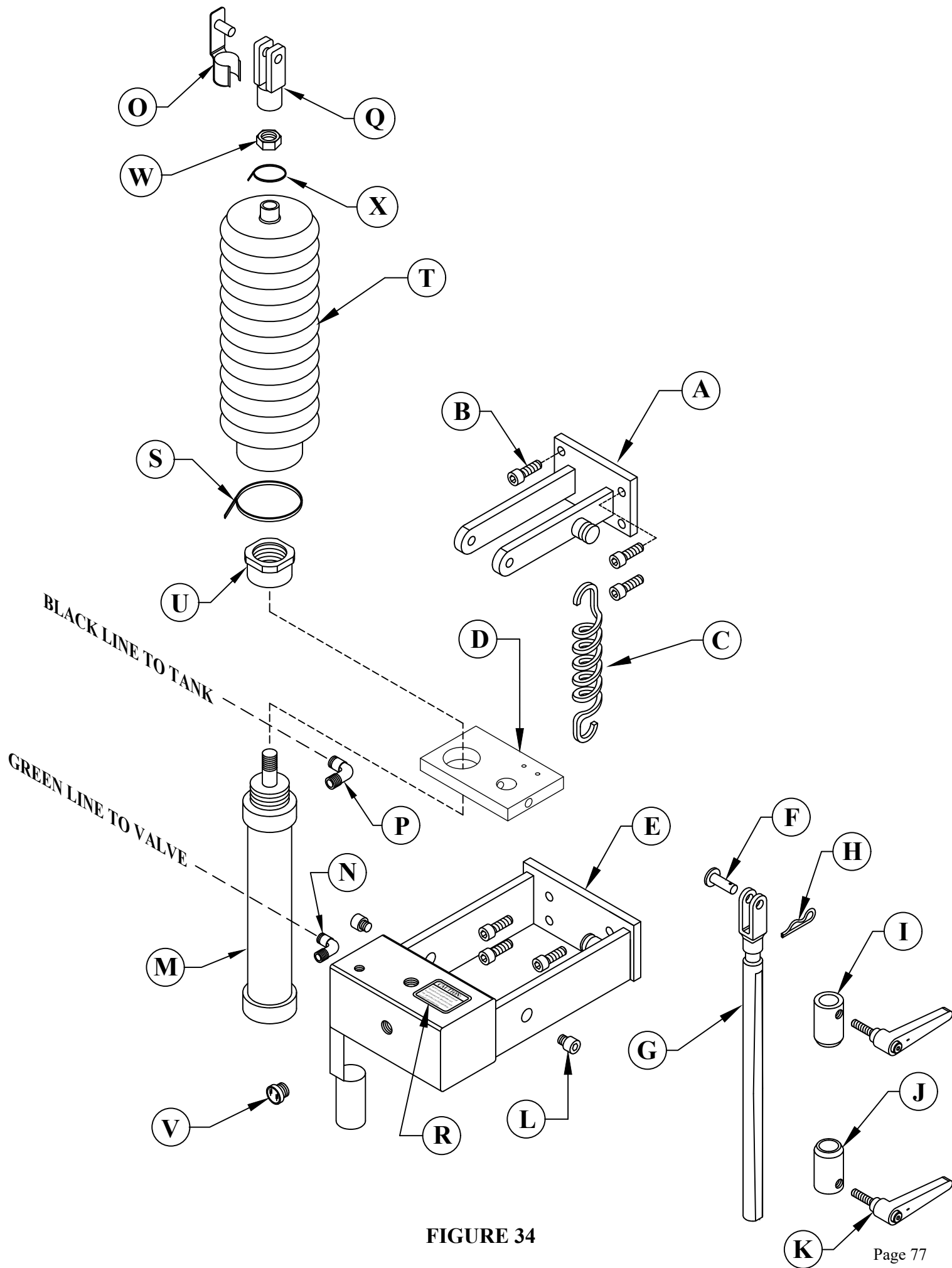


FIGURE 34

10.3 POWER DOWN FEED CONTROLS (S/N B64251003 & UP)

ITEM	PART #	DESCRIPTION
A	077736	PD Valve Mount
B	077183	Cord Connector
C	077738	90° Swivel
D	047535	Flow Control Valve
E	077746	1/4" NPT x 1/4" Street PL
F	078190	Regulator with Gauge
G	077770	1/8" Hex Nipple
H	077538	Gauge
I	077765	Brass Elbow
I 1	047251	Optional Air Regulator PK/PD Vise (Includes F, G, H, I & U)
J	045045	Breather
K	045667	Two Station Valve (DC) with Manifold <u>S/N B34200607 & Prior</u>
KA	045650	Solenoid 24VDC for 045655
KB	045655	Valve
K1	060040	Two Station Valve (AC) with Manifold <u>S/N B34310807 & After</u>
K1A	060039	Solenoid for 060040
K1B	060038	Valve for 060040
L	077777	3/8" NPT Plug
M	060104	Cord Grip
N	N/A	(provided by customer)
O	077719	Slide Valve
P	077737	1/4" NPT x 1/4" 90° Elbow
Q	077653	Modular FRL (Filt./Reg./Lub.) Assy.
Q1	077655	Bowl (Lubricator) for 077653
Q2	077654	Bowl (Filter) for 077653
R	221006	M4 x 50 SHCS
S	077740	3/8" 90° Male Swivel
T	045039	3600 x 4 Male Tee
U	077742	1/4" Male Swivel
V	077531	Check Valve
W	078455	Sight Glass
X	065025	Reservoir
Y	077701	Baffle
Z	073331	M5 X 45 SHCS
AA	046093	Limit Switch
BB	215010	M5 Greer Nut

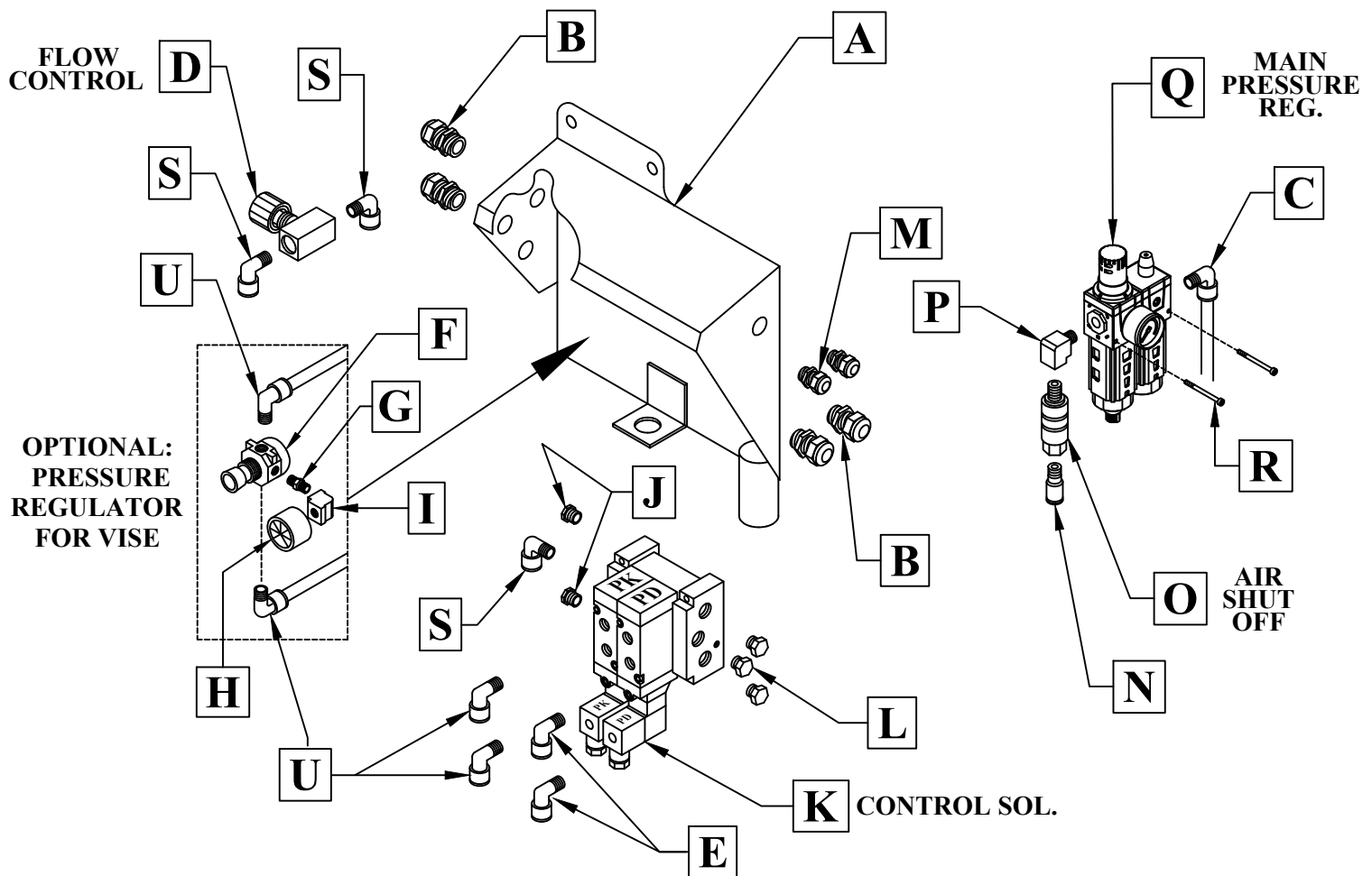
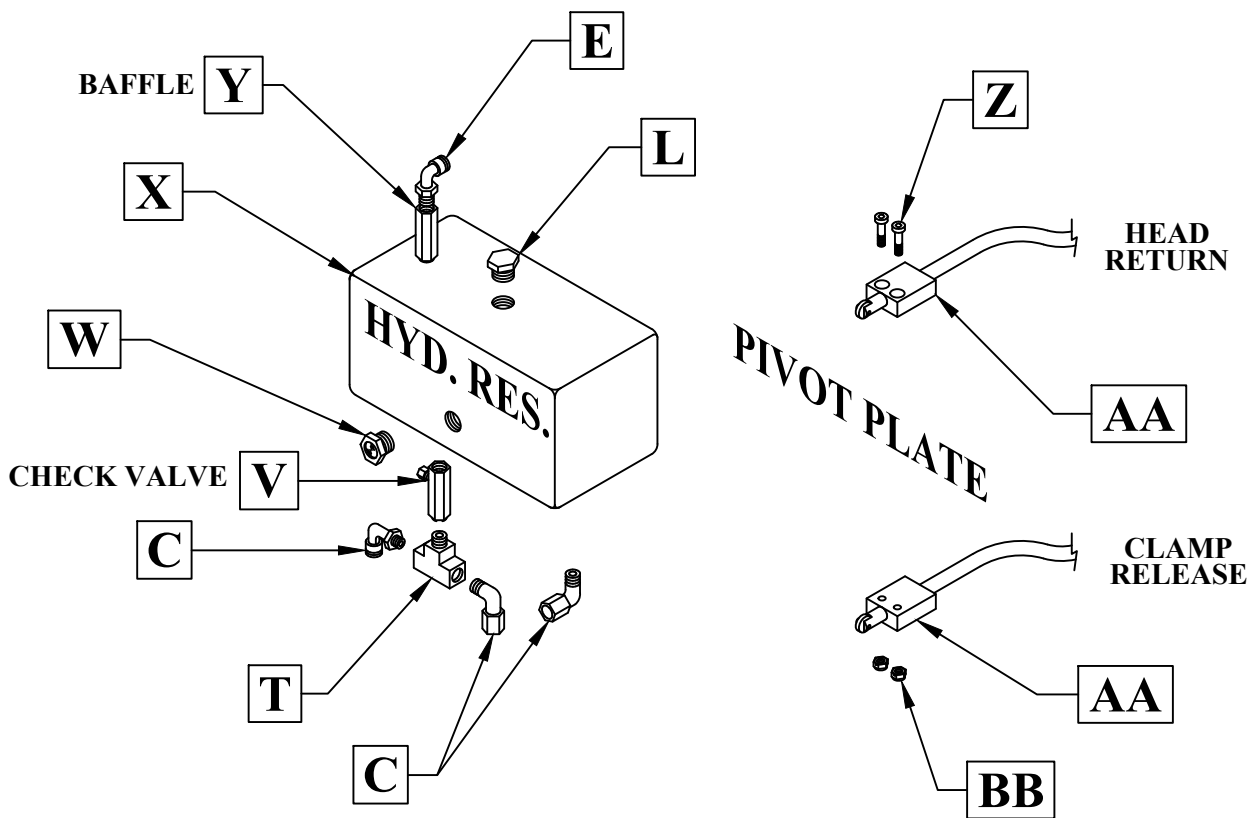


FIGURE 35

10.4 POWER DOWN FEED ELECTRICAL CONTROLS

S/N B64251003 & UP (See Note)

ITEM	PART #	DESCRIPTION
A	075250	Enclosure
B	077183	Cord Grip
C	075205	Mounting Plate
D	075210	Electric PD Mount Strip
E	073440	M4 x 6 SHCS
F	078104	End Bracket
I	078456	M4/6 Terminal Block
J	078457	Jumper Screw
K	060044	24 Volt Relay AC
K1*	048042	24 Volt Relay DC
L	060104	Cord Grip
M	660453	18/3 Cord - 90"
N	158202	5/16 Single Stud Bumper
O	115011	5/16 Nylon Lock Nuts
P**	562451	Micro Switch (SSC Foot Pedal)
	562452	Micro Switch (Linemaster Foot Pedal)
Q	078500	Complete Foot Switch
R*	048081	Glass Fuse
S*	048080	KM/AFR Fuse Term. Block - Obsolete
T*	221002	M4 X 12MM SHCS
U*	060053	Bridge Rectifier NTE5344

***These parts were used on saws from S/N B64251003 to B79960607. See inset on the next page.**

**** BEFORE ORDERING A MICROSWITCH: Inspect foot pedal to determine if it's a LINEMASTER or SSC.**

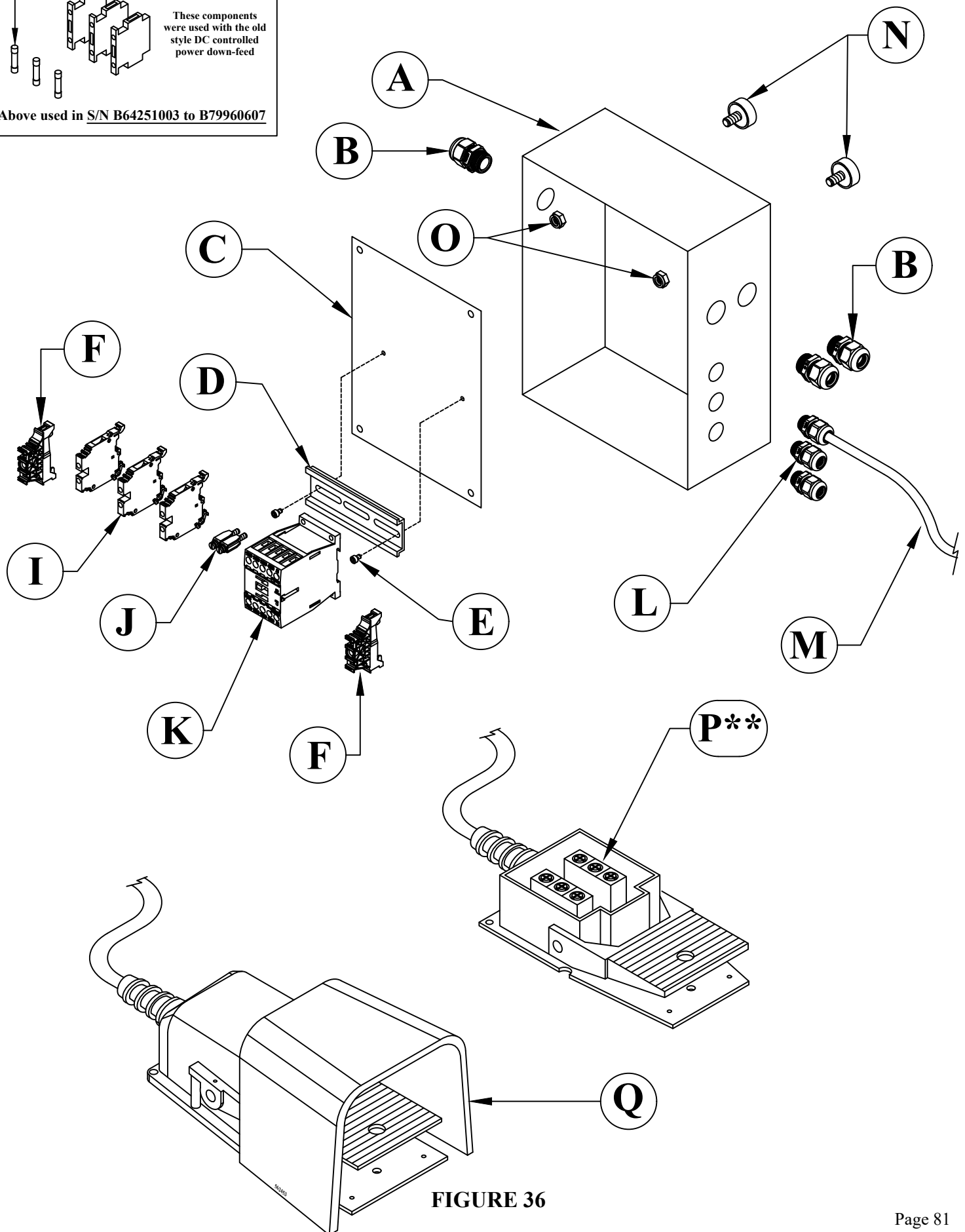
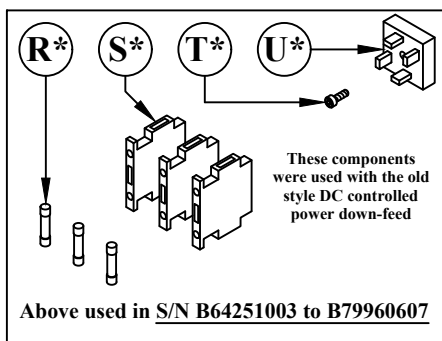


FIGURE 36

10.5 GUARD ASSEMBLY (POWER DOWN FEED)

ITEM	PART #	DESCRIPTION
A	077645	350 PD Guard Sales
B	073641	M10 x 65 SHCS
C	N/A	Movable Guard
D	077167	Snap Ring
E	077202	Spacer Nylon
F	078516	PD Guard Stop
G	073691	M6 x 12 Knob
H	077165	Ring
I	060140	9/16 x 85" Coolant Line
J	077154	Hose Barb
K	077155	Valve
L	073110	M10 Lock Washer
M	069999	Coolant Splitter
MA	069998	Coolant Splitter Ass'y (M, P, Q)
N	077864	M5 x 12 SHCS
O	214005	M5 Washer
P	070001	Coolant Tube (Right)
Q	070002	Coolant Tube (Left)
R	660350	Blade Rotation Decal
S	060345	Danger Decal
T	046300	Ratchet Hose Clamp

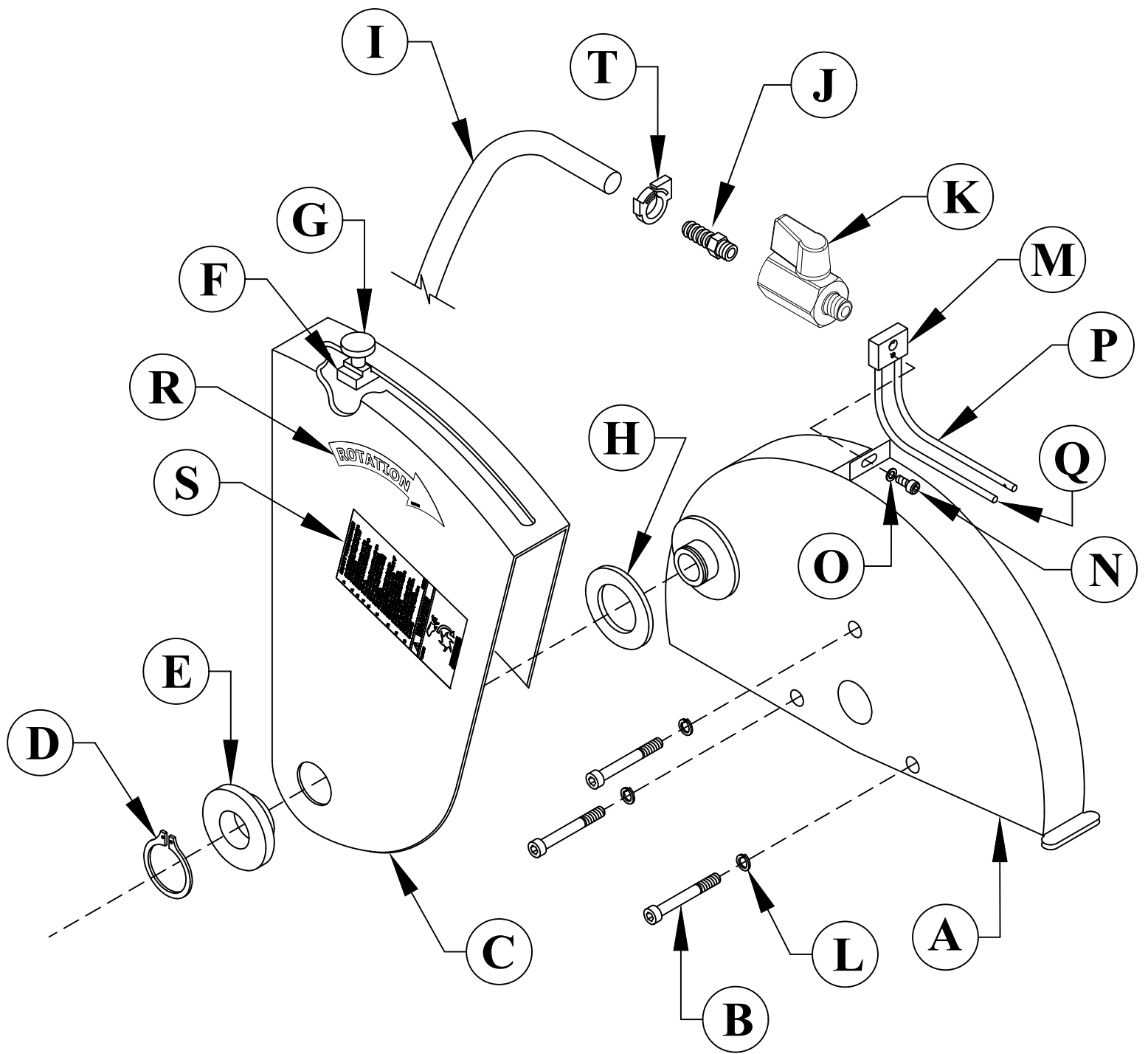


FIGURE 37

10.6 TEN FOOT (304 CM) SUPPLY TRACK

ITEM	QTY	PART #	DESCRIPTION
1	1	029242	10' Roller Conveyor W/Legs (Complete Assy.)
2	2	029243.1	10' Roller Conveyor Side
3	2	029244	Leg Roller Conveyor
4	10	029245	Replacement Rollers For #29243
5	1	029300	Conveyor Brace
6	3	029621	Conveyor Support Bracket
7	12	201205	M10 X 16MM DIN933 HHCS
8	12	208012	M10 DIN 934 Hex Nut
9	2	216015	M10 BN190 Flange Nut
10	2	224205	M10 X 16MM DIN-BN73 WLCS

076938 - 13" GUIDE ASSEMBLY (Optional)

ITEM	QTY	PART #	DESCRIPTION
1	2	026619	SMALL TEE NUT (M10)
2	2	043003	NYLON ROLLER 3.25
3	2	076941	BASE CLIP
4	1	076943	BASE MATERIAL GUIDE 13" PAINT
5	2	214012	M10 DIN125 REGULAR WASHER
6	2	221120	M8 X 25 DIN9121580 12.9 SHCS
7	2	229225	M10 X 12 X 70 SB912

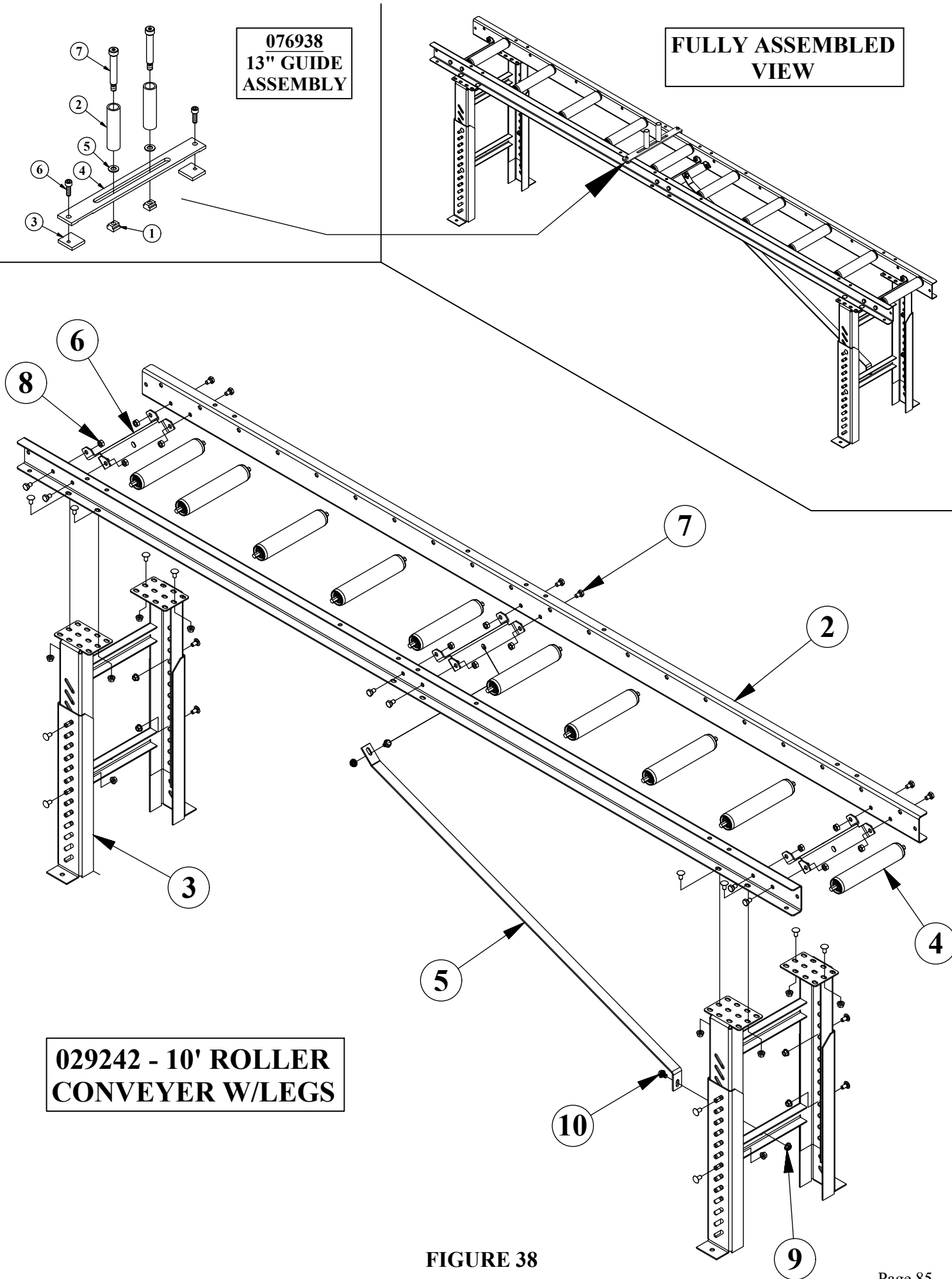


FIGURE 38

10.7 OVERVIEW OF SCOTCHMAN MEASURING SYSTEMS

Scotchman Ind. has several measuring systems available for our saws. They can be attached to our conveyors and increase speed and accuracy. All are AMERICAN MADE and below is the name and brief description of each.

QUICK-LOC:

- System includes: Rail, Handle, Quick-Loc Arm, Measuring Tape & Mounting Hardware
- Patented Loc-Stop System is manufactured from high quality stainless steel and provides a guarantee with teeth
- Teeth on the rail are in 1/16" increments
- The teeth on the stainless-steel stop align themselves to the teeth on the rail, to give you the exact measurement you set
- Guaranteed not to slip - In seconds, your work station is set up and locked in

MULTI-LOC:

- System includes: Rail, 3 Stops (Standard or Heavy Duty), Measuring Tape & Mounting Hardware
- Patented Loc-Stop System is manufactured from high quality stainless steel and provides a guarantee with teeth
- Teeth on the rail are in 1/16" increments
- The teeth on the stainless-steel stop align themselves to the teeth on the rail, to give you the exact measurement you set
- Guaranteed not to slip - In seconds, your work station is set up and locked in
- 3 Stops - standard stops (as close as 1" apart) or heavy duty stops (as close as 1-1/2" apart)

DIGITAL QUICK STOP:

Simply key in the desired cut length and press go; the positioner then moves to that desired length. Place your material up against the fence and make your cut.

- Digital Length Positioner
- Android tablet
- Works in inches, fractions, decimal or metric
- Fraction keys work as the "go" key
- Unlimited preset hot keys and cut list storage
- Keypad has large, easy to read numbers
- Adjustable reach stop
- Supports barcode operations and remote serial commands

PROGRAMMABLE LENGTH STOP OR MATERIAL PUSHER:

This programmable measuring system eliminates operator error and is easily programmed to measure length or has the ability to push material to a desired length. Simply enter a part length and the stop will advance to that position quickly and accurately. This Stop/Pusher System can move one length at a time or be programmed with a series of lengths & quantities for faster operation.

Please visit SCOTCHMAN.COM to see the above or call us at [1-800-843-8844](tel:1-800-843-8844) for more information.

Below is an example of our Quick-Loc Measuring System on a conveyor that is mounted to a saw. To change the measurement you grab the squeeze handle and move the stop to the desired length indicated on the measuring tape. The Quick-Loc Arm is designed to pivot up out of the way if needed.

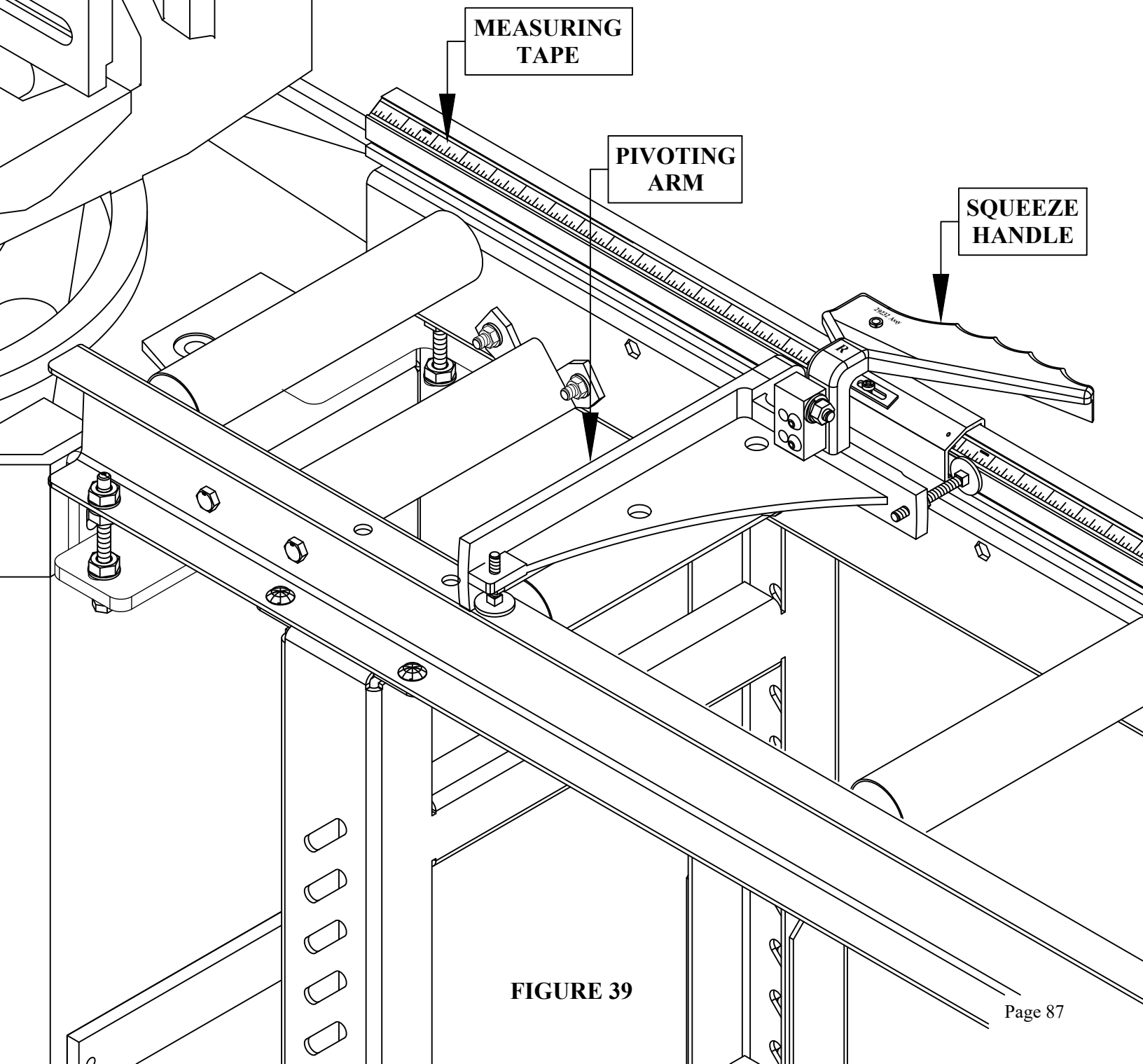


FIGURE 39

10.8 CUTTING COOLANTS AND LUBRICANTS

ITEM	PART #	DESCRIPTION
1 Gal.	075751	Synthetic Coolant
5 Gal.	075752	Synthetic Coolant
55 Gal.	075754	Synthetic Coolant
1 Gal.	075756	Stainless Coolant
5 Gal.	075757	Stainless Coolant
1 Qt.	075753	Air Line Lubricant
1 Gal.	075759	Air Line Lubricant
1 Gal.	075758	Gear Oil - Head
1 Qt.	060520	Hydraulic Oil for Power Down

10.9 MATERIAL STOP 30 INCH (76 CM)

ITEM	PART #	DESCRIPTION
A	677436	Stop Clamp (Includes D)
B	060315	Small Shaft
C	060310	Large Shaft
D	073460	M-10 x 16 SHCS
E	076930	Complete Ass'y (Includes all)
F	080193	M-8 Wrench
G	210016	M-16 Jam Nut

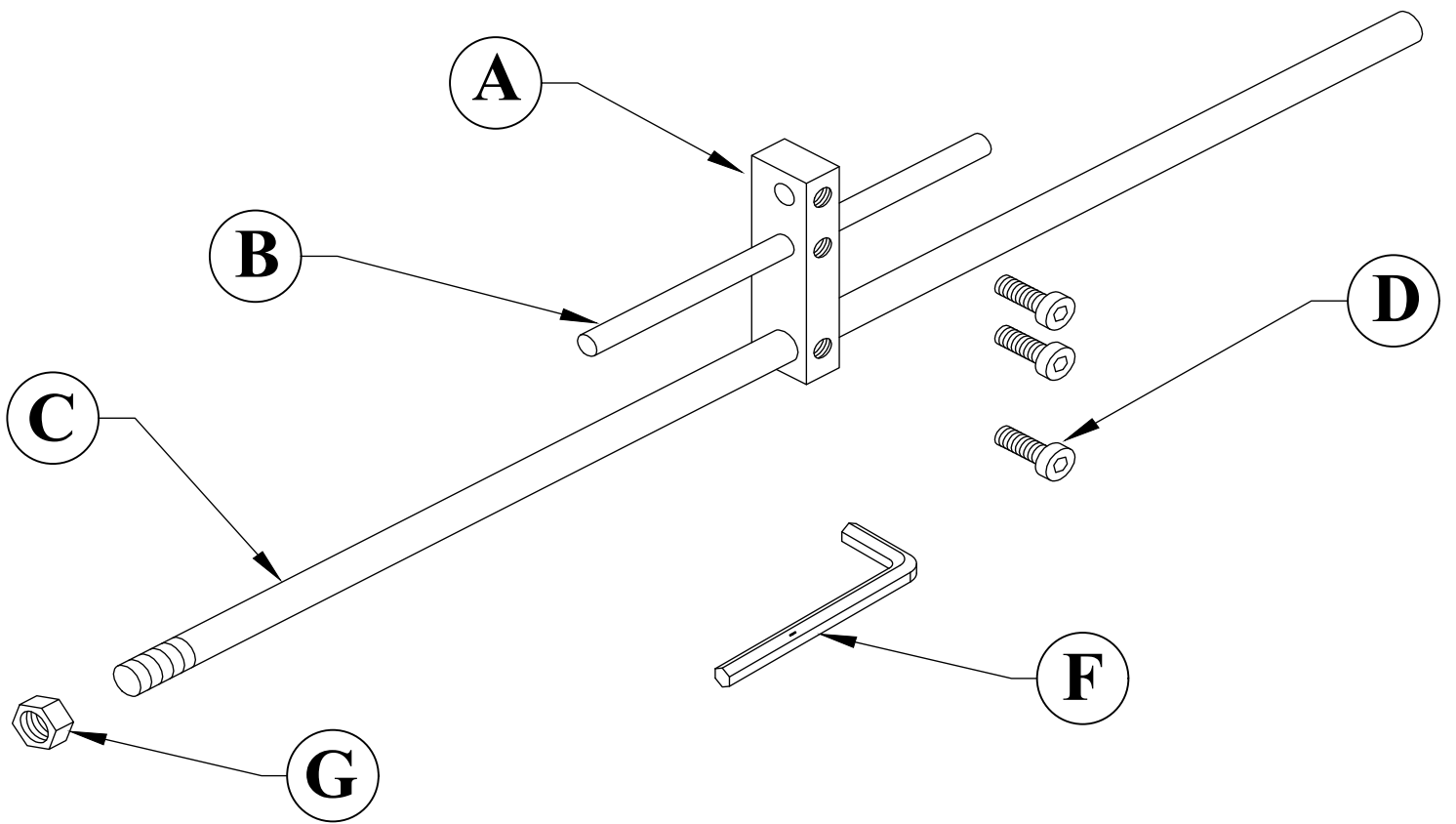
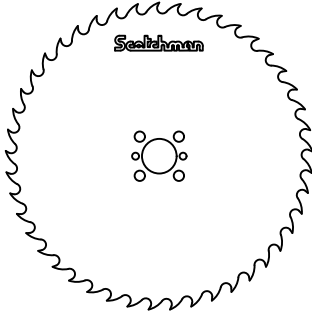


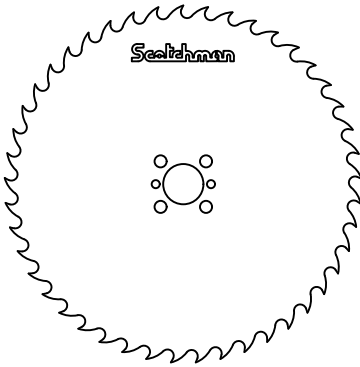
FIGURE 40

11.0 STOCK BLADES

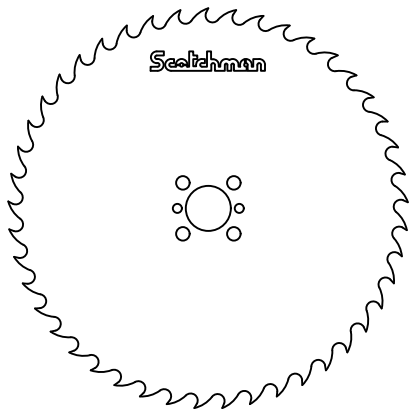
40 MM BORE HSS-DMo5 SAW BLADES (PIN SPACING 2/8/55 & 4/12/64)

ITEM	PART #	<u>10-3/4 INCH (275 MM)</u>	DESCRIPTION
A	074309		90 Tooth
B	074308		100 Tooth
C	074310		120 Tooth
D	074311		150 Tooth
E	074312		180 Tooth
F	074313		220 Tooth
G	074314		260 Tooth

12-1/2 INCH (315 MM)

A	074355		90 Tooth
B	074356		100 Tooth
C	074357		110 Tooth
D	074345		120 Tooth
E	074348		150 Tooth
F	074350		180 Tooth
G	074352		220 Tooth
H	074354		280 Tooth

14 INCH (350 MM)

A	074390		90 Tooth
B	074391		100 Tooth
C	074392		110 Tooth
D	074393		120 Tooth
E	074394		150 Tooth
F	074395		180 Tooth
G	074396		220 Tooth
H	074397		280 Tooth

THERE ARE THREE STYLES OF BLADES AVAILABLE:

STYLE 2: Has a round backed tooth with a square face and top. This style is designed for thin wall nonferrous tubes, plastics and synthetics.

STYLE 2A: Is an alternate top bevel grind. This grind is used on blades that have 240 teeth or more. Applications for this style are thin wall tubes, profiles with thin cross sections and nonferrous applications that require 240 teeth or more.

STYLE 3: Is a triple chip grind with a high/low tooth form. This grind is used on blades that have 220 teeth or less. This style is used for a wide range of materials from solid sections of nonferrous materials to heavy wall tubes and solid sections of steel and alloys.

The stock blades listed are oxide coated high speed steel. Blades with non-standard numbers of teeth can be custom ground and are considered a special saw blade. They can be provided with any number of teeth from 50 to 280. Our Power 2000 blades have a notch grind and have a titanium coating. We recommend them for cutting thin-walled tubing. Please call us for saw blade recommendations or go to www.scotchman.com for more information.

SEND YOUR BLADES BACK TO THE FACTORY FOR PROFESSIONAL RESHARPENING

TYPES OF BEVELS, NOTCHES, AND WHERE USED

ALTERNATE

Teeth are the same Height.
All the Teeth have a Bevel.
Every Other Tooth is Beveled on Every Other Side. Used on Blades with a 4.5 Pitch or Less, Generally, 220 Teeth or more.
More Teeth = Smaller Pitch
Used on thinner walled Tube & Angle and Small Solids.

TRIPLE CHIP

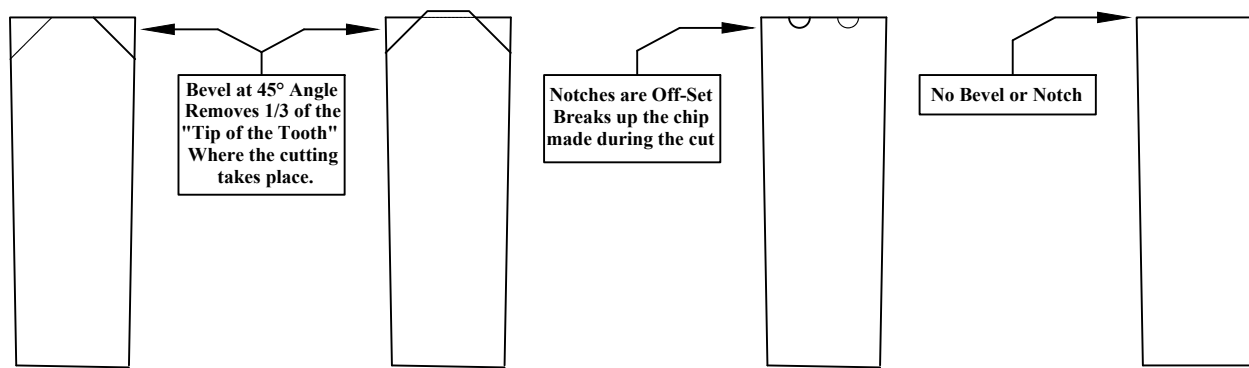
Every other Tooth is Taller.
Often said to have a "High-Low".
The High-Tooth has a Bevel on Both Sides. The Low Tooth has NO Bevel at all. Hi-Tooth cuts the Middle out of the Cut and Low Tooth is the Tooth that leaves Finish.
Generally used on Blades with a 4.5 Pitch or more.
Less Teeth = Larger Pitch.
Best for Solids and Thick Wall Tube.

NOTCH GRIND

Teeth are the same Height.
All the Teeth have a Notch.
Notches are Offset from Each Other and breaks up the Chip made during the Cut. Typically used on Blades with a 4.5 Pitch or Less, Generally, 220 Teeth or more.
More Teeth = Smaller Pitch
Best for thinner walled Tube & Blade RPM should be increased.

ROUND GRIND

Teeth are the same Height.
There is No Bevel. Typically used on Blades with a 3.5 Pitch or Less, as Teeth this small are extremely difficult to bevel. This works best for thinner walled Tube.
Also used for soft materials such as Nylon, Plastic, PVC, etc. in most any shape or size and with whatever tooth size works the best.



**Triple Chip and Alternate Grinds are by far the most common.
Power 2000 Blades have the Notch Grind & Titanium Coating.**

FIGURE 41